

Math 115AH-BH Notes

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Contents

1	Course A	3
1.1	Lecture 1	3
1.2	Lecture 2	5
1.3	Lecture 3	8
1.4	Lecture 4	10
1.5	Lecture 5	11
1.6	Lecture 6	14
1.7	Lecture 7	15
1.8	Lecture 8	16
1.9	Lecture 9	18
1.10	Lecture 10	19
1.11	Lecture 11	21
1.12	Lecture 12	23
1.13	Lecture 13	25
1.14	Lecture 14	27
1.15	Lecture 15	28
1.16	Lecture 16	29
1.17	Lecture 17	33
1.18	Lecture 18	35
1.19	Lecture 19	36
1.20	Lecture 20	37
1.21	Lecture 21	39
1.22	Lecture 22	40
1.23	Lecture 23	42
1.24	Lecture 24	43
1.25	Lecture 25	46
1.26	Lecture 26	48
1.27	Lecture 27	51
1.28	Extra Lectures	53

2	Course B	59
2.1	Lecture 1	59
2.2	Lecture 2	60
2.3	Lecture 3	62
2.4	Lecture 4	63
2.5	Lecture 5	64
2.6	Lecture 6	66
2.7	Lecture 7	68
2.8	Lecture 8	70
2.9	Lecture 9	73
2.10	Lecture 10	74
2.11	Lecture 11	75
2.12	Lecture 12	77
2.13	Lecture 13	78
2.14	Lecture 14	80
2.15	Lecture 15	82
2.16	Lecture 16	85
2.17	Lecture 17	87
2.18	Lecture 18	89
2.19	Lecture 19	92
2.20	Lecture 20	95
2.21	Lecture 21	97
2.22	Lecture 22	98
2.23	Lecture 23	100
2.24	Lecture 24	102
2.25	Lecture 25	103
2.26	Lecture 26	104
2.27	Lecture 27	107
2.28	Lecture 28	109

1 Course A

1.1 Lecture 1

Definition 1.1. A field is a set F with two maps $+: F \times F \rightarrow F$ and $\cdot: F \times F \rightarrow F$ satisfying the following axioms:

- For all $a, b, c \in F$, $(a + b) + c = a + (b + c)$.
- For all $a, b \in F$, $a + b = b + a$.
- There is an element $0 \in F$ such that $0 + a = a + 0 = a$ for all $a \in F$.
- For all $a \in F$, there is a $b \in F$ so that $a + b = b + a = 0$. We write $b = -a$.
- For all $a, b, c \in F$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- For all $a, b \in F$, $a \cdot b = b \cdot a$.
- There is an element $1 \neq 0 \in F$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in F$.
- For all $a \neq 0 \in F$, there is a $b \in F$ so that $a \cdot b = b \cdot a = 1$. We write $b = a^{-1}$.
- For all $a, b, c \in F$, $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(a + b) \cdot c = a \cdot c + b \cdot c$.

We can think of these axioms as ensuring the following:

- Axioms 1 and 2: Addition is nice (associative and commutative).
- Axiom 3: There is an element simulating 0 that does not change things when added to it.
- Axiom 4: We can subtract (if F is a field, and $a, b \in F$, then $(a + (-b)) + b = a + ((-b) + b) = a + 0 = a$, so $a + (-b)$ kind of like “subtracting” b from a).
- Axioms 5 and 6: Multiplication is nice (associative and commutative).
- Axiom 7: There is an element simulating 1 that does not change things when multiplied with them.
- Axiom 8: We can divide, but not by 0 (if F is a field and $a, b \in F$, then $(a \cdot b^{-1}) \cdot b = a \cdot (b^{-1} \cdot b) = a \cdot 1 = a$, so $a \cdot b^{-1}$ is kind of like “dividing” a by b).
- Axiom 9: Addition and multiplication behave well together (distributive laws).

From here on, we will drop using $\cdot$ to represent multiplication, and will stop using parenthesis when adding or multiplying more than 3 terms. We will also write $a - b$, rather than $a + (-b)$.

There are many examples of fields, such as $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$. Note however, that $\mathbb{Z}$ is not (for example, 2 does not have a multiplicative inverse in $\mathbb{Z}$). There are also finite fields of

prime order. If p is a prime, then we denote by $\mathbb{F}_p$ the field with elements $0, 1, 2, \dots, p-1$ with addition and multiplication defined $\pmod{p}$ (for example, suppose we wanted to find $1 + (p-1)$ inside $\mathbb{F}_p$. This is p normally, and $p \equiv 0 \pmod{p}$, so this is 0). It is a result in number theory that multiplicative inverses exist.

Now, to make sure that our field behaves nicely, we prove the following proposition.

Proposition 1.2. *Let F be a field and $a, b, c \in F$. The following are true.*

- (1) F is nonempty.
- (2) If $a + b = b + a = 0$ and $a + c = c + a = 0$, then $b = c$ (thus, the term $-a$ is well defined).
- (3) If $ab = ba = 1$ and $ac = ca = 1$, then $b = c$ (thus, the term a^{-1} is well defined).
- (4) $--a = a$ and $(a^{-1})^{-1} = a$.
- (5) If $a + b = a + c$, then $b = c$, and if $ab = ac$ with $a \neq 0$, then $b = c$.
- (6) $0a = a0 = 0$.
- (7) $(-a)(b) = (a)(-b) = -(ab)$.
- (8) $(-a)(-b) = ab$.
- (9) If $ab = 0$, then $a = 0$ or $b = 0$.

Proof.

- (1) $0, 1 \in F$, so $|F| \geq 2$.
- (2) Suppose that the conditions specified are met. Then, $b + a + c = (b + a) + c = 0 + c = c$, and $b + a + c = b + (a + c) = b + 0 = b$, so $b = c$.
- (3) Same as in (2).
- (4) Note that $a - a = -a + a = 0$. This can be interpreted as saying that a is the additive inverse of $-a$, rather than that $-a$ is the additive inverse of a . Thus, $a = --a$. Similarly for multiplication.
- (5) If $a + b = a + c$, then $-a + a + b = -a + a + c$, so that $b = b + 0 = c + 0 = c$. Similarly for multiplication.
- (6) $0a = (0 + 0)a = 0a + 0a$. Thus, $0a + 0 = 0a + 0a$, so (5) implies that $0a = 0$. Since addition is commutative, $a0 = 0$.
- (7) We show these are all additive inverses for ab , and (2) implies they are equal. First, $ab + (-a)(b) = (a - a)(b) = 0b = 0$. Second, $ab + (a)(-b) = a(b - b) = a0 = 0$. Finally, $ab - (ab) = 0$.
- (8) $(-a)(-b) = a(--b) = ab$.
- (9) If $a \neq 0$, then $a^{-1}ab = a^{-1}0 = 0$, so $b = 0$.

■

1.2 Lecture 2

Now suppose that $F = \{0, 1\}$, and nothing else. We want to make F into a field. We do the following:

$$\begin{array}{c|cc} + & 0 & 1 \\ \hline 0 & 0 & 1 \\ 1 & 1 & 0 \end{array} \quad \begin{array}{c|cc} \cdot & 0 & 1 \\ \hline 0 & 0 & 0 \\ 1 & 0 & 1 \end{array}$$

Note that this does make it into a field (in fact, this is $\mathbb{F}_2$). Note also that this is the only choice: all values are fixed by the previous proposition except $1 + 1$. If $1 + 1 = 1$, then $1 + 1 = 1 + 0$, so $1 = 0$.

Definition 1.3. Let F be a field. We define $F[t]$ to be the set of all formal polynomials in t , with addition and multiplication defined in the obvious way.

There is also the notion of a degree of a polynomial.

Definition 1.4. Suppose that $f \in F[t]$ is of the form $a_0 + a_1t + \cdots + a_nt^n$, with $a_n \neq 0$. We say that the degree of f is n , and write $\deg f = n$.

Two polynomials in $F[t]$ are equal if and only if they have the same coefficients. Note that $F[t]$ is not a field, as multiplicative inverses do not necessarily exist (for example, $1/t \notin F[t]$). To fix this, we do the following.

Definition 1.5. Let F be a field. We define $F(t) = \left\{ \frac{f}{g} : f, g \in F[t], g \neq 0 \right\}$ under the equivalence relation

$$\frac{f}{g} \sim \frac{h}{k} \iff fk = gh,$$

with addition

$$\frac{f}{g} + \frac{h}{k} = \frac{fk + gh}{gk}$$

and multiplication

$$\frac{f}{g} \cdot \frac{h}{k} = \frac{fh}{gk}.$$

This is a field, called the field of rational polynomials over F .

The zero of $F(t)$ is $\frac{0}{f}$ for any $f \in F[t]$, and the one is $\frac{f}{f}$ for any $f \in F[t]$. We can iterate this process. We define $F(t_1, t_2, \dots, t_n)$ as $F(t_1, t_2, \dots, t_{n-1})(t_n)$. Note that these are all fields, so we have in fact proved that there are an infinite number of fields. We can also talk about matrices.

Definition 1.6. Let F be a field. We define

$$\mathbb{M}_n(F) = \{A : A \text{ is an } n \times n \text{ matrix with entries in } F\}.$$

We give this an addition and multiplication as expected.

Once $n > 1$, multiplication is not commutative, and matrices do not necessarily have a multiplicative inverse, so this is not a field. However, it does have 0 as the matrix with all 0s as entries, and 1 as the $n \times n$ identity matrix.

We now discuss vector spaces.

Definition 1.7. Let F be a field. A vector space over F is a set V with two operations $+: V \times V \rightarrow V$ (called vector addition, or sometimes just addition) and $\cdot: F \times V \rightarrow V$ (called scalar multiplication) such that the following are true.

- For all $v_1, v_2, v_3 \in V$, $(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3)$.
- For all $v_1, v_2 \in V$, $v_1 + v_2 = v_2 + v_1$.
- There exists an element $0 \in V$ so that $v + 0 = 0 + v = v$ for all $v \in V$.
- For all $v \in V$, there exists a $w \in V$ so that $v + w = w + v = 0$. We write $w = -v$.
- For all $v \in V$, $1 \cdot v = v$.
- For all $v \in V$ and $\alpha, \beta \in F$, $(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v)$.
- For all $v \in V$ and $\alpha, \beta \in F$, $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$.
- For all $v_1, v_2 \in V$ and $\alpha \in F$, $\alpha \cdot (v_1 + v_2) = \alpha \cdot v_1 + \alpha \cdot v_2$.

The elements of V are called vectors, and the elements of F are called scalars.

For the same reasons as in fields, the zero of V is unique, and additive inverses in V are unique. To distinguish between them, we write 0_F for the zero of F , and 0_V for the zero of V . Similar reasoning also implies that $0_F \cdot v = 0_V$ and that $\alpha \cdot 0_V = 0_V$ for all $v \in V$ and $\alpha \in F$. It is also not too hard to check that $(-\alpha) \cdot v = -(\alpha \cdot v) = \alpha \cdot (-v)$. It is also not too hard to check that $(-1) \cdot v = -v$ for all $v \in V$. We will normally write vectors using Roman letters, and scalars using Greek letters.

Over a field F , the set $F^n = \{(x_1, x_2, \dots, x_n) : x_1, \dots, x_n \in F\}$ is a vector space over F , with addition defined as $(x_1, x_2, \dots, x_n) + (y_1, y_2, \dots, y_n) = (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$, and scalar multiplication defined as $\alpha \cdot (x_1, x_2, \dots, x_n) = (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$. In particular, $\mathbb{R}^n$ is a vector space over $\mathbb{R}$. This also shows that all fields are vector spaces over themselves. We can also find many other examples of vector spaces, however. For example, consider the set $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$. This is a vector space over $\mathbb{Q}$, with addition defined as normal addition $((a + b\sqrt{2}) + (c + d\sqrt{2}) = (a + c) + (b + d)\sqrt{2})$, and scalar multiplication defined as multiplication $(a \cdot (b + c\sqrt{2}) = ab + (ac)\sqrt{2})$. Now let I be an interval in $\mathbb{R}$. We let

$$\text{fcn } I = \{f : I \rightarrow \mathbb{R} : f \text{ is a function}\}.$$

We define addition on this pointwise as $(f + g)(x) = f(x) + g(x)$. We define scalar multiplication as $(\alpha \cdot f)(x) = \alpha f(x)$. This makes $\text{fcn } I$ into a vector space over $\mathbb{R}$.

Remark 1.8. It is only possible to add finitely many vectors, not infinitely many vectors.

Now let us investigate the following question: Can $\mathbb{Z}$ be a vector space over some field F ? Suppose that it was, and let $1_{\mathbb{Z}}$ denote $1 \in \mathbb{Z}$, and let 1_F denote $1 \in F$. Then $1_F \cdot 1_{\mathbb{Z}} = 1_{\mathbb{Z}}$. Let m_F denote $\underbrace{1_F + 1_F + \dots + 1_F}_{m \text{ times}}$, and $m_{\mathbb{Z}}$ denote $m \in \mathbb{Z}$. Now, $2_F \cdot 1_{\mathbb{Z}} = (1_F + 1_F) \cdot 1_{\mathbb{Z}} = 1_F \cdot 1_{\mathbb{Z}} + 1_F \cdot 1_{\mathbb{Z}} = 1_{\mathbb{Z}} + 1_{\mathbb{Z}} = 2_{\mathbb{Z}}$. Thus, 2_F must be nonzero, since $2_{\mathbb{Z}}$ is clearly nonzero. Since 2_F

is nonzero, it has a multiplicative inverse 2_F^{-1} . Now, $2_F^{-1} + 2_F^{-1} = 2_F^{-1}(1_F + 1_F) = 2_F^{-1}2_F = 1_F$. Then, $2_F^{-1} \cdot 1_{\mathbb{Z}} + 2_F^{-1} \cdot 1_{\mathbb{Z}} = (2_F^{-1} + 2_F^{-1}) \cdot 1_{\mathbb{Z}} = 1_F \cdot 1_{\mathbb{Z}} = 1_{\mathbb{Z}}$, meaning that $2_F^{-1} \cdot 1_{\mathbb{Z}} = \frac{1}{2}$. However, by the definition of a vector space, $\alpha \cdot v \in V$ for any $\alpha \in F$ and $v \in V$.

1.3 Lecture 3

Suppose that I is an interval. We define $C(I)$ to be the set of all continuous functions on I . We define $C^n(I)$ to be the set of all continuous functions on I whose n^{th} derivatives exist and are continuous. We can also define $C^\infty(I)$ to be the set of all functions on I whose n^{th} derivatives exist and are continuous for all $n \geq 0$. Finally, we define $C^\omega(I)$ to be the set of all functions on I to be the functions that converge to their Taylor Series in a neighborhood of every $x \in I$. We can also define $\text{Int}(I)$ to be the set of all integrable functions on I . These are all vector spaces over the field $\mathbb{R}$.

Another example of a vector space over a field F is $F[t]$ with regular addition, and scalar multiplication

$$\alpha \cdot (a_0 + a_1t + \cdots + a_nt^n) = \alpha a_0 + \alpha a_1t + \cdots + \alpha a_nt^n.$$

We can also define

$$F[t]_n = \{0 \in F[t]\} \cup \{f \in F[t] : \deg f \leq n\}.$$

Once again, this is a vector space over F (under the same operations of $F[t]$). We can also construct $F^{m \times n}$ to be the set of $m \times n$ matrices with entries in F . This is a vector space over F with addition defined as matrix addition, and scalar multiplication $\alpha \cdot A$ defined by multiplying each element by α .

Now, let F be a field and let V be a vector space over F . Let $S \neq \emptyset$ be a set, and let $\text{fcn}(S, V)$ denote the set of functions from S to V . We define an addition on $\text{fcn}(S, V)$ by $(f+g)(s) = f(s) + g(s)$, and define a scalar multiplication on $\text{fcn}(S, V)$ by $(\alpha \cdot f)(s) = \alpha \cdot f(s)$. This makes $\text{fcn}(S, V)$ into a vector space over F (for instance, its zero is the map $f(s) = 0_V$).

Definition 1.9. Suppose that F and K are fields with $F \subseteq K$. Then F is said to be a subfield of K , and K is said to be an extension of F . We write K/F .

One thing to notice is that K is a vector space over F , with addition defined as normal addition, and scalar multiplication defined as $\alpha \cdot k = \alpha k$, where $\alpha \in F$ and $k \in K$ (this fact is actually much more important than it first seems. For example, this is a key tool in showing that $\sqrt[3]{2}$ is not constructable with a straightedge and compass).

Suppose that $\alpha \cdot v = 0_V$ for some $\alpha \in F$ and $v \in V$. We claim either $\alpha = 0_F$ or $v = 0_V$. If $\alpha \neq 0_F$, then $\alpha^{-1} \cdot (\alpha \cdot v) = \alpha^{-1} \cdot 0_V = 0_V$, and $\alpha^{-1} \cdot (\alpha \cdot v) = (\alpha^{-1}\alpha) \cdot v = 1_F \cdot v = v$, so $v = 0$. This similarly gives that if $\alpha \cdot v = \alpha \cdot w$ and $\alpha \neq 0_F$, then $v = w$, and that if $\alpha \cdot v = \beta \cdot v$ and $v \neq 0_V$, then $\alpha = \beta$ (subtract to get one side to be 0_V , and apply the preceding property).

Definition 1.10. Let V be a vector space over a field F , and let $W \neq \emptyset \subseteq V$. If W is closed under addition (i.e. $w_1 + w_2 \in W$ for all $w_1, w_2 \in W$) and if W is closed under scalar multiplication (i.e. $\alpha \cdot w \in W$ for all $\alpha \in F$ and $w \in W$), then we say that W is a subspace of V .

Note that subspaces W of vector spaces V form vector spaces in their own right (every single property follows immediately for W because they are true in $V \supseteq W$). Note that $0_V \in W$, since if $w \in W$, then $(-1) \cdot w = -w \in W$, and so $w + (-w) = w - w = 0_V \in W$.

Proposition 1.11 (The Subspace Criterion). *Let V be a vector space over the field F , and let $W \subseteq V$ be nonempty. Then the following are equivalent.*

(1) W is a subspace of V .

(2) W is closed under addition and scalar multiplication.

(3) For all $w_1, w_2 \in W$ and $\alpha \in F$, $\alpha \cdot w_1 + w_2 \in W$.

Proof. We show (1) $\implies$ (2) $\implies$ (3) $\implies$ (1). That (1) $\implies$ (2) follows from the definition of a subspace. Now, if (2) holds, then for all $w_1, w_2 \in W$ and $\alpha \in F$, $\alpha \cdot w_1 \in W$ and $w_1 + w_2 \in W$. It follows that $\alpha \cdot w_1 + w_2 \in W$, so (3) holds. Finally, suppose (3). Take $\alpha = 1$ to get that $w_1 + w_2 \in W$ for all $w_1, w_2 \in W$. Take $\alpha = -1$ and $w_1 = w_2$ to get that $(-1) \cdot w_1 + w_1 = -w_1 + w_1 = 0_V \in W$. Finally, take $w_2 = 0_V \in W$ to get that $\alpha \cdot w_1 \in W$ for all $\alpha \in F$ and $w \in W$. Thus, W is closed under addition and scalar multiplication, so it is a subspace. ■

Now, if V is a vector space over a field F , then $0 = \{0_V\}$ is a subspace of V , and V is a subspace of itself. Another example of subspaces is that if $I \subseteq \mathbb{R}$ is an interval, then

$$C^\omega(I) \subsetneq C^\infty(I) \subsetneq \cdots \subsetneq C^n(I) \subsetneq \cdots \subsetneq C^1(I) \subsetneq C(I) \subsetneq \text{Int}(I) \subsetneq \text{fcn}(I)$$

is a chain of subspaces. Note that these are proper containments. For example,

$$f(x) = \begin{cases} e^{-1/x} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

is an element of $C^\infty(I)$, but not $C^\omega(I)$. From here on, we will drop the $\cdot$ in scalar multiplication.

1.4 Lecture 4

Suppose that $W_1 \subseteq W_2 \subseteq V$ where W_1, W_2 are subspaces of V . Then W_1 is a subspace of W_2 . Similarly, if $W_1 \subseteq W_2$ is a subspace, and $W_2 \subseteq V$ is a subspace, then $W_1 \subseteq V$ is a subspace.

Definition 1.12. Let V be a vector space over the field F . Let $v_1, v_2, \dots, v_n \in V$. We say that $v \in V$ is a linear combination of $v_1, v_2, \dots, v_n$ if there are $\alpha_1, \dots, \alpha_n \in F$ so that $v = \alpha_1 v_1 + \dots + \alpha_n v_n$. We say the span of $v_1, \dots, v_n$ is set of all linear combinations of $v_1, \dots, v_n$. We write $\text{Span}(v_1, \dots, v_n)$.

If $v_1, v_2, \dots, v_n \in V$, then $\text{Span}(v_1, \dots, v_n)$ is a subspace of V , since it is closed under addition and scalar multiplication. In fact, the span of $v_1, \dots, v_n$ is the (unique) smallest subspace of V containing $v_1, \dots, v_n$ in the sense that if W is a subspace of V containing $v_1, \dots, v_n$, then $\text{Span}(v_1, \dots, v_n) \subseteq W$. We set $\text{Span} \emptyset = \{0_V\} = 0$.

Definition 1.13. Let V be a vector space over the field F , and let $S \neq \emptyset \subseteq V$. Then we define $\text{Span } S$ to be the set of all finite linear combinations of vectors in S .

We also define $\text{Symm}_n(F)$ to be the set of all symmetric matrices in $\mathbb{M}_n(F)$. This is a subspace of $\mathbb{M}_n(F)$. Now let $A \in F^{m \times n}$. We define

$$\ker A = \{x \in F^{n \times 1} : Ax = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \in F^{m \times 1}\}.$$

We call this the kernel or null space of A . Now, if $S \subseteq V$, then

$$\text{Span } S = \bigcap_{\substack{W \subseteq V \text{ is a subspace} \\ S \subseteq W}} W.$$

(The intersection is a subspace containing S , so it contains $\text{Span } S$, but $\text{Span } S$ is one of the sets in the intersection, so the intersection is contained in $\text{Span } S$).

Definition 1.14. Let $W_1, W_2 \subseteq V$ be subspaces. We define

$$W_1 + W_2 = \{w_1 + w_2 : w_1 \in W_1, w_2 \in W_2\} = \text{Span}(W_1 \cup W_2).$$

More generally, if $W_i \subseteq V$ are subspaces indexed by I , we define

$$\sum_I W_i = \text{Span} \left(\bigcup_I W_i \right).$$

1.5 Lecture 5

Definition 1.15. Let V be a vector space over the field F , and let $v_1, \dots, v_n \in V$. We say that $\{v_1, \dots, v_n\}$ is linearly dependent if there are $\alpha_1, \dots, \alpha_n \in F$ at least one of which is nonzero, such that

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0_V.$$

The set $\{v_1, \dots, v_n\}$ is said to be linearly independent if, whenever there are $\alpha_1, \dots, \alpha_n \in F$ so that

$$\alpha_1 v_1 + \dots + \alpha_n v_n = 0_V,$$

then $\alpha_1 = \dots = \alpha_n = 0_F$.

Linear independence can be imagined as saying that you cannot write v_i as a linear combination of $v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n$. (If you could write $v_i = \alpha_1 v_1 + \dots + \alpha_{i-1} v_{i-1} + \alpha_{i+1} v_{i+1} + \dots + \alpha_n v_n$, then $\alpha_1 v_1 + \dots + \alpha_{i-1} v_{i-1} - v_i + \alpha_{i+1} v_{i+1} + \dots + \alpha_n v_n = 0$, and since $-1 \neq 0$, $v_1, v_2, \dots, v_n$ are not linearly independent. On the other hand, if $0 = \alpha_1 v_1 + \dots + \alpha_n v_n$ with $\alpha_i \neq 0$ for some i , then $v_i = -\alpha_1 \alpha_i^{-1} v_1 - \alpha_2 \alpha_i^{-1} v_2 - \dots - \alpha_{i-1} \alpha_i^{-1} v_{i-1} - \alpha_{i+1} \alpha_i^{-1} v_{i+1} - \dots - \alpha_n \alpha_i^{-1} v_n$). Another way that linear independence can be imagined is as making linear combinations unique in the sense that if $\alpha_1 v_1 + \dots + \alpha_n v_n = \beta_1 v_1 + \dots + \beta_n v_n$, then $\alpha_i = \beta_i$ for all i (to see this, subtract to get that $(\alpha_1 - \beta_1) v_1 + \dots + (\alpha_n - \beta_n) v_n = 0$, and use the definition to get that $\alpha_i - \beta_i = 0$ for all i , so $\alpha_i = \beta_i$). Note also that a set of vectors is linearly dependent if and only if the set is not linearly independent.

More generally, a nonempty set $S \subseteq V$ is called linearly dependent if some finite subset of S is linearly dependent/ We call S linearly independent if every finite subset of S is linearly independent.

Now suppose that $S \subseteq V$ is nonempty. If $0 \in S$, then S is linearly dependent: $1 \cdot 0 = 0$. In fact, we have the following.

Proposition 1.16. Let V be a vector space over the field F , and let $v_1, \dots, v_n \in V$. Then $v_1, \dots, v_n$ are linearly independent if and only if $v_i \neq 0$ for all i , and if $\alpha_1 v_1 + \dots + \alpha_n v_n = \beta_1 v_1 + \dots + \beta_n v_n$, then $\alpha_i = \beta_i$ for all i .

Proof. If $v_1, \dots, v_n$ are linearly independent, the statement follows from the remarks above. Now suppose that we have uniqueness in this sense. Suppose that $\alpha_1 v_1 + \dots + \alpha_n v_n = 0 = 0v_1 + \dots + 0v_n$. Then, by assumption, $\alpha_i = 0$ for all i (note that the fact $v_i \neq 0$ actually follows from the other assumption: if $v_1 = 0$, then $(\alpha_1)v_1 + \dots + \alpha_n v_n = (\alpha_1 + 1)v_1 + \dots + \alpha_n v_n$, so $\alpha_1 + 1 = \alpha_1$, a contradiction.). ■

Now note that $v_1, \dots, v_n$ are distinct, that $n > 2$, and that no $v_i = 0$. If no v_i is a scalar multiple of another v_j for $j \neq i$, it does not follow that $v_1, \dots, v_n$ are linearly independent. For example, take $v, w, w + v$, where v is not a scalar multiple of w . Note that $v + w$ cannot be a scalar multiple of v or w , since this implies that v is a scalar multiple of w (or the other way, but these two statements are equivalent). However, there is the linear dependence relation $(1)(w + v) + (-1)(w) + (-1)(v) = 0$.

If $S \subseteq V$ is nonempty and linearly independent, then $\emptyset \neq T \subseteq S$ is also linearly independent. Also, if T is linearly dependent, so is S . Note also that $\{v\}$ is linearly independent for all nonzero $v \in V$, since we have previously proved that if $\alpha v = 0$, then $\alpha = 0$ or $v = 0$. We call $Fv = \text{Span}(v)$ a line in V . Similarly, if $v, w \in V \setminus \{0\}$, and $v \notin \text{Span}(w)$, then $\{v, w\}$ is linearly independent (suppose that $\alpha_1 v + \alpha_2 w = 0$ is a linear dependence relation. If $\alpha_1 = 0$, then $\alpha_2 = 0$, so $\alpha_1 \neq 0$. Thus, $v = (-\alpha_2 \alpha_1^{-1})w \in \text{Span}(w)$), and $\text{Span}(v, w)$ is called a plane in V .

Suppose that $v_1, \dots, v_m \in \mathbb{R}^n$ are linearly dependent, and write $v_i = (a_{i,1}, \dots, a_{i,n})$. If $\alpha_1 v_1 + \dots + \alpha_m v_m$ is a linear dependence relation, write

$$A = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{pmatrix}.$$

Then, $B = A^\top$ has a nontrivial solution (namely, the vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$).

Now suppose that $f_1, \dots, f_n \in C^{n-1}(I)$, where $I = (\alpha, \beta) \subseteq \mathbb{R}$. Suppose that there are $\alpha_1, \dots, \alpha_n \in \mathbb{R}$ such that $\alpha_1 f_1 + \dots + \alpha_n f_n = 0$. Then we have the following equations

$$\begin{aligned} \alpha_1 f_1 + \dots + \alpha_n f_n &= 0 \\ \alpha_1 f'_1 + \dots + \alpha_n f'_n &= 0 \\ &\vdots \\ \alpha_1 f_1^{(n-1)} + \dots + \alpha_n f_n^{(n-1)} &= 0. \end{aligned}$$

In matrix form, we have that

$$\begin{pmatrix} f_1 & \cdots & f_n \\ f'_1 & \cdots & f'_n \\ \vdots & \ddots & \vdots \\ f_1^{(n-1)} & \cdots & f_n^{(n-1)} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

If the determinant of this matrix is nonzero, it is invertible. Thus, $\alpha_1 = \dots = \alpha_n = 0$.

Definition 1.17. Let $x \in \mathbb{R}$, and define the map

$$e_x : \mathbb{R}[t] \rightarrow \mathbb{R}$$

by

$$g = \sum a_i t^i \mapsto \sum a_i x^i.$$

This map is called evaluation at x .

Theorem 1.18. Let V be a vector space over the field F , and let $S \subseteq V$ be nonempty and linearly independent. Suppose that $v \in V \setminus \text{Span}(S)$. Then $S \cup \{v\}$ is linearly independent.

Proof. Suppose that $S \cup \{v\}$ is linearly dependent. Then we may find $v_1, \dots, v_{n+1} \in S \cup \{v\}$ such that there are $\alpha_1, \dots, \alpha_{n+1}$ not all zero so that $\alpha_1 v_1 + \dots + \alpha_{n+1} v_{n+1} = 0$. Note that one of the v_i 's must be v , as S is linearly independent, and that α_i must be nonzero for the same reason. Write $\alpha v + \alpha_1 v_1 + \dots + \alpha_n v_n = 0$. Then, $v = (-\alpha^{-1} \alpha_1) v_1 + \dots + (-\alpha^{-1} \alpha_n) v_n \in \text{Span}(S)$, a contradiction. ■

Corollary 1.19. *Let V be a vector space over the field F , and let $v_1, \dots, v_n \in V$ be linearly independent. If*

$$\text{Span}(v_1, \dots, v_n) \subsetneq V,$$

then there is $v_{n+1} \in V$ such that $v_1, \dots, v_n, v_{n+1}$ are linearly independent, and

$$\text{Span}(v_1, \dots, v_n) \subsetneq \text{Span}(v_1, \dots, v_{n+1}) \subseteq V.$$

1.6 Lecture 6

Definition 1.20. Let V be a nontrivial vector space over the field F . A basis for V is a linearly independent set B such that $V = \text{Span } B$. We say that V is a finite dimensional vector space over F if there is a basis B for V with finitely many elements.

If $V = 0$, we say that V has dimension zero.

Theorem 1.21. Let V be a finite dimensional vector space over F with basis $B = \{v_1, \dots, v_n\}$ and $v \in V$. Then there exist unique $\alpha_1, \dots, \alpha_n \in F$ so that $v = \alpha_1 v_1 + \dots + \alpha_n v_n$. We call $\alpha_1, \dots, \alpha_n$ the coordinates of v relative to the bases B .

Proof. The existence follows from the definition of a basis, and the uniqueness follows because if $\alpha_1 v_1 + \dots + \alpha_n v_n = \beta_1 v_1 + \dots + \beta_n v_n$, then $(\alpha_1 - \beta_1)v_1 + \dots + (\alpha_n - \beta_n)v_n = 0$, and since B is linearly independent, $\alpha_i - \beta_i = 0$ for all i . ■

Now suppose that V is spanned by a finite number (n) elements. Then V is finite dimensional. More specifically, if we have the set $\{v_1, \dots, v_n\}$ to be spanning, then a subset of this set is a basis. We prove this by induction. If $n = 1$, then this is trivially true. Now, if $v_1, \dots, v_n$ are linearly independent then we are done. If not, write $v_1 = \alpha_2 v_2 + \dots + \alpha_n v_n$ (note that this is possible by remarks from last lecture). For any v , we can write $v = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n = (\beta_1 \alpha_2 + \beta_2)v_2 + \dots + (\beta_1 \alpha_n + \beta_n)v_n$, so the set $\{v_2, \dots, v_n\}$ is spanning. This has $n - 1$ elements and is spanning so the result follows by induction.

Let us show that $F[t]$ is not finite dimensional. Suppose that it had a finite basis $\{v_1, \dots, v_n\}$, and let d_n denote the degree of v_n . Write $d = \max(d_1, \dots, d_n)$. Then, t^{d+1} cannot be in the span of $\{v_1, \dots, v_n\}$, so $F[t]$ cannot be finite dimensional. The same is not true for $F[t]_n$, since $\{1, t, \dots, t^n\}$ is a finite basis for this.

1.7 Lecture 7

Theorem 1.22. *Let V be a vector space over the field F , and let $\{v_1, \dots, v_n\}$ be a basis of V . Suppose that $v \in V$ satisfies $v = a_1v_1 + \dots + a_nv_n$, where $a_i \neq 0$ for some i . Then, $\{v_1, \dots, v_{i-1}, v, v_{i+1}, \dots, v_n\}$ is a basis of V .*

Proof. Without loss of generality, suppose that $i = 1$. Note that there cannot be a linear dependence relation between $v, v_2, \dots, v_n$, as this would imply a linear dependence relation on $v_1, \dots, v_n$ (just plug in $v = a_1v_1 + \dots + a_nv_n$). Now we show this spans. Suppose that $w \in V$ has the form $b_1v_1 + b_2v_2 + \dots + b_nv_n$. Note that $v_1 = a_1^{-1}v - a_1^{-1}a_2v_2 - \dots - a_1^{-1}a_nv_n$. Thus, we may plug this in to the expression for w to write w as a linear combination of $v, v_2, \dots, v_n$, meaning it is spanning. ■

Theorem 1.23. *Suppose that V is a vector space over the field F with $V = \text{Span}(v_1, \dots, v_n)$. Then any linearly independent subset of V has at most n elements.*

Proof. We know that a subset of $B = \{v_1, \dots, v_n\}$ is a basis for V , so we may assume that B is a basis for V . Now suppose that $\{w_1, \dots, w_m\} \subseteq V$ is a linearly independent set (in particular, each $w_i \neq 0$). We show that $m \leq n$. We show that, rearranging if necessary, for each $k \leq n$, $\{w_1, \dots, w_k, v_{k+1}, \dots, v_n\}$ is a basis. This suffices because, taking $k = n$, we get that $w_1, \dots, w_n$ is a basis, so there cannot be a w_{n+1} , because this can be written as a linear combination of $w_1, \dots, w_n$. Now we prove our claim by induction on k . If $k = 1$, we may write $w_1 = a_1v_1 + \dots + a_nv_n$. Rearranging if necessary, we may assume that $a_1 \neq 0$, so that the previous theorem implies that $\{w_1, v_2, \dots, v_n\}$ is a basis. Now suppose that the claim is true for $k < n$, so that $\{w_1, \dots, w_k, v_{k+1}, \dots, v_n\}$ is a basis for V . We may write $w_{k+1} = b_1w_1 + \dots + b_kw_k + a_{k+1}v_{k+1} + \dots + a_nv_n$. We have two cases. If $a_{k+1} = a_{k+2} = \dots = a_n = 0$, then $w_{k+1} \in \text{Span}(w_1, \dots, w_k)$, contradicting the fact that $w_1, \dots, w_m$ are linearly independent. Otherwise, there is some i so that $a_i \neq 0$. Rearranging, assume that $i = k + 1$. By the replacement theorem, $\{w_1, \dots, w_k, w_{k+1}, v_{k+2}, \dots, v_n\}$ is a basis for V . ■

Theorem 1.24. *Let V be a finite dimensional vector space over the field F . If B_1 and B_2 are two bases for V , then $|B_1| = |B_2|$. We call $|B_1|$ the dimension of V over F , and write $\dim_F V$. When F is clear, we write $\dim V$.*

Proof. Note that B_1 and B_2 are linearly independent. The theorem implies that $|B_1| \leq |B_2|$ and $|B_2| \leq |B_1|$, so $|B_1| = |B_2|$. ■

It is true that if V is not finite dimensional, then any two bases of V have the same cardinality (i.e. there exists a bijection between them).

1.8 Lecture 8

Proposition 1.25. *Let V be a finite dimensional vector space over F , and let $W \subseteq V$ be a subspace. Then every linearly independent subset in W is finite and part of a basis for W , which is a finite dimensional vector space over F .*

Proof. Any linearly independent set S in W is linearly independent in V , so $|S| \leq \dim V$. In particular, $\dim \text{Span } S \leq \dim V$. If $W = \text{Span } S$, we are done. If not, take $w_1 \in W \setminus \text{Span } S$, and let $S_1 = S \cup \{w_1\}$. This is linearly independent, and $|S_1| = |S| + 1$. Repeat this process. Because V is finite dimensional, this process must terminate. ■

Corollary 1.26. *Let V be a finite dimensional vector space over F . Then any linearly independent set in V can be extended to a basis for V .*

Theorem 1.27. *Let V be a vector space over F , and let $W_1, W_2 \subseteq V$ be finite dimensional subspaces. Then,*

- (1) $W_1 \cap W_2$ is a finite dimensional vector space over F .
- (2) $W_1 + W_2$ is a finite dimensional vector space over F .
- (3) $\dim W_1 + \dim W_2 = \dim(W_1 + W_2) + \dim(W_1 \cap W_2)$.

Proof.

- (1) Since $W_1 \cap W_2$ is a subspace of W_1 , it must be finite dimensional.
- (2) Let B_i be a basis for W_i . Then $W_1 + W_2 = \text{Span}(B_1 \cup B_2)$. Since $|B_1 \cup B_2| = |B_1| + |B_2| - |B_1 \cap B_2| \leq |B_1| + |B_2|$, there is a finite spanning set of $W_1 + W_2$. This can be reduced to a (finite) basis.
- (3) Let $B = \{v_1, \dots, v_n\}$ be a basis for $W_1 \cap W_2$. We extend B to a basis B_1 of W_1 , and to a basis B_2 of W_2 . Note that $B_1 \cap B_2 = B$. Thus, $\dim(W_1 + W_2) = \dim \text{Span}(B_1 \cup B_2) = |B_1| + |B_2| - |B_1 \cap B_2| = |B_1| + |B_2| - |B|$ (we only need to verify that $B_1 \cup B_2$ is linearly independent. If the coefficients of $B_1 \setminus B$ are zero, we have a contradiction, as B_2 is linearly independent. By similar reasoning, the coefficients of $B_2 \setminus B$ cannot all be zero. If both of these are satisfied, we will obtain a linear combination of $B_2 \setminus B$ that is equal to a linear combination of elements of B_1 , and so it must be an element of $W_1 \cap W_2$. However, since B_1 and B_2 are bases, we obtain that the coefficients of $B_1 \setminus B$ are zero, a contradiction). Rearranging, $\dim(W_1 + W_2) + \dim(W_1 \cap W_2) = \dim(W_1) + \dim(W_2)$. ■

If $W_1 \cap W_2 = \{0\}$, we write $W_1 \oplus W_2$ for $W_1 + W_2$. More generally, we write $\bigoplus_I W_i$ for $\sum_I W_i$ if $W_j \cap (\sum_{I \setminus \{j\}} W_i) = \{0\}$ for each j . This is equivalent to the statement that if $\sum_I w_i = 0$ for $w_i \in W_i$, then $w_i = 0$ for each i , and the statement that if $\sum_I w_i = \sum_I v_i$ for $w_i, v_i \in W_i$, then $w_i = v_i$. We now discuss linear transformations.

Definition 1.28. Let V and W be vector spaces over F . A map $T : V \rightarrow W$ is called a linear transformation if, for all $v_1, v_2 \in V$ and $\alpha \in F$, $T(v_1 + v_2) = T(v_1) + T(v_2)$ and $T(\alpha v_1) = \alpha T(v_1)$.

Note that an immediate implication of the definition of a linear transformation is that $T(0_V) = 0_W$, as $T(0_V) = T(0_V + 0_V) = T(0_V) + T(0_V)$. We may also write Tv rather than $T(v)$. Note also that T is linear if and only if $T(\alpha v_1 + v_2) = \alpha T(v_1) + T(v_2)$ for all $v_1, v_2 \in V$ and $\alpha \in F$. One direction is clear. To see the second, take $\alpha = 1$ and $v_2 = 0$ for (1) and (2) respectively.

We now consider examples of linear transformations. The zero map from V to W is defined as $Tv = 0_W$ for all $v \in V$, and the identity map from V to itself is defined by $Tv = v$ for all $v \in V$. A linear transformation $T : V \rightarrow V$ is called a linear operator. Now if $W \subseteq V$ is a subspace, we have the linear transformation defined by $T : w \in W \mapsto w \in V$ called the inclusion map. Restricting this to W gives the identity map.

1.9 Lecture 9

Definition 1.29. Let V and W be vector spaces over a field F , and let $T : V \rightarrow W$ be a linear transformation. We define $\mathcal{N}(T) = \ker(T) = \{v \in V : Tv = 0_W\}$. We call this the nullspace or kernel of T . We also define $\text{im}(T) = T(V) = \{w \in W : \exists v \in V, Tv = w\}$. This is called the image or the range.

We claim that $\ker T$ is a subspace of V and $\text{im } T$ is a subspace of W . To see this, notice that $0_V \in \ker T$, and if $v, w \in \ker T$, then $T(v + w) = Tv + Tw = 0_W + 0_W = 0_W$ and that $T(\alpha v) = \alpha Tv = \alpha 0_W = 0_W$. Notice also that if $v, w \in \text{im } T$ with $v = Tv_0$ and $w = Tw_0$, then $v + w = Tv_0 + Tw_0 = T(v_0 + w_0)$ and that $\alpha v = \alpha Tv_0 = T(\alpha v_0)$, and that $0_W \in \text{im } T$.

Theorem 1.30. Let V and W be vector spaces over a field F , and let V be finite dimensional. Let $T : V \rightarrow W$ be a linear transformation. Then $\text{im } T$ and $\ker T$ are finite dimensional vector spaces over F , and $\dim V = \dim \ker T + \dim \text{im } T$.

Proof. Let $n = \dim V$. Since $\ker T$ is a subspace of V , it must be a finite dimensional vector space. Write $m = \dim \ker T$. Let $B_0 = \{v_1, \dots, v_m\}$ be a basis for $\ker T$ and extend it to a basis for V $B = \{v_1, \dots, v_m, \dots, v_n\}$. We claim that $C = \{Tv_{m+1}, \dots, Tv_n\}$ are a basis for $\text{im } T$. Take $w = Tv$, and write $v = a_1v_1 + \dots + a_nv_n$. Then, $Tv = a_1Tv_1 + \dots + a_mTv_m + \dots + a_nTv_n$. Since $v_1, \dots, v_m \in \ker T$, we have that $w = a_{m+1}Tv_{m+1} + \dots + a_nTv_n$, so C spans. Now suppose that $a_{m+1}Tv_{m+1} + \dots + a_nTv_n = 0_W$. Thus, $0_W = T(a_{m+1}v_{m+1} + \dots + a_nv_n)$, and so $a_{m+1}v_{m+1} + \dots + a_nv_n \in \ker T$, say $a_1v_1 + \dots + a_mv_m$. Thus, $-a_1v_1 - \dots - a_mv_m + a_{m+1}v_{m+1} + \dots + a_nv_n = 0$, and since B is a basis, we have that $-a_1 = \dots = -a_m = a_{m+1} = \dots = a_n = 0_V$, and so C is linearly independent. Thus, C is a basis for $\text{im } T$. By our construction of B and C , $\dim V = \dim \ker T + \dim \text{im } T$. ■

Now, if V and W are vector spaces, let $L(V, W)$ be the set of all linear transformations from V to W . Note that $L(V, W)$ is a vector space over F with $(T + U)(v) = Tv + Uv$ and $(\alpha T)v = \alpha(Tv)$.

Definition 1.31. Let V be a vector space over the field F , and let $T : V \rightarrow V$ be a linear operator. We define

$$T^n = \begin{cases} 1_V & n = 0 \\ T \circ T^{n-1} & n \in \mathbb{Z}^+ \end{cases}.$$

1.10 Lecture 10

Definition 1.32. Let V and W be vector spaces over the field F . Let $T : V \rightarrow W$ be linear.

- (1) We say that T is a monomorphism or nonsingular if T is injective.
- (2) We say that T is an epimorphism if T is surjective.
- (3) We say that T is an isomorphism if T is bijective and $T^{-1} : W \rightarrow V$ is linear.

We say that V and W are isomorphic if there exists an isomorphism $T : V \rightarrow W$. In this case, we write $V \cong W$, and we will write the isomorphism as $T : V \xrightarrow{\sim} W$.

Note that isomorphism is an equivalence relation on the set of vector spaces over F . We claim that a linear map T is an isomorphism if and only if it is bijective. Clearly, isomorphisms are bijective. Now, let $w_1 = Tv_1$ and $w_2 = Tv_2$. Then,

$$T^{-1}(\alpha w_1 + w_2) = T^{-1}(\alpha Tv_1 + Tv_2) = T^{-1}(T(\alpha v_1 + v_2)) = \alpha v_1 + v_2 = \alpha T^{-1}(w_1) + T^{-1}(w_2),$$

so T^{-1} is linear.

Definition 1.33. Let $T : V \rightarrow W$ be linear. We say that T takes linearly independent sets to linearly independent sets if, whenever $(v_i)_{i \in I}$ are linearly independent in V , $(Tv_i)_{i \in I}$ are distinct and linearly independent in W .

Theorem 1.34. Let $T : V \rightarrow W$ be a linear transformation. Then the following are equivalent.

- (1) T is injective.
- (2) T takes linearly independent sets in V to linearly independent sets in W .
- (3) $\ker T = \{0\}$.
- (4) $\dim \ker T = 0$.

Proof. Suppose that T is injective, and that $v_1, \dots, v_n \in V$ are linearly independent. Suppose that $0_W = \alpha_1 Tv_1 + \dots + \alpha_n Tv_n$. Then $T(0_V) = T(\alpha_1 v_1 + \dots + \alpha_n v_n)$, so $\alpha_1 v_1 + \dots + \alpha_n v_n = 0_V$, and so $\alpha_1 = \dots = \alpha_n = 0_F$. Now suppose that T takes linearly independent sets in V to linearly independent sets in W , and let $v \in \ker T$ be nonzero. Then, $\{v\}$ is linearly independent, and so $\{Tv\}$ must be linearly independent. However, $Tv = 0_W$ is linearly dependent, a contradiction. Thus, no nonzero elements exist in the kernel, and so $\ker T = \{0\}$. If $\ker T = \{0\}$, clearly $\dim \ker T = 0$. Finally, suppose that $\dim \ker T = 0$, and that $Tv = Tw$. Then, $T(v - w) = 0_W$, and so $v - w \in \ker T$. Since $\dim \ker T = 0$, we have that $\ker T = \{0\}$, and so $v - w = 0$. Thus, $v = w$, and so T is injective. ■

Theorem 1.35. Suppose that V and W are finite dimensional vector spaces over F with $\dim V = \dim W$. Let $T : V \rightarrow W$ be a linear transformation. Then the following are equivalent.

- (1) T is an isomorphism.

(2) T is a monomorphism.

(3) T is an epimorphism.

(4) If $B = \{v_1, \dots, v_n\}$ is a basis for V , then $\{Tv_1, \dots, Tv_n\}$ is a basis for W .

(5) There exists a basis B of V that maps to a basis of W .

Proof. That (1) $\Rightarrow$ (2) is clear. Suppose that T is a monomorphism, so that $\dim \ker T = 0$. Thus, $\dim \operatorname{im} T = \dim V = \dim W$. Since $\dim \operatorname{im} T$ is a subspace of W , it must be equal to all of W , so that T is an epimorphism. Now, if T is an epimorphism, then $\{Tv_1, \dots, Tv_n\}$ spans, so we only need to show that it is linearly independent. If it was linearly dependent, then (rearranging if necessary), $\{Tv_1, \dots, Tv_{n-1}\}$ spans, which is impossible (it implies linearly independent sets of size n cannot exist, which they must by the definition of dimension). That (4) $\Rightarrow$ (5) is clear. Now, if $\{v_1, \dots, v_n\}$ is a basis of V mapping to a basis of W , then T is clearly surjective (if $w = a_1Tv_1 + \dots + a_nTv_n$, then $w = T(a_1v_1 + \dots + a_nv_n)$). Now if $Tv = Tw$, then $T(v - w) = 0_W$. Write $v = a_1v_1 + \dots + a_nv_n$ and $w = b_1v_1 + \dots + b_nv_n$. Then $v - w = (a_1 - b_1)v_1 + \dots + (a_n - b_n)v_n$, and so $0_W = T(v - w) = (a_1 - b_1)Tv_1 + \dots + (a_n - b_n)Tv_n$. Since $\{Tv_1, \dots, Tv_n\}$ is a basis, $a_1 - b_1 = \dots = a_n - b_n = 0$, and so $a_i = b_i$ for all i . Thus, $v = w$ and T is injective, and so bijective, and so a linear transformation. ■

Theorem 1.36 (Universal Property of Vector Spaces). *Let V be a finite dimensional vector space over F , and let $B = \{v_1, \dots, v_n\}$ be a basis for V . Let W be a vector space over F , and let $w_1, \dots, w_n \in W$ (not necessarily distinct). Then there exists a unique linear transformation T with $T : v_i \mapsto w_i$.*

Proof. Define T by

$$T : \alpha_1v_1 + \dots + \alpha_nv_n \mapsto \alpha_1w_1 + \dots + \alpha_nw_n.$$

This satisfies $v_i \mapsto w_i$, and is linear (note that this is well defined and is defined everywhere, as $v_1, \dots, v_n$ is a basis). Thus, such a linear transformation exists. Now suppose that T is a linear transformation satisfying $Tv_i = w_i$. Then,

$$T(\alpha_1v_1 + \dots + \alpha_nv_n) = \alpha_1Tv_1 + \dots + \alpha_nTv_n = \alpha_1w_1 + \dots + \alpha_nw_n,$$

so this T is unique. ■

If we have $f v_i = w_i$ for all i , we say that $T : V \rightarrow W$ defined by $\sum \alpha_i v_i \mapsto \sum \alpha_i w_i$ extends f to a linear transformation. This property holds for general vector spaces.

1.11 Lecture 11

Theorem 1.37 (Classification of Finite Dimensional Vector Spaces). *Let V and W be finite dimensional vector spaces over the field F . Then $V \cong W$ if and only if $\dim V = \dim W$.*

Proof. Suppose that $V \cong W$ and $T : V \rightarrow W$ is an isomorphism. Let $B = \{v_1, \dots, v_n\}$ be a basis for V . Then $C = \{Tv_1, \dots, Tv_n\}$ is linearly independent in W , and since $|C| = n$ and $\text{Span } C = W$ (since T is surjective), C is a basis for W and $\dim V = \dim W$. Now suppose that $n = \dim V = \dim W$. Let $B = \{v_1, \dots, v_n\}$ be a basis for V and $C = \{w_1, \dots, w_n\}$ be a basis for W . Choose the unique linear transformation $T : V \rightarrow W$ with $v_i \mapsto w_i$. Then T takes the basis B of V to the basis C of W , and so T is an isomorphism. ■

Similarly, if V and W are (arbitrary) vector spaces over F with bases B and C respectively such that $|B| = |C|$, then $V \cong W$, since we may extend our bijection to a linear transformation, which is necessarily an isomorphism, and if $V \cong W$, have bases B and C , then $|B| = |C|$.

Let V and W be finite dimensional vector spaces over F . Let $B = \{v_1, \dots, v_n\}$ be an ordered basis for V and $C = \{w_1, \dots, w_m\}$ be an ordered basis for W . If $v \in V$ with $v = \alpha_1 v_1 + \dots + \alpha_n v_n$, write

$$[v]_B = \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}.$$

We call $[v]_B$ the coordinate matrix of v relative to the ordered basis B . Now, $[v_i]_B$ has all zeros except in the i -th component, which is 1. The set of all of these is a basis for $F^{n \times 1}$, and since $v \mapsto [v]_B$ is a linear transformation, it must be an isomorphism (it sends a basis to a basis). Thus, $V \cong F^{n \times 1}$. Similarly, $W \cong F^{m \times 1}$. Now let $T : V \rightarrow W$ be linear. Then, there are $\alpha_{i,j} \in F$ for each i, j such that

$$Tv_j = \sum_{i=1}^m \alpha_{i,j} w_i.$$

Let

$$A = \begin{pmatrix} \alpha_{1,1} & \cdots & \alpha_{1,n} \\ \vdots & \ddots & \vdots \\ \alpha_{m,1} & \cdots & \alpha_{m,n} \end{pmatrix}.$$

Then the j^{th} column of A is $[Tv_j]_C$. Now, A is a linear transformation from $F^{n \times 1} \rightarrow F^{m \times 1}$ such that $A[v]_B = [Tv]_C$ (which is necessarily unique by the universal property of vector spaces). In other words, the following diagram commutes

$$\begin{array}{ccc}
 V & \xrightarrow{[\cdot]_B} & F^{n \times 1} \\
 \downarrow T & & \downarrow A \\
 W & \xrightarrow{[\cdot]_C} & F^{m \times 1}
 \end{array}$$

This matrix A is called the matrix representation of T relative to the ordered bases B and C . We write $A = [T]_{B,C}$. In the case $V = W$ and $B = C$, we just write $[T]_B$.

1.12 Lecture 12

Let us find the matrix representation of

$$\begin{aligned} D : \mathbb{R}[x]_3 &\rightarrow \mathbb{R}[x]_3 \\ f(x) &\mapsto f'(x) \end{aligned}$$

using the standard basis $\{1, t, t^2, t^3\}$. We have that $D(1) = 0$, $D(t) = 1$, $D(t^2) = 2t$, and $D(t^3) = 3t^2$. Thus, the matrix is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

As another example, consider $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ being the counterclockwise rotation by an angle of θ , with the standard basis $\{e_1 = (1, 0), e_2 = (0, 1)\}$. It can be shown that $T_\theta e_1 = \cos(\theta)e_1 + \sin(\theta)e_2$ and that $T_\theta e_2 = (-\sin(\theta))e_1 + \cos(\theta)e_2$, so that the matrix representation is

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}.$$

Note that in the case $\theta = 0$, this is the identity.

Theorem 1.38 (Matrix Theory Theorem). *Let V and W be finite dimensional vector spaces of dimension n and m over F respectively, and let B and C be ordered bases for V and W . Then the map*

$$\begin{aligned} \varphi : L(V, W) &\rightarrow F^{m \times n} \\ T &\mapsto [T]_{B,C} \end{aligned}$$

is an isomorphism. In particular, $\dim L(V, W) = mn$.

Proof. It is not hard to check that φ is linear. Now, if $\varphi(T) = 0$, then $T(v_i) = 0$ for all $v_i \in B$. Thus, $T(v) = 0$ for all $v \in V$, so $T = 0$, so that $\ker \varphi = \{0\}$. Thus, φ is injective. Now given an element of $F^{m \times n}$, we map v_i to its i^{th} column, and extend this linearly. We then have a transformation mapping to it, so φ is surjective, and thus an isomorphism. ■

Theorem 1.39. *Let V, W, U be finite dimensional vector spaces over F with ordered bases B, C, D respectively. If $T : V \rightarrow W$ and $S : W \rightarrow U$ are linear, then*

$$[S \circ T]_{B,D} = [S]_{C,D} \cdot [T]_{B,C}.$$

Proof. We have that, for all $v \in V$,

$$[S]_{C,D} \cdot [T]_{B,C} [v]_B = [S]_{C,D} [Tv]_C = [S(Tv)]_D = [(S \circ T)(v)]_D = [S \circ T]_{B,D} [v]_B.$$

■

Note also that, if $T : V \rightarrow W$ is a linear transformation, then T is an isomorphism if and only if $[T]_{B,C}$ is invertible. To see this, notice that, if $[T]_{B,C}$ is invertible, then $[T]_{B,C}[T]_{B,C}^{-1} = 1$ and that $[T]_{B,C}^{-1}[T]_{B,C} = 1$, so the map $T' : W \rightarrow V$ with $[T']_{C,B} = [T]_{B,C}^{-1}$ must have $T'T = 1_V$ and $TT' = 1_W$. Thus, T is an isomorphism. Conversely, if T is an isomorphism, then $[T]_{B,C}[T^{-1}]_{C,B} = [TT^{-1}]_C = [1_W]_C = 1$ and $[T^{-1}]_{C,B}[T]_{B,C} = [T^{-1}T]_B = [1_V]_B = 1$, so that $[T]_{B,C}$ is invertible.

Now, if V is a finite dimensional vector space over F with $\dim n$ and ordered basis B , then

$$\begin{aligned}\varphi : L(V, V) &\rightarrow \mathbb{M}_n(F) \\ T &\mapsto [T]_B\end{aligned}$$

satisfies $\varphi(T + S) = \varphi(T) + \varphi(S)$, $\varphi(T \circ S) = \varphi(T)\varphi(S)$, $\varphi(0_V) = 0_{F^{n \times 1}}$, and $\varphi(1_V) = 1_{F^{n \times 1}}$.

Definition 1.40. Let V be a finite dimensional vector space over F with ordered bases B and C . Then the invertible matrix $[1_V]_{B,C}$ is called a change of basis matrix.

Note that the inverse of $[1_V]_{B,C}$ is $[1_V]_{C,B}$.

Theorem 1.41 (Change of Basis Theorem). *Let V and W be finite dimensional vector spaces over F with ordered bases B, B' for V and C, C' for W . Let $T : V \rightarrow W$ be linear. Then*

$$\begin{aligned}[T]_{B,C} &= [1_W]_{C',C}[T]_{B',C'}[1_V]_{B,B'} \\ &= [1_W]_{C,C'}^{-1}[T]_{B',C'}[1_V]_{B',B}^{-1} \\ &= [1_W]_{C',C}[T]_{B',C'}[1_V]_{B',B}^{-1}\end{aligned}$$

Proof. We have that

$$[1_W]_{C',C}^{-1} = [1_W]_{C,C'} \quad [1_V]_{B,B'} = [1_V]_{B',B}^{-1}.$$

Since

$$[1_W]_{C',C}[T]_{B',C'}[1_V]_{B,B'} = [1_W \circ T]_{B',C}[1_V]_{B,B'} = [1_W \circ T \circ 1_V]_{B,C} = [T]_{B,C},$$

the result follows. ■

1.13 Lecture 13

Let S be the standard basis of $\mathbb{R}^2$, and let $B = \{v_1 = (1, 1), v_2 = (2, 1)\}$. Let us find $[T]_S$, where $T(1, 1) = 6(1, 1) + (2, 1)$ and $T(2, 1) = -2(1, 1) + (2, 1)$. Then,

$$[T]_B = \begin{pmatrix} 6 & -2 \\ 1 & 1 \end{pmatrix}.$$

We have that

$$[1_{\mathbb{R}^2}]_{B,S} = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix},$$

so that

$$[T]_S = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 6 & -2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}^{-1},$$

which can be computed to be

$$\begin{pmatrix} -8 & 16 \\ -8 & 15 \end{pmatrix}.$$

As another example, let us find the matrix of a rotation counterclockwise of an angle θ about $(1, 1, 1)$ in $\mathbb{R}^3$. Take the basis $B = \{v_1 = (\sqrt{3}/3, \sqrt{3}/3, \sqrt{3}/3), v_2 = (0, \sqrt{2}/2, -\sqrt{2}/2), v_3 = (-\sqrt{6}/3, \sqrt{6}/6, \sqrt{6}/6)\}$. Then we have that $Tv_1 = v_1$, $Tv_2 = \cos(\theta)v_2 + \sin(\theta)v_3$, and $Tv_3 = -\sin(\theta)v_2 + \cos(\theta)v_3$. Thus,

$$[T]_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}.$$

Since

$$[1_V]_{B,S} = \begin{pmatrix} \sqrt{3}/3 & 0 & -\sqrt{6}/3 \\ \sqrt{3}/3 & \sqrt{2}/2 & \sqrt{6}/6 \\ \sqrt{3}/3 & -\sqrt{2}/2 & \sqrt{6}/6 \end{pmatrix}.$$

From here, we can compute $[T]_S$.

Definition 1.42. Let $A, B \in \mathbb{M}_n(F)$. We say A is similar to B , and write $A \sim B$, if there exists an invertible $C \in \mathbb{M}_n(F)$ such that

$$A = C^{-1}BC.$$

Clearly, $A \sim A$ and if $A \sim B$, then $B \sim A$. Now if $A \sim B$ and $B \sim C$ with $A = DBD^{-1}$ and $B = ECE^{-1}$, then $A = (DE)C(DE)^{-1}$, so $A \sim C$. Thus, $\sim$ is an equivalence relation.

Theorem 1.43. Let $A, B \in \mathbb{M}_n(F)$. Then $A \sim B$ if and only if there exists a vector space V over F with $\dim V = n$ such that there is a linear transformation $T : V \rightarrow V$ with $A = [T]_{\mathcal{B}}$ and $B = [T]_{\mathcal{C}}$ for some ordered bases $\mathcal{B}$ and $\mathcal{C}$ for V .

Proof. If $A = [T]_{\mathcal{B}}$ and $B = [T]_{\mathcal{C}}$, then $C = [1_V]_{\mathcal{B}, \mathcal{C}} \in \mathbb{M}_n(F)$ is invertible with $A = C^{-1}BC$. Now suppose that $C \in \mathbb{M}_n(F)$ is invertible with $A = C^{-1}BC$. Take $V = F^n$ and define $T : V \rightarrow V$ by

$$Te_j = \sum_{i=1}^n A_{i,j}e_i,$$

where $\mathcal{S} = \{e_1, \dots, e_n\}$ is the standard basis. Then, $[T]_{\mathcal{S}} = A = C^{-1}BC$. Let $w_j = \sum_{i=1}^n (C^{-1})_{i,j}e_i$. Take $\mathcal{B} = \{w_1, \dots, w_n\}$. Then $C^{-1} = [1_V]_{\mathcal{B}, \mathcal{S}}$, so that $A = C^{-1}[T]_{\mathcal{B}}C$. Thus, $B = [T]_{\mathcal{B}}$. ■

1.14 Lecture 14

Definition 1.44. Let V be a nontrivial vector space over F , and let $T : V \rightarrow V$ be a linear operator (an element of $L(V, V)$). Let $\lambda \in F$. We define

$$S_\lambda = T - \lambda 1_V$$

to be a linear operator on V . We write $E_T(\lambda) = \ker S_\lambda$. We say that λ is an eigenvalue of T if $E_T(\lambda) \neq 0$. If $\lambda \neq 0$, we call $E_T(\lambda)$ an eigenspace of V relative to T and λ , and we call any $v \in E_T(\lambda)$ an eigenvector of T and λ .

Note that v is an eigenvector of $T : V \rightarrow V$ with eigenvalue λ if and only if $Tv = \lambda v$. Note also that if λ is an eigenvalue of T , then $E_T(\lambda)$ is a subspace of V . Now suppose that $\lambda \neq \mu$, and that $v \in E_T(\lambda) \cap E_T(\mu)$. Then, $\lambda v = Tv = \mu v$, so that $\lambda = \mu$ or $v = 0$. Since $\lambda \neq \mu$, we have that $v = 0$, so $E_T(\lambda) \cap E_T(\mu) = 0$.

Let $B = \{v_1, \dots, v_n\}$ be a basis for V , which we now suppose is finite dimensional, and suppose that $Tv_i = \lambda_i v_i$. Then,

$$[T]_B = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}.$$

We say that T is diagonalizable if there is an ordered basis B for V such that $[T]_B$ is diagonal. If T is diagonalizable with basis B such that $[T]_B$ is diagonal, then the elements of B are eigenvectors of T .

Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. By the isomorphism theorem, λ is an eigenvalue of T if and only if S_λ is not injective, if and only if $[S_\lambda]_B$ is not invertible for some basis B of V . This occurs if and only if $\det([S_\lambda]_B) = 0 = \det(\lambda I - [T]_B)$. If $A \in \mathbb{M}_n(F)$, we call

$$f_A(t) = \det(tI - A)$$

the characteristic polynomial of A .

Proposition 1.45. If $A, B \in \mathbb{M}_n(F)$ are similar, then $f_A = f_B$.

Proof. If $A = C^{-1}BC$, then $f_A = \det(C^{-1}(tI - B)C) = \det(C^{-1})\det(tI - B)\det(C) = \det(tI - B) = f_B$. ■

It follows that $f_{[T]_B} = f_{[T]_C}$ for any linear operator $T : V \rightarrow V$ and bases B and C of V . We call this the characteristic polynomial of T , and write f_T .

Theorem 1.46. Let V be a finite dimensional vector space over F , and let $T : V \rightarrow V$ be linear. Then the eigenvalues of T are precisely the roots of f_T .

Proof. Let $\lambda \in F$, and let B be an ordered basis for V . Write $A = [T]_B$, so that $f_T = \det(tI - A)$. Then λ is a root of f_T if and only if $f_T(\lambda) = \det(tI - A)|_{t=\lambda} = 0$. ■

1.15 Lecture 15

Theorem 1.47. *Let $T : V \rightarrow V$ be linear, and let $\lambda_1, \dots, \lambda_n \in F$ be distinct eigenvalues of T . Take $0 \neq v_i \in E_T(\lambda_i)$. Then $\{v_1, \dots, v_n\}$ is linearly independent.*

Proof. We show this by induction on n . If $n = 1$, we are automatically done, as $v_1 \neq 0$. Now suppose that

$$0 = a_1 v_1 + \dots + a_n v_n$$

for $a_1, \dots, a_n \in F$. Applying $S_{\lambda_n} = T - \lambda_n 1_V$, we get

$$0 = a_1(\lambda_1 - \lambda_n)v_1 + \dots + a_n(\lambda_{n-1} - \lambda_n)v_{n-1},$$

as $S_{\lambda_n}(v_i) = Tv_i - \lambda_n v_i = \lambda_i v_i - \lambda_n v_i$. By the induction hypothesis, $a_i(\lambda_i - \lambda_n) = 0$ for $1 \leq i \leq n-1$. Since $\lambda_i - \lambda_n \neq 0$, we have that $a_i = 0$ for $1 \leq i \leq n-1$. Thus, $a_n v_n = 0$. Since $v_n \neq 0$, we have that $a_n = 0$, so $v_1, \dots, v_n$ are linearly independent. ■

We look at some consequences of this.

Corollary 1.48. *Let V be finite dimensional vector space over F with $\dim V = n$. If $T : V \rightarrow V$ is linear and has n distinct eigenvalues, then T is diagonalizable.*

Corollary 1.49. *If V is a finite dimensional vector space over F with $\dim V = n$, and $T : V \rightarrow V$ is linear, then T has at most n distinct eigenvalues.*

Corollary 1.50. *Let V be a vector space over F , and let $T : V \rightarrow V$ be linear, where $\lambda_1, \dots, \lambda_n$ are distinct eigenvalues of T . Let*

$$W = E_T(\lambda_1) + \dots + E_T(\lambda_n).$$

If $v_i \in E_T(\lambda_i)$ are chosen such that $v_1 + \dots + v_n = 0$, then $v_i = 0$ for all i .

1.16 Lecture 16

From here, we will assume that all our fields are subfields of $\mathbb{C}$, and that our fields are closed under complex conjugation.

Definition 1.51. Let F be a subfield of $\mathbb{C}$ that is closed under complex conjugation, and let V be a vector space over F . Suppose we have a map

$$\langle \cdot, \cdot \rangle : V \times V \rightarrow F$$

satisfying

- $\langle v_1 + v_2, v_3 \rangle = \langle v_1, v_3 \rangle + \langle v_2, v_3 \rangle$
- $\langle v_1, v_2 \rangle = \overline{\langle v_2, v_1 \rangle}$
- $\langle \alpha v_1, v_2 \rangle = \alpha \langle v_1, v_2 \rangle = \langle v_1, \overline{\alpha} v_2 \rangle$
- $\langle v, v \rangle \in \mathbb{R}^{\geq 0}$, and $\langle v, v \rangle = 0$ if and only if $v = 0$.

Then, we call V an inner product space over F .

We write $\|v\| = \sqrt{\langle v, v \rangle}$ and call this the length, norm, or magnitude of v . Note that $\langle v, v \rangle \in \mathbb{R}$ follows from the second axiom.

Proposition 1.52. Let V be an inner product space over F . Take $\alpha \in F$ and $v_1, v_2, v_3 \in V$.

- (1) $\langle 0, v_1 \rangle = \langle v_1, 0 \rangle = 0$.
- (2) $\langle \alpha v_1 + v_2, v_3 \rangle = \alpha \langle v_1, v_3 \rangle + \langle v_2, v_3 \rangle$.
- (3) $\langle v_1, \alpha v_2 + v_3 \rangle = \overline{\alpha} \langle v_1, v_2 \rangle + \langle v_1, v_3 \rangle$.
- (4) $\langle v_1, v_2 + v_3 \rangle = \langle v_1, v_2 \rangle + \langle v_1, v_3 \rangle$.

Proof.

- (1) We have that $\langle 0, v_1 \rangle = \langle 0 + 0, v_1 \rangle = \langle 0, v_1 \rangle + \langle 0, v_1 \rangle$, so that $\langle 0, v_1 \rangle = 0$. We also have that $\langle v_1, 0 \rangle = \overline{\langle 0, v_1 \rangle} = 0$.
- (2) We have that

$$\langle \alpha v_1 + v_2, v_3 \rangle = \langle \alpha v_1, v_3 \rangle + \langle v_2, v_3 \rangle = \alpha \langle v_1, v_3 \rangle + \langle v_2, v_3 \rangle.$$

- (3) Notice that

$$\langle v_1, \alpha v_2 + v_3 \rangle = \overline{\langle \alpha v_2 + v_3, v_1 \rangle} = \overline{\alpha \langle v_2, v_1 \rangle + \langle v_3, v_1 \rangle} = \overline{\alpha} \overline{\langle v_2, v_1 \rangle} + \overline{\langle v_3, v_1 \rangle} = \overline{\alpha} \langle v_1, v_2 \rangle + \langle v_1, v_3 \rangle.$$

- (4) Take $\alpha = 1$ in (3).

■

Definition 1.53. Let V be an inner product space over F . If $F \subseteq \mathbb{R}$ define the angle θ between $v_1, v_2 \neq 0 \in V$ with $0 \leq \theta \leq 2\pi$ by

$$\cos \theta = \frac{\langle v_1, v_2 \rangle}{||v_1|| \cdot ||v_2||}.$$

If $F \not\subseteq \mathbb{R}$, define θ by

$$\cos \theta = \frac{|\langle v_1, v_2 \rangle|}{||v_1|| \cdot ||v_2||}.$$

Note that if $\langle v, w \rangle = 0$ for all $w \in V$, then $v = 0$ (take $w = v$). Now, fix $v \in V$. Then the map $\langle \cdot, v \rangle : V \rightarrow F$ is a linear transformation. However, $\langle v, \cdot \rangle : V \rightarrow F$ is linear if and only if $F \subseteq \mathbb{R}$.

Definition 1.54. Let V and W be inner product spaces over F . If $f : V \rightarrow W$ is a map, we say that f is sesquilinear if

$$f(\alpha v + v') = \bar{\alpha}f(v) + f(v')$$

for all $\alpha \in F$ and $v, v' \in V$.

The map $\langle v, \cdot \rangle : V \rightarrow F$ is sesquilinear.

If $V = F^n$, we make V into an inner product space over F as follows. Take $v = (v_1, \dots, v_n)$ and $w = (w_1, \dots, w_n)$ to be elements of V . We define

$$v \cdot w = \langle v, w \rangle = \sum_{i=1}^n v_i \bar{w}_i.$$

If $I = [\alpha, \beta] \subseteq \mathbb{R}$ and $V = C(I)$, we define

$$\langle f, g \rangle = \int_{\alpha}^{\beta} f(x)g(x)dx.$$

Definition 1.55. Let $A \in \mathbb{M}_n(F)$. We define the adjoint of A to be

$$A^* = \begin{pmatrix} \overline{a_{1,1}} & \overline{a_{2,1}} & \cdots & \overline{a_{n,1}} \\ \overline{a_{1,2}} & \overline{a_{2,2}} & \cdots & \overline{a_{n,2}} \\ \vdots & \vdots & \ddots & \vdots \\ \overline{a_{1,n}} & \overline{a_{2,n}} & \cdots & \overline{a_{n,n}} \end{pmatrix}.$$

If $F \subseteq \mathbb{R}$, then $A^* = A^{\top}$. If $A \in F^{m \times n}$, we can still define A^* by $(A^*)_{i,j} = \overline{A_{j,i}}$. We make $V = \mathbb{M}_n(F)$ into an inner product space under

$$\langle A, B \rangle = \text{Tr}(AB^*),$$

where $\text{Tr}(C) = \sum_{i=1}^n C_{i,i}$.

Theorem 1.56. *Let V be an inner product space over F . Then, for all $v_1, v_2 \in V$ and $\alpha \in F$, we have*

$$(1) \quad \|v_1\| \in \mathbb{R} \text{ with } \|v_1\| \geq 0 \text{ and } \|v_1\| = 0 \text{ if and only if } v_1 = 0.$$

$$(2) \quad \|\alpha v_1\| = |\alpha| \cdot \|v_1\|.$$

$$(3) \quad |\langle v_1, v_2 \rangle| \leq \|v_1\| \cdot \|v_2\|.$$

$$(4) \quad \|v_1 + v_2\| \leq \|v_1\| + \|v_2\|.$$

Proof.

(1) This is immediate by definition.

(2) We have that

$$\|\alpha v_1\| = \sqrt{\langle \alpha v_1, \alpha v_1 \rangle} = \sqrt{\alpha \langle v_1, \alpha v_1 \rangle} = \sqrt{\alpha \bar{\alpha} \langle v_1, v_1 \rangle} = \sqrt{|\alpha|^2 \langle v_1, v_1 \rangle} = |\alpha| \cdot \|v_1\|.$$

(3) If v_1 or v_2 is 0, this is done, so we assume $v_1, v_2 \neq 0$. Let

$$v = v_2 - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1.$$

We claim that $\langle v, \alpha v_1 \rangle = 0$ for all $\alpha \in F$. Note that

$$\begin{aligned} \langle v, \alpha v_1 \rangle &= \left\langle v_2 - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1, \alpha v_1 \right\rangle = \langle v_2, \alpha v_1 \rangle - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} \langle v_1, \alpha v_1 \rangle \\ &= \bar{\alpha} \langle v_2, v_1 \rangle - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} \cdot \bar{\alpha} \langle v_1, v_1 \rangle = \bar{\alpha} \langle v_2, v_1 \rangle - \bar{\alpha} \langle v_2, v_1 \rangle = 0. \end{aligned}$$

Thus, we have

$$\begin{aligned} 0 \leq \langle v, v \rangle &= \left\langle v_2 - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1, v_2 - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1 \right\rangle = \langle v_2, v_2 \rangle + \left\langle v_2 - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1, -\frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1 \right\rangle = \langle v_2, v_2 \rangle \\ &= \left\langle v_2 - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} v_1, v_2 \right\rangle = \langle v_2, v_2 \rangle - \frac{\langle v_2, v_1 \rangle}{\|v_1\|^2} \langle v_1, v_2 \rangle = \|v_2\|^2 - \frac{\overline{\langle v_1, v_2 \rangle}}{\|v_1\|^2} \langle v_1, v_2 \rangle \\ &= \|v_2\|^2 - \frac{|\langle v_1, v_2 \rangle|^2}{\|v_1\|^2} = \frac{\|v_1\|^2 \cdot \|v_2\|^2 - |\langle v_1, v_2 \rangle|^2}{\|v_1\|^2}. \end{aligned}$$

Thus,

$$0 \leq \|v_1\|^2 \cdot \|v_2\|^2 - |\langle v_1, v_2 \rangle|^2 \implies |\langle v_1, v_2 \rangle|^2 \leq \|v_1\|^2 \cdot \|v_2\|^2 \implies |\langle v_1, v_2 \rangle| \leq \|v_1\| \cdot \|v_2\|.$$

(4) We have that

$$\begin{aligned} \|v_1 + v_2\|^2 &= \langle v_1 + v_2, v_1 + v_2 \rangle = \|v_1\|^2 + \langle v_1, v_2 \rangle + \langle v_2, v_1 \rangle + \|v_2\|^2 \\ &= \|v_1\|^2 + \langle v_1, v_2 \rangle + \overline{\langle v_1, v_2 \rangle} + \|v_2\|^2 = \|v_1\|^2 + 2\Re\langle v_1, v_2 \rangle + \|v_2\|^2 \\ &\leq \|v_1\|^2 + 2\sqrt{(\Re\langle v_1, v_2 \rangle)^2 + (\Im\langle v_1, v_2 \rangle)^2} + \|v_2\|^2 \\ &= \|v_1\|^2 + 2|\langle v_1, v_2 \rangle| + \|v_2\|^2 \leq \|v_1\|^2 + 2\|v_1\| \cdot \|v_2\| + \|v_2\|^2 = (\|v_1\| + \|v_2\|)^2. \end{aligned}$$

Thus,

$$\|v_1 + v_2\| \leq \|v_1\| + \|v_2\|.$$

■

1.17 Lecture 17

Definition 1.57. Let V be an inner product space over F . Define the distance between $v_1, v_2 \in V$ by $d(v_1, v_2) = \|v_1 - v_2\|$. This satisfies

- $d(v, w) \in \mathbb{R}^{\geq 0}$ for all $v, w \in V$.
- $d(v, w) = 0 \iff v = w$ for all $v, w \in V$.
- $d(v, w) = d(w, v)$ for all $v, w \in V$.
- $d(v, x) \leq d(v, w) + d(w, x)$ for all $v, w, x \in V$.

For example, using the dot product in $\mathbb{R}^n$ gives the usual Euclidean notion of distance.

Definition 1.58. Let V be an inner product space over F . We say that $v, w \in V$ are orthogonal if $\langle v, w \rangle = 0$. We write $v \perp w$ in this case.

More generally, we say that a set S is orthogonal if $v \perp w$ for all $v \neq w \in S$.

Definition 1.59. Let V be an inner product space over F , and let B be a basis for V . We say that B is an orthogonal basis if $v \perp w$ for all $v \neq w \in B$. If additionally, $\|v\| = 1$ for all $v \in B$, we say that B is an orthonormal basis.

Similarly, if S is an orthogonal set with the added condition that $\|s\| = 1$ for all $s \in S$, then we say that S is orthonormal. Note that $0 \perp x$ for all $x \in V$ and that $x \perp y \iff y \perp x$.

Theorem 1.60. Let V be an inner product space over F and let $S \subseteq V$ be orthogonal. Suppose that $0 \notin S$. Then S is linearly independent. If V is finite dimensional and $|S| = \dim V$, then S is an orthogonal basis for V .

Proof. Let $v \in \text{Span}(S)$. Then there are distinct $v_1, \dots, v_n \in S$ and $\alpha_1, \dots, \alpha_n \in F$ such that $v = \sum_{i=1}^n \alpha_i v_i$. Then $\langle v, v_j \rangle = \alpha_j \cdot \|v_j\|^2$ for all j . Taking $v = 0$, we get that $\alpha_j \|v_j\|^2 = 0$ for all j . Since $v_j \neq 0$, we have that $\alpha_j = 0$ for all j , meaning that S is linearly independent. ■

We record the following result, which is an easy corollary of the proof.

Theorem 1.61. Let $\{v_1, \dots, v_n\}$ be an orthogonal set, with $v_i \neq 0$ for all $1 \leq i \leq n$. If

$$v = \alpha_1 v_1 + \dots + \alpha_n v_n,$$

then

$$\alpha_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2} \quad 1 \leq i \leq n.$$

In particular, if $\{v_1, \dots, v_n\}$ is orthonormal, then

$$\alpha_i = \langle v, v_i \rangle \quad 1 \leq i \leq n.$$

Now, if $S = (s_i)_{i \in I}$ is orthogonal, we claim that $S' = (s_i/\|s_i\|)_{i \in I}$ is orthonormal. This is orthogonal, as

$$\left\langle \frac{s_i}{\|s_i\|}, \frac{s_j}{\|s_j\|} \right\rangle = \frac{1}{\|s_i\| \cdot \|s_j\|} \langle s_i, s_j \rangle = 0$$

and

$$\left\| \frac{s_i}{\|s_i\|} \right\| = \frac{\|s_i\|}{\|s_i\|} = 1.$$

Theorem 1.62. *Let V be an inner product space over F , and let B be an orthogonal basis for V . Take $v \in V$. Then $\langle v, w \rangle = 0$ for all but finitely many $w \in B$ and $v = \sum_B \frac{\langle v, w \rangle}{\|w\|^2} w$ is a finite sum. If, in addition, B is orthonormal, then this becomes $v = \sum_B \langle v, w \rangle w$.*

Proof. Write $v = \sum_B \alpha_w w$. Recall all but finitely many α_w 's are 0. By the previous theorem, all but finitely many $\langle v, w \rangle$'s are 0. ■

Corollary 1.63 (Parseval's Equation). *Let V be a finite dimensional inner product space over F with an orthonormal basis $\{v_1, \dots, v_n\}$. Let $v, w \in V$. Then,*

$$\langle v, w \rangle = \sum_{i=1}^n \langle v, v_i \rangle \overline{\langle w, v_i \rangle}.$$

In particular,

$$\|v\|^2 = \sum_{i=1}^n |\langle v, v_i \rangle|^2.$$

Proof. Write $v = \sum_{i=1}^n \langle v, v_i \rangle v_i$ and $w = \sum_{i=1}^n \langle w, v_i \rangle v_i$. Then,

$$\begin{aligned} \langle v, w \rangle &= \left\langle \sum_{i=1}^n \langle v, v_i \rangle v_i, \sum_{j=1}^n \langle w, v_j \rangle v_j \right\rangle = \sum_{i=1}^n \langle v, v_i \rangle \left\langle v_i, \sum_{j=1}^n \langle w, v_j \rangle v_j \right\rangle \\ &= \sum_{i=1}^n \sum_{j=1}^n \langle v, v_i \rangle \overline{\langle w, v_j \rangle} \langle v_i, v_j \rangle = \sum_{i=1}^n \langle v, v_i \rangle \overline{\langle w, v_i \rangle}. \end{aligned}$$

Take $v = w$ for the special case. ■

1.18 Lecture 18

Theorem 1.64 (Gram-Schmidt). *Let V be an inner product space over F , and let $\emptyset \neq S_n = \{v_1, \dots, v_n\} \subseteq V$ be a linearly independent set. Then there exists $y_1, \dots, y_n \in V$ such that*

- $y_1 = v_1$.
- $T_n = \{y_1, \dots, y_n\}$ is an orthogonal linearly independent set.
- $\text{Span}(T_n) = \text{Span}(S_n)$.

Proof. We prove this by induction on n . The case $n = 1$ is clear. Now assume that this is true for n . Take $\{v_1, \dots, v_{n+1}\}$ be linearly independent. Apply the induction hypothesis to get $\{y_1, \dots, y_n, v_{n+1}\}$, where the y_i 's satisfy our conditions. Take

$$y_{n+1} = v_{n+1} - \sum_{k=1}^n \frac{\langle v_{n+1}, y_k \rangle}{\|y_k\|^2} y_k.$$

We claim that $y_{n+1} \notin \text{Span}\{y_1, \dots, y_n\}$. If it was, then $v_{n+1} \in \text{Span}\{y_1, \dots, y_n\} = \text{Span}\{v_1, \dots, v_n\}$, so $\{v_1, \dots, v_{n+1}\}$ is not linearly independent. Thus, $\{y_1, \dots, y_{n+1}\}$ is linearly independent. Further, we claim that $\langle y_{n+1}, y_j \rangle = 0$. We have that

$$\begin{aligned} \langle y_{n+1}, y_j \rangle &= \left\langle v_{n+1} - \sum_{k=1}^n \frac{\langle v_{n+1}, y_k \rangle}{\|y_k\|^2} y_k, y_j \right\rangle \\ &= \langle v_{n+1}, y_j \rangle - \sum_{k=1}^n \frac{\langle v_{n+1}, y_k \rangle}{\|y_k\|^2} \langle y_k, y_j \rangle \\ &= \langle v_{n+1}, y_j \rangle - \sum_{k=1}^n \frac{\langle v_{n+1}, y_k \rangle}{\|y_k\|^2} \delta_{k,j} \|y_j\|^2 \\ &= \langle v_{n+1}, y_j \rangle - \langle v_{n+1}, y_j \rangle = 0 \end{aligned}$$

Thus, $\{y_1, \dots, y_{n+1}\}$ is a linearly independent orthogonal set. Finally,

$$\text{Span}\{v_1, \dots, v_{n+1}\} = \text{Span}\{y_1, \dots, y_n, v_{n+1}\}.$$

Since $v_{n+1} \in \text{Span}\{y_1, \dots, y_{n+1}\}$,

$$\text{Span}\{y_1, \dots, y_n, v_{n+1}\} \subseteq \text{Span}\{y_1, \dots, y_{n+1}\}.$$

Conversely, $y_{n+1} \in \text{Span}\{y_1, \dots, y_n, v_{n+1}\}$, so we have equality. The theorem follows by induction. ■

Theorem 1.65. *Let V be a finite dimensional inner product space over F . Then V has an orthogonal basis. If $F = \mathbb{R}$ or $F = \mathbb{C}$, then V has an orthonormal basis.*

Proof. We have proven that any basis can be converted into an orthogonal basis by the Gram-Schmidt Theorem. If $F = \mathbb{R}$ or $F = \mathbb{C}$, we may divide by the magnitudes of these vectors to get an orthonormal basis for V . ■

1.19 Lecture 19

Recall that $d(v_1, v_2) = \|v_1 - v_2\|$. Given a set $\emptyset \neq S \subseteq V$ and a vector $v \in V$, we generalize this by defining

$$d(v, S) = \inf_{w \in S} \{d(v, w)\}.$$

Definition 1.66. Let V be an inner product space over F , and let $\emptyset \neq S \subseteq V$ and $v \in V$. We say that v is orthogonal to S , and write $v \perp S$ if $\langle s, v \rangle = 0$ for all $s \in S$. We define $S^\perp = \{v \in V : v \perp S\}$.

Note that $S^\perp$ is always a subspace of V , regardless of whether S is. Note also that $S \subseteq (S^\perp)^\perp = S^{\perp\perp}$. Suppose that $v \in S \cap S^\perp$. Then, in particular, $\langle v, v \rangle = 0$, so $v = 0$. Thus, $S + S^\perp = S \oplus S^\perp$. Since every element of $S^\perp$ is orthogonal to S , we write $S \perp S^\perp$ for this (we call this the orthogonal sum).

Theorem 1.67. Let V be an inner product space over F , and let $S \subseteq V$ be a finite dimensional subspace. Take $v \in V$. Then we may uniquely write $v = s + s^\perp$, where $s \in S$ and $s^\perp \in S^\perp$. In particular, $V = S \perp S^\perp$. Moreover, $\|v\|^2 = \|s\|^2 + \|s^\perp\|^2$. If, additionally, V is finite dimensional, $\dim V = \dim S + \dim S^\perp$.

Proof. Take an orthogonal basis $\{v_1, \dots, v_n\}$ for S . Take $v \in V$, and define

$$s = \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i.$$

Let $s^\perp = v - s$. Note that

$$\begin{aligned} \langle s^\perp, v_j \rangle &= \left\langle v - \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i, v_j \right\rangle \\ &= \langle v, v_j \rangle - \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} \langle v_i, v_j \rangle \\ &= \langle v, v_j \rangle - \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} \delta_{i,j} \|v_j\|^2 \\ &= \langle v, v_j \rangle - \langle v, v_j \rangle = 0. \end{aligned}$$

Thus, $s^\perp \perp S$, so $s^\perp \in S^\perp$. Since $v = s + s^\perp$, we have proven existence. Take $s, s_1 \in S$ and $t, t_1 \in S^\perp$, and suppose that $s + t = s_1 + t_1$. Then $s - s_1 = t_1 - t$. The left hand side is in S , and the right hand side is in $S^\perp$, so $s - s_1 \in S \cap S^\perp = 0$. Thus, $s = s_1$ and $t_1 - t = 0$, so $t = t_1$. Hence, this decomposition is unique. Now,

$$\begin{aligned} \|v\|^2 &= \langle s + s^\perp, s + s^\perp \rangle = \langle s, s + s^\perp \rangle + \langle s^\perp, s + s^\perp \rangle \\ &= \langle s, s \rangle + \langle s, s^\perp \rangle + \langle s^\perp, s \rangle + \langle s^\perp, s^\perp \rangle = \langle s, s \rangle + \langle s^\perp, s^\perp \rangle = \|s\|^2 + \|s^\perp\|^2. \end{aligned}$$

Finally, if V is finite dimensional, then $\dim V = \dim(S + S^\perp) = \dim S + \dim S^\perp - \dim(S \cap S^\perp) = \dim S + \dim S^\perp$. ■

1.20 Lecture 20

Note that if $v \perp w$, then $\|v + w\|^2 = \|v\|^2 + \|w\|^2$. We have the following corollary.

Corollary 1.68. *Let V be an inner product space over F , and let $B = \{v_1, \dots, v_n\}$ be an orthogonal set in V with $0 \notin B$. Let $v \in V$. Then*

$$\sum_{i=1}^n \frac{|\langle v, v_i \rangle|^2}{\|v_i\|^2} \leq \|v\|^2$$

with equality if and only if

$$v = \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i.$$

Proof. Take

$$s = \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i$$

and $s^\perp = v - s$. Then,

$$\begin{aligned} \|v\|^2 &= \|s\|^2 + \|s^\perp\|^2 \\ &= \left\| \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i \right\|^2 + \|s^\perp\|^2 \geq \left\| \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i \right\|^2 \\ &= \sum_{i=1}^n \left\| \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i \right\|^2 = \sum_{i=1}^n \frac{|\langle v, v_i \rangle|^2}{\|v_i\|^4} \|v_i\|^2 \\ &= \sum_{i=1}^n \frac{|\langle v, v_i \rangle|^2}{\|v_i\|^2}. \end{aligned}$$

The equality here occurs when $s^\perp = 0$, so that $v = s = \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i$. ■

If V is an inner product space over F and $S \subseteq V$ is a finite dimensional subspace, then for any $v \in V$, we can write $v = s + s^\perp$ for $s \in S$ and $s^\perp \in S^\perp$. We call s the orthogonal projection of v on S , and denote it by v_S . The uniqueness of this value implies that

$$v_S = \sum_{i=1}^n \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i$$

for an orthogonal basis $v_1, \dots, v_n$ of S .

Theorem 1.69. *Let V be an inner product space over F , and let $S \subseteq V$ be a finite dimensional subspace, and $v \in V$. Then $d(v, v_S) = \|v - v_S\| \leq \|v - r\| = d(v, r)$ for all $r \in S$. Equivalently, $d(v, S) = d(v, v_S)$. Moreover, if $r \in S$, then $\|v - v_S\| = \|v - r\|$ if and only if $r = v_S$.*

Proof. Write $v = s + s^\perp$ with $s = v_S$ and $s^\perp \in S^\perp$. Let $r \in S$. Then $v - r = (v - v_S) + (v_S - r) = s^\perp + (v_S - r)$. Since $v_S - r \in S$, $\|v - r\|^2 = \|v - v_S\|^2 + \|v_S - r\|^2 \geq \|v - v_S\|^2$ with equality if and only if $v_S = r$. ■

Definition 1.70. Let V be an inner product space over F , and let $S \subseteq V$ be a finite dimensional subspace. The value $\|v - v_S\|$ is called the error of v not being in S .

Definition 1.71. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. We say that T is hermitian or self adjoint if $\langle Tv, w \rangle = \langle v, Tw \rangle$ for all $w, v \in V$.

Theorem 1.72. Let V and W be finite dimensional inner product spaces over F with orthonormal bases, and let $T : V \rightarrow W$ be linear. Then there exists a unique linear transformation $T^* : V \rightarrow W$ such that $\langle Tv, w \rangle = \langle v, T^*w \rangle$ for all $v \in V$ and $w \in W$. T^* is called the adjoint of T .

Proof. Let $\mathcal{B}$ and $\mathcal{C}$ be ordered orthonormal bases for V and W , and let $A = [T]_{\mathcal{B}, \mathcal{C}}$. Consider the transformation T^* given by the matrix A^* . Let $n = \dim V$ and $m = \dim W$, and write

$$A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{pmatrix}.$$

Note that we only need to show that $\langle Tv_i, w_j \rangle = \langle v_i, T^*w_j \rangle$ for $v_i \in \mathcal{B}$ and $w_j \in \mathcal{C}$. Now,

$$\langle Tv_i, w_j \rangle = \langle a_{1,i}w_1 + \cdots + a_{m,i}w_m, w_j \rangle = a_{1,i}\langle w_1, w_j \rangle + \cdots + a_{m,i}\langle w_m, w_j \rangle = a_{j,i}$$

and

$$\langle v_i, T^*w_j \rangle = \langle v_i, \overline{a_{j,1}}v_1 + \cdots + \overline{a_{j,n}}v_n \rangle = a_{j,1}\langle v_i, v_1 \rangle + \cdots + a_{j,n}\langle v_i, v_n \rangle = a_{j,i}.$$

Thus, T^* satisfies this. Now suppose that there are two linear transformations S and R satisfying our assumptions. Then $\langle v, Sw \rangle = \langle v, Rw \rangle$ for all $v \in V$ and $w \in W$. Then $\langle v, (S - R)w \rangle = 0$ for all $v \in V$ and $w \in W$. Fix w , so that $\langle v, (S - R)w \rangle = 0$ for all v . In particular, $\langle (S - R)w, (S - R)w \rangle = 0$, so $(S - R)w = 0$. Since w was arbitrary, $S - R = 0$, so $S = R$. Hence, we have proven uniqueness. ■

Thus, T is self adjoint if T is equal to its adjoint. Note that the same is true of matrices: the linear transformation given by A is hermitian if and only if $A = A^*$.

1.21 Lecture 21

Theorem 1.73. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. Let λ be an eigenvalue of T , and $0 \neq v \in E_T(\lambda)$. Then

$$\lambda = \frac{\langle Tv, v \rangle}{\|v\|^2} \quad \bar{\lambda} = \frac{\langle v, Tv \rangle}{\|v\|^2}.$$

In particular, $\lambda \in \mathbb{R}$ if and only if $\langle Tv, v \rangle = \langle v, Tv \rangle$.

Proof. By assumption, $Tv = \lambda v$, so $\langle Tv, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle = \lambda \|v\|^2$, and $\langle v, Tv \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle = \bar{\lambda} \|v\|^2$. As $v \neq 0$, the result follows. ■

Corollary 1.74. Let V be an inner product space over F , and $T : V \rightarrow V$ be a linear hermitian operator. Then any eigenvalue of T is real.

Proof. We have that $\langle Tv, v \rangle = \langle v, Tv \rangle$ for all $v \in V$ by definition. Thus, if λ is an eigenvalue of T , then it must be real. ■

Let $f \in F[t]$. Notice that $f(\alpha) = 0$ if and only if $(t - \alpha) \mid f(t)$ (one way is clear, to see the other, write $f(t) = (t - \alpha)h(t) + C$, where $\deg C < \deg(t - \alpha) = 1$, so C is constant, and evaluate at α to get $C = 0$). Note that, in particular, f cannot have more than $\deg f$ roots, counting multiplicity.

Definition 1.75. Let V be a vector space over F , and let $W \subseteq V$ be a subspace. Take $T : V \rightarrow V$ to be linear. We say that W is T -invariant if $Tw \in W$ for all $w \in W$.

Lemma 1.76. Let V be an inner product space over F , let $T : V \rightarrow V$ be a hermitian linear operator, and let $S \subseteq V$ be a T -invariant subspace. Then

- (1) $S^\perp$ is T -invariant.
- (2) $T|_{S^\perp} : S^\perp \rightarrow S^\perp$ is hermitian.

Proof.

- (1) Let $w \in S^\perp$. We must show that $Tw \in S^\perp$. Take $v \in S$. Then, $\langle v, Tw \rangle = \langle Tv, w \rangle$. Since $Tv \in S$, this is 0, so $Tw \in S^\perp$.
- (2) We have seen that $S^\perp$ is T -invariant, so this is a valid map. Since T is linear and hermitian, $T|_{S^\perp}$ is as well. ■

1.22 Lecture 22

Theorem 1.77. *Let $F = \mathbb{R}$ or $F = \mathbb{C}$, and let V be a finite dimensional inner product space over F with $\dim V = n$. Take $T : V \rightarrow V$ to be a hermitian operator. Then there exists an orthonormal basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V , where each v_i is an eigenvector for some $\lambda_i \in \mathbb{R}$. In particular, T is diagonalizable.*

Proof. We prove this by induction on n . Suppose that $n = 1$, so that $V = \text{Span}(v)$ for any nonzero $v \in V$. Since $Tv \in \text{Span}(v)$, we have that $Tv = \alpha v$, so that $v \in E_T(\alpha)$. Since T is hermitian, $\alpha \in \mathbb{R}$. Then take $\mathcal{B} = \{v/\|v\|\}$. Now suppose that the statement is true for $n - 1$. Let $\mathcal{C}$ be an orthonormal basis for V , with $\dim V = n$. Let $A = [T]_{\mathcal{C}} \in \mathbb{M}_n(F) \subseteq \mathbb{M}_n(\mathbb{C})$. Now, $A = A^*$, and so $Ax \cdot y = x \cdot Ay$ for all $x, y \in \mathbb{C}^{n \times 1}$, where $\cdot$ denotes the dot product. Hence, $A : \mathbb{C}^{n \times 1} \rightarrow \mathbb{C}^{n \times 1}$ is hermitian. Hence, there is an eigenvalue λ of A , which is necessarily real. Now, $f_T = f_A$, so f_T has λ as a root. Let $v \in E_T(\lambda)$ be nonzero, and let $F_v = \text{Span}(v)$. Then F_v is T -invariant. Write $V = F_v + (F_v)^\perp$ and $\dim V = \dim F_v + \dim (F_v)^\perp = 1 + \dim (F_v)^\perp$. Then $\dim (F_v)^\perp = n - 1$. By previous results, $(F_v)^\perp$ is T -invariant and $T : F_v^\perp \rightarrow F_v^\perp$ is hermitian. By the induction hypothesis, $F_v^\perp$ has an orthonormal basis $\{v_2, \dots, v_n\}$ that are eigenvectors of T with real eigenvalues. Let $v_1 = v/\|v\|$, which is an eigenvector of T with a real eigenvalue. Then $\langle v_1, v_i \rangle = 0$ for $i \neq 1$, and $\langle v_1, v_1 \rangle = 1$, so $\{v_1, \dots, v_n\}$ is an orthonormal basis for V where each v_i is an eigenvector of T with a real eigenvalue. ■

Definition 1.78. Let $T : V \rightarrow W$ be a linear transformation. We say that T is an isometry if T is an isomorphism, and if $\langle Tv_1, Tv_2 \rangle = \langle v_1, v_2 \rangle$ for all v_1, v_2 .

Proposition 1.79. *Let $T : V \rightarrow W$ be a linear transformation, where $\dim V = \dim W$ and V and W are inner product spaces over $F = \mathbb{R}$ or $\mathbb{C}$. The following are equivalent.*

- (1) $\langle Tv_1, Tv_2 \rangle = \langle v_1, v_2 \rangle$ for all v_1, v_2 .
- (2) T is an isometry.
- (3) T takes every orthonormal basis of V to an orthonormal basis of W .
- (4) T takes some orthonormal basis of V to an orthonormal basis of W .
- (5) $\|Tv\| = \|v\|$ for each $v \in V$.

Proof. We show (1) $\implies$ (2) $\implies$ (3) $\implies$ (4) $\implies$ (5) $\implies$ (1). Suppose that $\langle Tv_1, Tv_2 \rangle = \langle v_1, v_2 \rangle$ for all v_1, v_2 . We show that T is injective, and hence bijective. Suppose that $Tv = 0$. Then $0 = \langle Tw, 0 \rangle = \langle Tw, Tv \rangle = \langle w, v \rangle$ for all $w \in W$. Taking $w = v$, $v = 0$, so T is injective. Suppose that T is an isometry, and let $\{v_1, \dots, v_n\}$ be an orthonormal basis. Then, $\{Tv_1, \dots, Tv_n\}$ is necessarily a basis, as T is bijective. We have that

$$\langle Tv_i, Tv_j \rangle = \langle v_i, v_j \rangle = \delta_{i,j},$$

so that this is an orthonormal basis. Hence, each orthonormal basis of V is mapped to an orthonormal basis of W . That (3) $\implies$ (4) is trivial. Let $\{v_1, \dots, v_n\}$ be an orthonormal

basis of V mapping to an orthonormal basis of W . Take $v \in V$ and write $v = a_1v_1 + \cdots + a_nv_n$. Then,

$$\|Tv\|^2 = \|a_1Tv_1 + \cdots + a_nTv_n\|^2 = |a_1|^2 + \cdots + |a_n|^2 = \|a_1v_1 + \cdots + a_nv_n\|^2 = \|v\|^2.$$

Hence, $\|Tv\| = \|v\|$. Finally, suppose that $\|Tv\| = \|v\|$ for all $v \in V$. Let $v \in V$, and suppose that $\|v\| = 1$. Write $w = \alpha v + \beta v^\perp$, where $v^\perp \in (\text{Span}(v))^\perp$ has norm 1. Then,

$$|\alpha|^2 + \beta\bar{\alpha}\langle Tv^\perp, Tv \rangle + \alpha\bar{\beta}\langle Tv, Tv^\perp \rangle + |\beta|^2 = \langle \alpha Tv + \beta Tv^\perp, \alpha Tv + \beta Tv^\perp \rangle = \|Tw\|^2 = \|w\|^2 = |\alpha|^2 + |\beta|^2.$$

Then $\beta\bar{\alpha}\langle Tv^\perp, Tv \rangle + \alpha\bar{\beta}\langle Tv, Tv^\perp \rangle = 0$, so that $\beta\bar{\alpha}\langle Tv^\perp, Tv \rangle = a_i$ for $a \in \mathbb{R}$. Now, $\langle Tv^\perp, Tv \rangle$ is fixed, so we can arrange for α, β to not satisfy this unless $a = 0$. Hence, $\beta\bar{\alpha}\langle Tv^\perp, Tv \rangle = 0$, so that $\langle Tv^\perp, Tv \rangle = 0$. Now take $v_1, v_2 \in V$, and suppose without loss of generality that $\|v_1\| = 1$. Write $v_2 = \alpha v_1 + \beta v_1^\perp$ for $v_1^\perp \in \text{Span}(v_1)^\perp$ with norm 1. Then,

$$\begin{aligned} \langle Tv_1, Tv_2 \rangle &= \langle Tv_1, \alpha Tv_1 + \beta Tv_1^\perp \rangle = \bar{\alpha}\langle Tv_1, Tv_1 \rangle + \bar{\beta}\langle Tv_1, Tv_1^\perp \rangle \\ &= \bar{\alpha}\langle v_1, v_1 \rangle = \langle v_1, \alpha v_1 \rangle + \langle v_1, \beta v_1^\perp \rangle = \langle v_1, \alpha v_1 + \beta v_1^\perp \rangle = \langle v_1, v_2 \rangle. \end{aligned}$$

■

Theorem 1.80. *Let V be a finite dimensional inner product space over $F = \mathbb{R}$ or $\mathbb{C}$. Let $\mathcal{B}$ and $\mathcal{C}$ be ordered orthonormal bases for V . Then $[1_V]_{\mathcal{B}, \mathcal{C}} : F^{n \times 1} \rightarrow F^{n \times 1}$ is an isometry.*

Proof. Write

$$[1_V]_{\mathcal{B}, \mathcal{C}} = \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix}.$$

By definition, $b_i = a_{1,i}c_1 + \cdots + a_{n,i}c_n$, so that $1 = \|b_i\|^2 = |a_{1,i}|^2 + \cdots + |a_{n,i}|^2$. Note that the columns of this matrix are linearly independent (and hence spanning), as if not, it would imply a linear dependence relation among the b_i 's. For $i \neq j$, note also that

$$\begin{aligned} 0 &= \langle b_i, b_j \rangle = \langle a_{1,i}c_1 + \cdots + a_{n,i}c_n, a_{1,j}c_1 + \cdots + a_{n,j}c_n \rangle = \sum_{k=1}^n \sum_{\ell=1}^n a_{k,i} \overline{a_{k,j}} \delta_{k,\ell} \\ &= a_{1,i} \overline{a_{1,j}} + \cdots + a_{n,i} \overline{a_{n,j}}. \end{aligned}$$

Hence the columns of the matrix are also orthogonal, and thus form an orthonormal basis. We therefore have that the transformation sends an orthonormal basis to another orthonormal basis, so that it is an isometry. ■

Proposition 1.81. *Let T be a hermitian operator, and let λ_1, λ_2 be distinct eigenvalues of T . Take $v \in E_T(\lambda_1)$ and $w \in E_T(\lambda_2)$. Then $v \perp w$.*

Proof. We have that

$$\lambda_1 \langle v, w \rangle = \langle \lambda_1 v, w \rangle = \langle Tv, w \rangle = \langle v, Tw \rangle = \langle v, \lambda_2 w \rangle = \lambda_2 \langle v, w \rangle.$$

Since $\lambda_1 \neq \lambda_2$, $\langle v, w \rangle = 0$. ■

1.23 Lecture 23

Notice that $V = E_T(\lambda_1) \perp \cdots \perp E_T(\lambda_k)$, where T is hermitian and the λ_i 's are the eigenvalues of T (T is diagonalizable, and if $v_1 + \cdots + v_k = 0$ for $v_i \in V_i$, then $\langle v_i, v_i \rangle = \langle v_1 + \cdots + v_k, v_i \rangle = 0$). Now, $V = E_T(\lambda_i) \perp E_T(\lambda_i)^\perp$. Since we have a direct sum, we have uniqueness in the decompositions, so that

$$v = v_{E_T(\lambda_1)} + \cdots + v_{E_T(\lambda_k)}.$$

Now define

$$P_{\lambda_i}(v) = v_{E_T(\lambda_i)}.$$

Note that $\text{im } P_{\lambda_i} = E_T(\lambda_i)$ and $\ker P_{\lambda_i} = E_T(\lambda_i)^\perp$. We also have that

$$P_{\lambda_i} \circ P_{\lambda_j} = \delta_{i,j} P_{\lambda_i}.$$

We call the P_{λ_i} 's the orthogonal idempotents. For $v \in V$,

$$\begin{aligned} 1_V v = v &= v_{E_T(\lambda_1)} + \cdots + v_{E_T(\lambda_k)} \\ &= P_{\lambda_1}(v) + \cdots + P_{\lambda_k}(v) = (P_{\lambda_1} + \cdots + P_{\lambda_k})(v), \end{aligned}$$

so that $1_V = P_{\lambda_1} + \cdots + P_{\lambda_k}$. We also have

$$\begin{aligned} T &= T \circ 1_V = T \circ (P_{\lambda_1} + \cdots + P_{\lambda_k}) \\ &= TP_{\lambda_1} + \cdots + TP_{\lambda_k} \\ &= \lambda_1 P_{\lambda_1} + \cdots + \lambda_k P_{\lambda_k}. \end{aligned}$$

Hence,

$$Tv = \sum_{i=1}^k \lambda_i v_{E_T(\lambda_i)}.$$

Note that, in particular, $P_{\lambda_i} T = T P_{\lambda_i}$. Now, reorder the eigenvectors of T to a basis B' , where the first n_1 elements have eigenvalue λ_1 , the next n_2 have eigenvalue λ_2 , and so on (where $\dim V = n_1 + \cdots + n_k$).

Theorem 1.82 (Spectral Theorem for Hermitian Operators). *Let $F = \mathbb{R}$ or $F = \mathbb{C}$, and let V be a finite dimensional inner product space over F with $\dim V = n$. Take $T : V \rightarrow V$ to be a hermitian operator. Then there exists an orthonormal basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V , where each v_i is an eigenvector for some $\lambda_i \in \mathbb{R}$. In particular, T is diagonalizable. If $\lambda_1, \dots, \lambda_k$ are the distinct eigenvalues of T , then $E_T(\lambda_i) \perp E_T(\lambda_j)$ for $i \neq j$, and*

$$V = \bigoplus_{i=1}^k E_T(\lambda_i).$$

1.24 Lecture 24

Suppose that $T, T' : V \rightarrow W$, and that $S : W \rightarrow X$. note that

$$\langle (S \circ T)v, x \rangle = \langle Tv, S^*x \rangle = \langle v, (T^* \circ S^*)x \rangle,$$

so that $(S \circ T)^* = T^* \circ S^*$. Note also that

$$\langle (T + T')v, w \rangle = \langle Tv, w \rangle + \langle T'v, w \rangle = \langle v, T^*w \rangle + \langle v, T'^*w \rangle = \langle v, (T^* + T'^*)w \rangle,$$

so that $(T + T')^* = T^* + T'^*$. Finally, if $\lambda \in F$, note that

$$\langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \langle v, T^*w \rangle = \langle v, (\bar{\lambda}T^*)w \rangle,$$

so that $(\lambda T)^* = \bar{\lambda}T^*$. Note the immediate consequence of this that

$$\begin{aligned} \varphi : L(V, W) &\rightarrow L(W, V) \\ T &\mapsto T^* \end{aligned}$$

is sesquilinear (note also by existence and uniqueness, φ is bijective).

Lemma 1.83. *Let V be a finite dimensional inner product space over F , and let $T : V \rightarrow V$ be linear. Suppose that W has an orthonormal basis, and that $W \subseteq V$ is a T -invariant subspace. Then $W^\perp \subseteq V$ is T^* -invariant.*

Proof. Let $w^\perp \in W^\perp$ and $x \in W$ be arbitrary. Then,

$$\langle x, T^*w^\perp \rangle = \langle Tx, w^\perp \rangle = 0,$$

as $Tx \in W$ by assumption. Thus, $T^*w \in W^\perp$. ■

Definition 1.84. Let V be a finite dimensional vector space over F , and let $T : V \rightarrow V$ be linear. We say that T is triangularizable if there exists an ordered basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}$ is upper triangular ($([T]_{\mathcal{B}})_{i,j} = 0$ whenever $i > j$).

Theorem 1.85 (Schur). *Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be linear. Then T is triangularizable. Moreover, there exists an ordered orthonormal basis $\mathcal{B}$ for T such that $[T]_{\mathcal{B}}$ is upper triangular.*

Proof. We use induction on $n = \dim V$. If $n = 1$, take any orthonormal basis (which must exist, as we are working over $\mathbb{C}$) to achieve this. Suppose that $n > 1$. Consider the characteristic polynomial f_{T^*} , which has a root $\lambda \in \mathbb{C}$. Let $0 \neq v \in E_{T^*}(\lambda)$, with $\|v\| = 1$. Write $V = (\mathbb{C}v) \perp (\mathbb{C}v)^\perp$. Now, $n = \dim V = \dim \mathbb{C}v + \dim (\mathbb{C}v)^\perp = 1 + \dim (\mathbb{C}v)^\perp$. Hence, $\dim (\mathbb{C}v)^\perp = n - 1$. Since $\mathbb{C}v$ is T^* -invariant, $(\mathbb{C}v)^\perp$ is T -invariant. By the induction hypothesis, there exists an orthonormal basis $\mathcal{C} = \{v_1, \dots, v_{n-1}\}$ of $(\mathbb{C}v)^\perp$ such that $[T]_{\mathcal{C}}$ is upper triangular (viewing T as an operator on $(\mathbb{C}v)^\perp$). Let $\mathcal{B} = \{v_1, \dots, v_{n-1}, v\}$, which is an ordered orthonormal basis for V . Then,

$$[T]_{\mathcal{B}} = \begin{pmatrix} & & & | \\ & [T|_{(\mathbb{C}v)^\perp}]_{\mathcal{C}} & & [Tv]_{\mathcal{B}} \\ & \dots & & | \\ 0 & & 0 & \end{pmatrix},$$

which is upper triangular. ■

Corollary 1.86. *Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be linear, and let $\mathcal{C}$ be an ordered orthonormal basis for V . Then there exists an ordered orthonormal basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}$ is upper triangular, and $[T]_{\mathcal{B}} = [1_V]_{\mathcal{C},\mathcal{B}}[T]_{\mathcal{C}}[1_V]_{\mathcal{C},\mathcal{B}}^*$, with $[1_V]_{\mathcal{C},\mathcal{B}}^{-1} = [1_V]_{\mathcal{C},\mathcal{B}}^*$.*

Proof. We know that there exists such a $\mathcal{B}$ such that $[T]_{\mathcal{B}}$ is upper triangular, so it suffices to prove that $[1_V]_{\mathcal{C},\mathcal{B}}^{-1} = [1_V]_{\mathcal{C},\mathcal{B}}^*$. We show instead that $I = [1_V]_{\mathcal{C},\mathcal{B}}^*[1_V]_{\mathcal{C},\mathcal{B}}$. Write

$$[1_V]_{\mathcal{C},\mathcal{B}} = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{pmatrix}.$$

Then,

$$[1_V]_{\mathcal{C},\mathcal{B}}^*[1_V]_{\mathcal{C},\mathcal{B}} = \begin{pmatrix} \overline{a_{1,1}} & \cdots & \overline{a_{n,1}} \\ \vdots & \ddots & \vdots \\ \overline{a_{1,n}} & \cdots & \overline{a_{n,n}} \end{pmatrix} \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{pmatrix}.$$

Write B for this matrix. Then, $(B)_{i,j} = \overline{a_{1,i}}a_{1,j} + \cdots + \overline{a_{n,i}}a_{n,j}$. Now, $c_i = a_{1,i}b_1 + \cdots + a_{n,i}b_n$, so that

$$\begin{aligned} \delta_{i,j} &= \langle c_j, c_i \rangle = \langle a_{1,j}b_1 + \cdots + a_{n,j}b_n, a_{1,i}b_1 + \cdots + a_{n,i}b_n \rangle \\ &= a_{1,j}\langle b_1, a_{1,i}b_1 + \cdots + a_{n,i}b_n \rangle + \cdots + a_{n,j}\langle b_n, a_{1,i}b_1 + \cdots + a_{n,i}b_n \rangle \\ &= a_{1,j}\overline{a_{1,i}}\langle b_1, b_1 \rangle + \cdots + a_{1,j}\overline{a_{n,i}}\langle b_1, b_n \rangle + \cdots + a_{n,j}\overline{a_{1,i}}\langle b_n, b_1 \rangle + \cdots + a_{n,j}\overline{a_{n,i}}\langle b_n, b_n \rangle \\ &= a_{1,j}\overline{a_{1,i}} + \cdots + a_{n,j}\overline{a_{n,i}} \\ &= (B)_{i,j}. \end{aligned}$$

Hence, $B = I$. ■

Note that the above proof shows in fact that $[1_V]_{\mathcal{C},\mathcal{B}}^{-1} = [1_V]_{\mathcal{C},\mathcal{B}}^*$ for any ordered orthonormal bases $\mathcal{C}, \mathcal{B}$.

Definition 1.87. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. Suppose that $T^* : V \rightarrow V$ exists. Then we say that T is a normal operator if $TT^* = T^*T$.

For example, every hermitian operator is normal.

Lemma 1.88. *Let V be a finite dimensional inner product space over F , and let $T : V \rightarrow V$ be normal, and take $\lambda \in F$. Let $0 \neq v \in V$. Then $v \in E_T(\lambda)$ if and only if $v \in E_{T^*}(\overline{\lambda})$. In other words, $Tv = \lambda v$ if and only if $T^*v = \overline{\lambda}v$.*

Proof. We have that

$$\|Sv\|^2 = \langle Sv, Sv \rangle = \langle v, S^*Sv \rangle = \langle v, SS^*v \rangle = \langle S^*v, S^*v \rangle = \|S^*v\|^2$$

for some $S : V \rightarrow V$ normal. Set $S = T - \lambda 1_V$, so that λ is an eigenvalue of T if and only if $\ker S \neq 0$. Notice $S^* = (T - \lambda 1_V)^* = T^* - \overline{\lambda} 1_V$, so that S is normal. Now, $Tv = \lambda v$ if and only if $Sv = 0$, which happens if and only if $S^*v = 0$, if and only if $T^*v = \overline{\lambda}v$. ■

Theorem 1.89. *Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be normal. Then there exists an ordered orthonormal basis $\mathcal{C}$ for V consisting of eigenvectors of T . In particular, T is diagonalizable. Moreover, if $\mathcal{B}$ is an ordered orthonormal basis for V , then*

$$[T]_{\mathcal{C}} = [1_V]_{\mathcal{B}, \mathcal{C}} [T]_{\mathcal{B}} [1_V]_{\mathcal{B}, \mathcal{C}}^*.$$

Proof. We use induction on $n = \dim V$. If $n = 1$, this is clear. Suppose that $n > 1$, and take $\bar{\lambda} \in \mathbb{C}$ to be an eigenvalue of T^* . Let $0 \neq v \in E_{T^*}(\bar{\lambda})$. Then, $v \in E_T(\lambda)$. Hence, $\mathbb{C}v$ is T -invariant and T^* -invariant, so that $(\mathbb{C}v)^\perp$ is both T - and T^* -invariant. We must verify that $(T|_{(\mathbb{C}v)^\perp})^* = T^*|_{(\mathbb{C}v)^\perp}$. By definition, for all $x, y \in (\mathbb{C}v)^\perp$,

$$\langle x, (T|_{(\mathbb{C}v)^\perp})^* y \rangle = \langle T|_{(\mathbb{C}v)^\perp} x, y \rangle = \langle Tx, y \rangle = \langle x, T^* y \rangle = \langle x, T^*|_{(\mathbb{C}v)^\perp} y \rangle.$$

By uniqueness,

$$T^*|_{(\mathbb{C}v)^\perp} = (T|_{(\mathbb{C}v)^\perp})^*.$$

Hence, $T|_{(\mathbb{C}v)^\perp}$ is also normal. Since $\dim(\mathbb{C}v)^\perp = \dim V - \dim(\mathbb{C}v) = n - 1$, we may apply our induction hypothesis to obtain an orthonormal basis $\mathcal{C}_0 = \{v_2, \dots, v_n\}$ for $(\mathbb{C}v)^\perp$ of eigenvectors of $T|_{(\mathbb{C}v)^\perp}$, which are therefore eigenvectors of T . Hence,

$$\mathcal{C} = \left\{ \frac{v}{\|v\|}, v_2, \dots, v_n \right\}$$

is an orthonormal basis for V consisting of eigenvectors of T . We have seen that $[1_V]_{\mathcal{C}, \mathcal{B}}^{-1} = [1_V]_{\mathcal{C}, \mathcal{B}}^*$ for all orthonormal bases $\mathcal{C}$ and $\mathcal{B}$, so we are done by change of basis. $\blacksquare$

1.25 Lecture 25

Theorem 1.90. *Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be linear. Then T is normal if and only if there is an orthonormal basis $\mathcal{B}$ for V consisting of eigenvectors of T . In particular, T is diagonalizable if and only if it is normal.*

Proof. By our previous result, if T is normal, there is an orthonormal basis of eigenvectors of T . Now, let $\mathcal{B}$ be an ordered orthonormal basis for V consisting of eigenvectors of T . Then,

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix},$$

where $n = \dim V$. Since $\mathcal{B}$ is orthonormal,

$$[T^*]_{\mathcal{B}} = [T]_{\mathcal{B}}^* = \begin{pmatrix} \bar{\lambda}_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \bar{\lambda}_n \end{pmatrix}.$$

Thus,

$$[T^*T]_{\mathcal{B}} = [T^*]_{\mathcal{B}}[T]_{\mathcal{B}} = \begin{pmatrix} |\lambda_1|^2 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & |\lambda_n|^2 \end{pmatrix} = [T]_{\mathcal{B}}[T^*]_{\mathcal{B}} = [TT^*]_{\mathcal{B}}.$$

Since matrix representations are unique, $T^*T = TT^*$. ■

Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $\mathcal{C}$ be an ordered basis of V . Let T be normal with eigenvalues $\lambda_1, \dots, \lambda_n$, and let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . We first claim that $E_T(\lambda_i) \perp E_T(\lambda_j)$ for $i \neq j$. Take $v \in E_T(\lambda_i)$ and $w \in E_T(\lambda_j)$, and assume that both v and w are nonzero. Then,

$$\lambda_i \langle v, w \rangle = \langle \lambda_i v, w \rangle = \langle Tv, w \rangle = \langle v, T^*w \rangle = \langle v, \bar{\lambda}_j w \rangle = \bar{\lambda}_j \langle v, w \rangle,$$

so that $\langle v, w \rangle = 0$, as $i \neq j$. Now, notice that

$$V = E_T(\lambda_1) \oplus \cdots \oplus E_T(\lambda_k),$$

as if

$$v_1 + \cdots + v_k = 0$$

for $v_i \in E_T(\lambda_i)$, then

$$\langle v_i, v_i \rangle = \langle v_1, v_i \rangle + \cdots + \langle v_n, v_i \rangle = \langle v_1 + \cdots + v_n, v_i \rangle = 0.$$

Since we also have that $V = E_T(\lambda_i) + E_T(\lambda_i)^\perp$, we have that $E_T(\lambda_i)^\perp = E_T(\lambda_1) + \cdots + E_T(\lambda_{i-1}) + E_T(\lambda_{i+1}) + \cdots + E_T(\lambda_k)$. Now, if we define $P_{\lambda_i}(v) = v_{E_T(\lambda_i)}$, we have that $\text{im } P_{\lambda_i} = E_T(\lambda_i)$, that $\ker P_{\lambda_i} = E_T(\lambda_i)^\perp$, and that $P_{\lambda_i} \circ P_{\lambda_j} = \delta_{i,j} P_{\lambda_i}$. We call $P_{\lambda_1}, \dots, P_{\lambda_k}$ the orthogonal idempotents. We once again have that

$$v = P_{\lambda_1}v + \cdots + P_{\lambda_k}v,$$

and so that

$$1_V = P_{\lambda_1} + \cdots + P_{\lambda_k},$$

so that

$$T = T \circ P_{\lambda_1} + \cdots + T \circ P_{\lambda_k} = \lambda_1 P_{\lambda_1} + \cdots + \lambda_k P_{\lambda_k},$$

meaning in particular that

$$T \circ P_{\lambda_i} = P_{\lambda_i} \circ T$$

for each i .

1.26 Lecture 26

Theorem 1.91 (Spectral Theorem for Normal Operators). *Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$. Then T is normal if and only if it is diagonalizable. If T is normal (and hence diagonalizable) with distinct eigenvalues $\lambda_1, \dots, \lambda_k$, then $E_T(\lambda_i) \perp E_T(\lambda_j)$ for $i \neq j$, and*

$$V = \bigoplus_{i=1}^k E_T(\lambda_i).$$

Further, T^ is normal with distinct eigenvalues $\overline{\lambda_1}, \dots, \overline{\lambda_k}$, and $E_T(\lambda_i) = E_{T^*}(\overline{\lambda_i})$ for each i .*

Notice that the orthogonal idempotents $P_{\overline{\lambda_1}}, \dots, P_{\overline{\lambda_k}}$ for T^* satisfy $P_{\lambda_i} = P_{\overline{\lambda_i}}$. Thus, $T^* = \overline{\lambda_1}P_{\lambda_1} + \dots + \overline{\lambda_k}P_{\lambda_k}$. Now, using the fact that $P_{\lambda_i} \circ P_{\lambda_j} = \delta_{i,j}P_{\lambda_i}$, we have that

$$T^2 = (\lambda_1 P_{\lambda_1} + \dots + \lambda_k P_{\lambda_k})(\lambda_1 P_{\lambda_1} + \dots + \lambda_k P_{\lambda_k}) = \lambda_1^2 P_{\lambda_1} + \dots + \lambda_k^2 P_{\lambda_k}.$$

By induction,

$$T^m = \lambda_1^m P_{\lambda_1} + \dots + \lambda_k^m P_{\lambda_k}.$$

If $f = a_m t^m + \dots + a_1 t + a_0$ is a polynomial, then

$$f(T) = a_m T^m + \dots + a_1 T + a_0 1_V = \sum_{i=1}^k f(\lambda_i) P_{\lambda_i}.$$

Define

$$f_i = \prod_{\substack{j=1 \\ j \neq i}}^k \frac{t - \lambda_j}{\lambda_i - \lambda_j},$$

so that $f_i(\lambda_j) = \delta_{i,j}$. If $g = \overline{\lambda_1} f_1 + \dots + \overline{\lambda_k} f_k$, then $g(\lambda_i) = \overline{\lambda_i}$, and

$$g(T) = \sum_{i=1}^k g(\lambda_i) P_{\lambda_i} = \overline{\lambda_1} P_{\lambda_1} + \dots + \overline{\lambda_k} P_{\lambda_k} = T^*.$$

Proposition 1.92. *Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be linear.*

- (1) *T is normal if and only if there is $g \in \mathbb{C}[t]$ such that $T^* = g(T)$.*
- (2) *T is an isometry if and only if T is normal and $|\lambda| = 1$ for every eigenvalue λ of T .*
- (3) *If T is normal, then T is hermitian if and only if every eigenvalue of T is real.*

Proof.

- (1) We have already seen the existence of such a g if T is normal. If such a g exists, then

$$T^*T = g(T)T = Tg(T) = TT^*,$$

so T is normal.

- (2) If T is an isometry, then $T^* = T^{-1}$ as

$$\langle v_1, T^*Tv_2 \rangle = \langle Tv_1, Tv_2 \rangle = \langle v_1, v_2 \rangle,$$

so that $T^*T = 1_V$. Hence, T is normal. We also have that $\|Tv\| = \|v\|$, so if $Tv = \lambda v$, $|\lambda| = 1$. Conversely, if T is normal and $|\lambda| = 1$ for all eigenvalues λ of T , then there is an orthonormal basis $v_1, \dots, v_n$ of V consisting of eigenvectors of T , and for $v = a_1v_1 + \dots + a_nv_n$,

$$\|Tv\| = \|\lambda_1a_1v_1 + \dots + \lambda_na_nv_n\| = |\lambda_1|^2|a_1|^2 + \dots + |\lambda_n|^2|a_n|^2 = |a_1|^2 + \dots + |a_n|^2 = \|v\|,$$

so T is an isometry.

- (3) Clearly, if T is hermitian every eigenvalue is real. If every eigenvalue of T is real, then T^* has the same eigenvalues and eigenspaces. Thus, if $v_1, \dots, v_n$ is an orthonormal basis of V consisting of eigenvectors of T , then T and T^* agree on this basis, and thus agree on V .

■

We call a matrix A unitary if $A^* = A^{-1}$.

Theorem 1.93 (Singular Value Theorem for Matrices). *Let $F = \mathbb{R}$ or $\mathbb{C}$, and let $A \in F^{m \times n}$. Then there are unitary $U \in \mathbb{M}_n(F)$ $X \in \mathbb{M}_m(F)$ such that*

$$X^*AU = D = \begin{pmatrix} \mu_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \mu_r & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Further, we can choose $\mu_1 \geq \dots \geq \mu_r > 0$, and r is the rank of A .

Proof. We have that $A^*A \in \mathbb{M}_n(F)$ is hermitian with non-negative real eigenvalues: it is clearly hermitian, and if $A^*Av = \lambda v$, then $\lambda \langle v, v \rangle = \langle A^*Av, v \rangle = \langle Av, Av \rangle$, so λ cannot be negative, as this would mean that $\|Av\|^2$, clearly a nonnegative number, is negative. Let $\lambda_1, \dots, \lambda_r$ be the positive eigenvalues ordered such that $\lambda_1 \geq \dots \geq \lambda_r > 0$. Take unitary U such that

$$U^*(A^*A)U = \begin{pmatrix} \lambda_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_r & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix},$$

which is possible by change of basis. Let $C = AU \in F^{m \times n}$, so that $C^*C = (AU)^*(AU) = U^*(A^*A)U \in \mathbb{M}_n(F)$. Write $\lambda_i = \mu_i^2$, where $\mu_i > 0$. Then $\mu_1 \geq \dots \geq \mu_r > 0$. Write $E^{(k)}$ for the k -th column of the matrix E . Then,

$$\lambda_i \delta_{i,j} = (C^*C)_{i,j} = \sum_{\ell=1}^n (C^*)_{i,\ell} C_{\ell,j} = \sum_{\ell=1}^n \overline{C_{\ell,i}} C_{\ell,j} = \sum_{\ell=1}^n C_{\ell,j} \overline{C_{\ell,i}} = \langle C^{(j)}, C^{(i)} \rangle.$$

Hence,

$$C = [C^{(1)} \ \dots \ C^{(r)} \ 0 \ \dots \ 0] \in F^{m \times n}$$

satisfies $\mathcal{C}_0 = \{C^{(1)}, \dots, C^{(r)}\}$ is an orthogonal set in $F^{m \times n}$. Hence, the rank of C is r . We also have that

$$\|C^{(i)}\|^2 = \langle C^{(i)}, C^{(i)} \rangle = \lambda_i = \mu_i^2.$$

As U is invertible, the rank of A is the same as the rank of $AU = C$, which is r . Thus, $\text{rank } A = r$. Now define $X^{(i)} = C^{(i)}/\mu_i$. Then $\mathcal{B}_0 = \{X^{(1)}, \dots, X^{(r)}\}$ is an orthonormal set. Extend this to an ordered orthonormal basis $\mathcal{B} = \{X^{(1)}, \dots, X^{(m)}\}$. Write

$$X = [X^{(1)} \ \dots \ X^{(m)}] = [1_{F^{m \times 1}}]_{\mathcal{B}, \mathcal{S}},$$

where $\mathcal{S}$ is the standard basis. Since $\mathcal{B}$ and $\mathcal{S}$ are orthonormal, $XX^* = I$. Set

$$D = \begin{pmatrix} \mu_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \mu_r & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix}.$$

Then,

$$XD = [X^{(1)} \ \dots \ X^{(m)}]D = [\mu_1 X^{(1)} \ \dots \ \mu_r X^{(r)} \ 0 \ \dots \ 0] = C = AU.$$

Thus,

$$X^*AU = D.$$

■

Note that D is not diagonal.

1.27 Lecture 27

Definition 1.94. Let $A \in F^{m \times n}$ where $F = \mathbb{R}$ or $\mathbb{C}$. Write $A = XDU^*$, where X , D and U are as in the Singular Value Theorem for Matrices. Then this is called a Singular Value Decomposition for A . We call $\mu_1, \dots, \mu_r$ the singular values of A .

Corollary 1.95. *The singular values of $A \in F^{m \times n}$, where $F = \mathbb{R}$ or $\mathbb{C}$, are unique (including multiplicity) up to order.*

Proof. Let $A = XDU^*$ be a singular value decomposition of A . Then

$$A^*A = (XDU^*)^*(XDU^*) = UD^*X^*XDU^* = UD^*DU.$$

Thus A^*A and D^*D are similar, and so have the same eigenvalues. If $D_{i,i} = \alpha_i$ for $i \leq r$ (and $D_{i,j} = 0$ everywhere else), then the eigenvalues of D^*D are the α_i^2 's. Hence, the singular values of A are the positive square roots of the eigenvalues of A^*A counting multiplicity, which is unique up to ordering. ■

Theorem 1.96 (Singular Value Theorem for Linear Transformations). *Let $F = \mathbb{R}$ or $\mathbb{C}$, and let V and W be finite dimensional inner product spaces over F , and $T : V \rightarrow W$ be linear of rank r . Then there are orthonormal bases $\mathcal{B} = \{v_1, \dots, v_n\}$ for V and $\mathcal{C} = \{w_1, \dots, w_m\}$ for w , along with $\mu_1 \geq \dots \geq \mu_r > 0 \in \mathbb{R}$ satisfying*

$$Tv_i = \begin{cases} \mu_i w_i & 1 \leq i \leq r \\ 0 & i > r \end{cases}.$$

Proof. Fix orthonormal bases $\mathcal{S}_V$ and $\mathcal{S}_W$ for V and W respectively. Write $A = [T]_{\mathcal{S}_V, \mathcal{S}_W}$. Let $\lambda_1 \geq \dots \geq \lambda_r > 0$ be the positive eigenvalues of A^*A , and take U as in the proof of the Singular Value Theorem for Matrices. Let $v'_1, \dots, v'_n$ be an orthonormal set of eigenvectors of A^*A (in the correct order), and $\mathcal{B}_0 = \{v'_1, \dots, v'_n\}$. Take $[v_i]_{\mathcal{S}_V} = v'_i$, and $\mathcal{B} = \{v_1, \dots, v_n\}$. Then $U = [1_V]_{\mathcal{B}, \mathcal{S}_V}$. Continue to construct X , and write $w'_i = X^{(i)}$. Write $\mathcal{C}_0 = \{w'_1, \dots, w'_m\}$. Set $[w_i]_{\mathcal{S}_W} = w'_i$ with $\mathcal{C} = \{w_1, \dots, w_m\}$. Then $X = [1_W]_{\mathcal{C}, \mathcal{S}_W}$. We also have that the w_i 's are orthonormal, and $X^*AU = D$. Hence, $A = XDU^*$. Denote by $\mathcal{S}_i^V$ $[(\mathcal{S}_V)_i]_{\mathcal{S}_V}$, and similarly for W . Now,

$$Av'_i = XDU^{-1}v'_i = XD\mathcal{S}_i^V = \begin{cases} X(\mu_i \mathcal{S}_i^W) & 1 \leq i \leq r \\ X(0) & i > r \end{cases} = \begin{cases} \mu_i w'_i & 1 \leq i \leq r \\ 0 & i > r \end{cases}.$$

Converting back to linear transformations,

$$Tv_i = \begin{cases} \mu_i w_i & 1 \leq i \leq r \\ 0 & i > r \end{cases},$$

as desired. ■

Notice that, in the Singular Value Theorem for Matrices, $D^* = D^\top$, so that $U^*A^*X = D^\top$. In particular, A and A^* have the same singular values. Applying this to our proof of the Singular Value Theorem for Linear Transformations, we obtain that

$$T^*w_i = \begin{cases} \mu_i v_i & 1 \leq i \leq r \\ 0 & i > r \end{cases}.$$

Theorem 1.97 (Polar Decomposition). *Let $F = \mathbb{R}$ or $\mathbb{C}$, and let $A \in \mathbb{M}_n(F)$. Then there is some unitary $\tilde{U} \in \mathbb{M}_n(F)$ and $N \in \mathbb{M}_n(F)$ hermitian with nonnegative eigenvalues such that $A = \tilde{U}N$.*

Proof. By the Singular Value Theorem for Matrices, decompose $A = XDU^*$, where X, U are unitary elements of $\mathbb{M}_n(F)$. We have that $D = D^*$, and D has nonnegative eigenvalues. Thus,

$$A = XDU^* = XU^*UDU = (XU^*)(UDU^*).$$

Define $\tilde{U} = XU^*$ and $N = UDU^*$. We have that

$$(XU^*)^*(XU^*) = UX^*XU^* = UU^* = I.$$

Since N is clearly hermitian and has nonnegative eigenvalues, we are done. ■

1.28 Extra Lectures

Definition 1.98. Let V be a vector space over F . The vector space $V^* = L(V, F)$ is called the dual space of V .

Let V be a nontrivial vector space over the field F with basis B . For each $v_0 \in B$, we define $f_{v_0} : V \rightarrow F$ being the unique linear map with $v_0 \mapsto 1$ and $v \mapsto 0$ for all $v \in B \setminus \{v_0\}$. Note that $\text{im } f_{v_0} = F$, and that $\ker f_{v_0} = \text{Span}(B \setminus \{v_0\})$. Note also that f_{v_0} sends w to the v_0 -coordinate of w . We write

$$\delta_{i,j} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Now, we have a map

$$\begin{aligned} D_B : V &\rightarrow V^* \\ v \in B &\mapsto f_v. \end{aligned}$$

We claim that D_B is injective. Suppose that $D_B(w) = 0$ for $w = \sum_{v \in B} \alpha_v v$. Then, $\sum_{v \in B} \alpha_v f_v = 0$, so that

$$0 = \left(\sum_{v \in B} \alpha_v f_v \right) (v') = \sum_{v \in B} \alpha_v \delta_{v,v'} = \alpha_{v'}$$

for all $v' \in B$, so that $\alpha_{v'} = 0$ for all $v' \in B$. Thus, $\ker D_B = \{0\}$, so D_B is injective. Write $B^* = D_B(B) = \{f_v : v \in B\}$. Note that we have also proved that B^* is linearly independent.

Definition 1.99. Let V be a vector space over F . Any element in V^* is called a linear functional.

Now let C be a basis for V^* . Then, we have the map

$$D_C : V^* \rightarrow (V^*)^* = V^{**}$$

induced by $f_0 \in C \mapsto G_{f_0}$, where $G_{f_0}(f) = \delta_{f,f_0}$ for $f \in C$. We call V^{**} the double dual of V .

Both of our previous maps depend on the basis we choose. We wish to find a map independent of basis. We claim that there is such a monomorphism

$$L : V \rightarrow V^{**}.$$

For each $v \in V$, define

$$L_v : V^* \rightarrow F$$

by $L_v(f) = f(v)$. If we show that L_v is linear. We have that

$$L_v(\alpha f + g) = (\alpha f + g)(v) = \alpha f(v) + g(v) = \alpha L_v(f) + L_v(g),$$

so that L_v is linear. Now define

$$\begin{aligned} L : V &\rightarrow V^{**} \\ v &\mapsto L_v. \end{aligned}$$

We claim that L is linear. We have that

$$L(\alpha v + v')(f) = L_{\alpha v + v'}(f) = f(\alpha v + v') = \alpha f(v) + f(v') = \alpha L_v(f) + L_{v'}(f) = \alpha L(v)(f) + L(v')(f).$$

Now suppose that $L(v) = 0$, so that $f(v) = 0$ for all f . If $v \neq 0$, choose $f = f_v$ so that $f(v) = 1$ to obtain a contradiction. Therefore, $\ker L = \{0\}$, so L is injective. Thus, we have proven the following.

Theorem 1.100. *Let V be a vector space over F . Then there is a canonical injective linear transformation*

$$L : V \rightarrow V^{**}.$$

Now suppose that V is finite dimensional, and let $B = \{v_1, \dots, v_n\}$ be a basis for V . Then $B^* = \{f_1, \dots, f_n\}$ defined by $f_i(v_j) = \delta_{i,j}$ is linearly independent. We have an injective linear transformation D_B , but $\dim V^* = \dim L(V, F) = \dim V \dim F = \dim V$, so D_B is an isomorphism, meaning $V \cong V^*$. Note also that $V \cong V^{**}$ in this case. Now if

$$B^{**} = \{L_{v_1}, \dots, L_{v_n}\},$$

then $L : V \rightarrow V^{**}$ is a natural isomorphism, and so when V is finite dimensional, we can identify L_v and v . Note also that we may use our bases B , B^* , B^{**} as follows. If $v \in V$, write

$$v = \alpha_1 v_1 + \dots + \alpha_n v_n,$$

so

$$f_i(v) = \alpha_i,$$

meaning

$$v = \sum_{i=1}^n f_i(v) v_i = \sum_{i=1}^n L_v(f_i) v_i.$$

Now if $f \in V^*$ with

$$f = \beta_1 f_1 + \dots + \beta_n f_n,$$

then

$$L_{v_j}(f) = \beta_1 f_1(v_j) + \dots + \beta_n f_n(v_j) = \beta_j,$$

so

$$f = \sum_{i=1}^n f(v_i) f_i = \sum_{i=1}^n L_{v_i}(f) f_i.$$

Let V and W be vector spaces over F , and let $T : V \rightarrow W$ be linear. If $g \in W^*$, then $g \circ T \in V^*$.

Definition 1.101. Let V and W be vector spaces over F , and let $T : V \rightarrow W$ be linear. Define the transpose of T by

$$\begin{aligned} T^\top : W^* &\rightarrow V^* \\ g &\mapsto g \circ T. \end{aligned}$$

We claim that $T^\top$ is linear. Let $g, g' \in W^*$ and $\alpha \in F$. Then,

$$T^\top(\alpha g + g') = (\alpha g + g') \circ T = \alpha(g \circ T) + (g' \circ T) = \alpha T^\top(g) + T^\top(g').$$

Theorem 1.102. Let V and W be finite dimensional vector spaces over F , and let B and C be ordered bases for V and W respectively. Then,

$$[T]_{B,C}^\top = [T^\top]_{C^*,B^*}.$$

Proof. Write

$$B = \{v_1, \dots, v_n\}, \quad B^* = \{f_1, \dots, f_n\}, \quad C = \{w_1, \dots, w_m\}, \quad C^* = \{g_1, \dots, g_m\}.$$

Let $[T]_{B,C} = (\alpha_{i,j})$ and $[T^\top]_{C^*,B^*} = (\beta_{i,j})$. Then,

$$Tv_k = \sum_{i=1}^m \alpha_{i,k} w_i \in W \tag{1.1}$$

$$T^\top g_j = \sum_{i=1}^n \beta_{i,j} f_i \in V^*. \tag{1.2}$$

Computing,

$$(T^\top g_j)(v_k) = g_j(Tv_k) = g_j\left(\sum_{i=1}^m \alpha_{i,k} w_i\right) = \sum_{i=1}^m \alpha_{i,k} g_j(w_i) = \sum_{i=1}^m \alpha_{i,k} \delta_{i,j} = \alpha_{j,k}.$$

We also have

$$(T^\top g_j)(v_k) = \left(\sum_{i=1}^n \beta_{i,j} f_i\right)(v_k) = \sum_{i=1}^n \beta_{i,j} f_i(v_k) = \sum_{i=1}^n \beta_{i,j} \delta_{i,k} = \beta_{k,j}.$$

Thus, $\alpha_{j,k} = \beta_{k,j}$ for all j, k . ■

Definition 1.103. Let V be a vector space over F , and let $\emptyset \neq S \subseteq V$. Then the set

$$S^0 = \{f \in V^* : \forall s \in S, f(s) = 0\}$$

is called the annihilator of S .

Proposition 1.104. Let V be a vector space over F , and $\emptyset \neq S \subseteq V$. Then S^0 is a subspace of V^* .

Proof. Since the zero function is in S^0 , it is nonempty. Let $f, g \in S^0$ and $\alpha \in F$. Then,

$$(\alpha f + g)(s) = \alpha f(s) + g(s) = 0$$

for all $s \in S$. ■

Note that $\ker T^\top = (\operatorname{im} T)^0$.

Proposition 1.105. *Let V be a finite dimensional vector space over F , and let $W \subseteq V$ be a subspace. Then $\dim V = \dim W + \dim W^0$.*

Proof. Let $\{v_1, \dots, v_k\}$ be a basis for W , and extend it to a basis $B = \{v_1, \dots, v_n\}$ for V . Let $B^* = \{f_1, \dots, f_n\}$ be the dual basis of B . We claim that $C = \{f_{k+1}, \dots, f_n\}$ is a basis for W^0 . Clearly, $\operatorname{Span}(C) \subseteq W^0$. Let $f \in W^0$. Write $f = \sum_{i=1}^n \beta_i f_i = \sum_{i=1}^n f(v_i) f_i = \sum_{i=k+1}^n f(v_i) f_i$. Thus, $W^0 \subseteq \operatorname{Span}(C)$, so $W^0 = \operatorname{Span}(C)$. Thus, C is a basis for W^0 . ■

Corollary 1.106. *Let V be a finite dimensional vector space over F , and let $W \subseteq V$ be a subspace. Identifying V and V^{**} by $v \leftrightarrow L_v$, we have that $W = (W^0)^0 = W^{00}$.*

Proof. We have that $W^0 \subseteq V^*$ and $W^{00} \subseteq V^{**} = V$ are subspaces. Now, $\dim V = \dim W + \dim W^0$, $\dim V = \dim W^*$, and $\dim V^* = \dim W^0 + \dim W^{00}$. Thus, $\dim W + \dim W^0 = \dim W^0 + \dim W^{00}$, so that $\dim W = \dim W^{00}$. Now we show that $W \subseteq W^{00}$. We have that $L_w(f) = f(w) = 0$ for all $f \in W^0$, so that $w = L_w \in W^{00}$. Thus, $W \subseteq W^{00}$, and so $W = W^{00}$. ■

Theorem 1.107. *Let V and W be finite dimensional vector spaces over F and that $T : V \rightarrow W$ is linear. Then $\dim \operatorname{im} T = \dim \operatorname{im} T^\top$.*

Proof. We know that $\dim W = \dim \operatorname{im} T + \dim (\operatorname{im} T)^0$ and that $\dim W^* = \dim \operatorname{im} T^\top + \dim \ker T^\top$. Since $\dim W = \dim W^*$, $\dim \operatorname{im} T - \dim \operatorname{im} T^\top = \dim \ker T^\top - \dim (\operatorname{im} T)^0$. Recall that $(\operatorname{im} T)^0 = \ker T^\top$, so that this is 0. Thus, $\dim \operatorname{im} T = \dim \operatorname{im} T^\top$. ■

Now let $A \in F^{m \times n}$. Define the row rank of A to be the dimension of the span of the rows of A , and similarly for the columns. Then the row rank of A is the same as the column rank of A (taking transposes and using previous results establishes this). We write $\operatorname{rank} A$ for this value.

Proposition 1.108. *If V, W, X are finite dimensional vector spaces over F with $T : V \rightarrow W$ and $S : W \rightarrow X$ linear, then*

$$\dim \operatorname{im} S + \dim \operatorname{im} T - \dim W \leq \dim \operatorname{im} S \circ T \leq \min\{\dim \operatorname{im} T, \dim \operatorname{im} S\}.$$

Proof. Note that $\operatorname{im}(S \circ T) \subseteq \operatorname{im} S$, so that $\dim \operatorname{im}(S \circ T) \leq \dim \operatorname{im} S$. We also have that $\ker(S \circ T) \supseteq \ker T$, so $\dim \ker(S \circ T) \geq \dim \ker T$. Now,

$$\dim \operatorname{im} T = \dim V - \dim \ker T \geq \dim V - \dim \ker(S \circ T) = \dim \operatorname{im}(S \circ T).$$

We also have

$$\begin{aligned} \dim \operatorname{im} T &= \dim \operatorname{im} S|_{\operatorname{im} T} + \dim \ker S|_{\operatorname{im} T} = \dim \operatorname{im}(S \circ T) + \dim \ker S|_{\operatorname{im} T} \\ &\leq \dim \operatorname{im}(S \circ T) + \dim \ker S = \dim \operatorname{im}(S \circ T) + \dim W - \dim \operatorname{im} S. \end{aligned}$$

■

Let F be a field with $|F| \geq n + 1$. Given $\alpha_0, \dots, \alpha_n$ and $\beta_0, \dots, \beta_n$, where the α_i 's are distinct, we show that there exists a unique $f \in F[x]_n$ such that $f(\alpha_i) = \beta_i$ for all i . Define

$$f_i(t) = \frac{(t - \alpha_0) \cdots (t - \alpha_{i-1})(t - \alpha_{i+1}) \cdots (t - \alpha_n)}{(\alpha_i - \alpha_0) \cdots (\alpha_i - \alpha_{i-1})(\alpha_i - \alpha_{i+1}) \cdots (\alpha_i - \alpha_n)}.$$

Then, the polynomial $\sum \beta_i f_i$ satisfies this, so we have proven existence. Now, the f_i 's are linearly independent, so they must span $F[x]_n$, as $\dim F[x]_n = n + 1$. Similarly, the e_{α_i} 's are linearly independent (where $e_a(f) = f(a)$), and thus must span $F[x]_n^*$. Notice further that $e_{\alpha_i}(f_j) = \delta_{i,j}$, so that $\{f_0, \dots, f_n\}$ and $\{e_{\alpha_0}, \dots, e_{\alpha_n}\}$ are dual bases. Thus, if

$$f = \sum \gamma_i f_i,$$

then

$$\gamma_i = f(\alpha_i),$$

so we have proven uniqueness.

Let F be a subfield of $\mathbb{C}$ closed under complex conjugation, and let $B = \{v_1, \dots, v_n\}$ be an orthonormal basis for the finite dimensional inner product space V over F . Define $f_i = \langle \cdot, v_i \rangle \in V^*$. Then $f_i(v_j) = \delta_{i,j}$, so B and $B^* = \{f_1, \dots, f_n\}$ are dual bases. In particular, if $v \in V$, then there are unique $\alpha_i \in F$ such that

$$v = \sum_{i=1}^n \alpha_i v_i,$$

by the definition of a basis. Since the f_i 's are the coordinate functions of the v_i 's, we have that

$$v = \sum_{i=1}^n f_i(v) v_i = \sum_{i=1}^n \langle v, v_i \rangle v_i.$$

More generally, suppose that $C = \{w_1, \dots, w_n\}$ is an orthogonal basis. Define

$$g_i = \left\langle \cdot, \frac{w_i}{\|w_i\|^2} \right\rangle,$$

which is linear. We once again have that $g_i(w_j) = \delta_{i,j}$, so C and $C^* = \{g_1, \dots, g_n\}$ are dual bases. Now let $g \in V^*$ be arbitrary. Then

$$g = \sum_{i=1}^n g(w_i) g_i = \sum_{i=1}^n g(w_i) \left\langle \cdot, \frac{w_i}{\|w_i\|^2} \right\rangle = \left\langle \cdot, \sum_{i=1}^n \overline{g(w_i)} \frac{w_i}{\|w_i\|^2} \right\rangle.$$

Write

$$z = \sum_{i=1}^n \overline{g(w_i)} \frac{w_i}{\|w_i\|^2}.$$

Then we have $g = \langle \cdot, z \rangle$. Thus, every linear transformation $g : V \rightarrow F$ is of the form $g(v) = \langle v, z \rangle$ for some $z \in V$. Thus, the map

$$H : V \rightarrow V^* \quad z \mapsto \langle \cdot, z \rangle$$

is surjective (in fact, bijective: if $\langle \cdot, v \rangle = \langle \cdot, w \rangle$ for some v, w , then $\langle \cdot, v - w \rangle = 0$, so we must have that $v - w = 0$ by plugging in $v - w$ for $\cdot$). Note that H is not linear in general. This is because, if $z, z' \in V$ and $\alpha \in F$,

$$H(\alpha z + z') = \langle \cdot, \alpha z + z' \rangle = \bar{\alpha} \langle \cdot, z \rangle + \langle \cdot, z' \rangle = \bar{\alpha} H(z) + H(z').$$

Thus, H is only linear if $F \subseteq \mathbb{R}$.

2 Course B

2.1 Lecture 1

We recall the following definition.

Definition 2.1. Let V be a vector space over F , and let $W_i \subseteq V$ for $i \in I$ be subspaces. We define

$$\sum_{i \in I} W_i = \text{Span} \left(\bigcup_{i \in I} W_i \right).$$

Definition 2.2. Let V be a vector space over F , and $W_i \subseteq V$ be a subspace for $i \in I$. Let $W \subseteq V$ be a subspace. We say that W is the internal direct sum of the W_i for $i \in I$, denoted $W = \bigoplus_{i \in I} W_i$ if, for all $w \in W$, there are unique $w_i \in W_i$, only finitely many of which are nonzero, such that $w = \sum_{i \in I} w_i$.

We also define a ring to be a field, where we do not require commutativity of multiplication and multiplicative inverses. If multiplication is commutative in a ring, we say the ring is a commutative ring, and if multiplicative inverses exist, we say it is a division ring. Thus, a commutative division ring is a field.

2.2 Lecture 2

Definition 2.3. Let V be a vector space over F , and $W_i \subseteq V$ be subspaces for $i \in I$. We say that the W_i are independent if, whenever $w_i \in W_i$, with only finitely many nonzero, satisfy $\sum_{i \in I} w_i = 0$, then $w_i = 0$ for all $i \in I$.

Theorem 2.4. Let V be a vector space over F , and $W_i \subseteq V$ be a subspace for $i \in I$. For a subspace $W \subseteq V$, the following are equivalent

$$(1) \ W = \bigoplus_{i \in I} W_i.$$

$$(2) \ W = \sum_{i \in I} W_i \text{ and } W_i \cap \sum_{j \in I \setminus \{i\}} W_j = 0 \text{ for all } i.$$

$$(3) \ W = \sum_{i \in I} W_i, \text{ and the } W_i \text{ are independent.}$$

Proof. We show that (1) $\implies$ (2) $\implies$ (3) $\implies$ (1). Suppose (1). Fix i and take $x \in W_i \cap \sum_{j \in I \setminus \{i\}} W_j$. Write $x = w_i = \sum_{j \in I \setminus \{i\}} w_j$ for $w_i \in W_i$ and $w_j \in W_j$, where all but finitely many of the w_j are 0. Then, $0 = w_i - \sum_{j \in I \setminus \{i\}} w_j$. We also have that $0 = \sum_{i \in I} 0$. By uniqueness, $w_i = 0$, so $x = 0$. Now suppose (2), and suppose that $\sum_{i \in I} w_i = 0$ for finitely many of $w_i \in W_i$ nonzero. Fix k , and write $w_k = -\sum_{i \in I \setminus \{k\}} w_i$. This is in $W_k \cap \sum_{i \in I \setminus \{k\}} W_i = 0$. Thus, $w_k = 0$. As k was arbitrary, the W_i are independent. Finally, suppose (3), and suppose that $\sum_{i \in I} w_i = \sum_{i \in I} w'_i$ for finitely many $w_i \in W_i$ nonzero. Then, $\sum_{i \in I} (w_i - w'_i) = 0$, so that $w_i - w'_i = 0$ for all i . Thus, $w_i = w'_i$, so (1) holds. ■

Corollary 2.5. Let V be a vector space over F , and $W_i \subseteq V$ be a subspace for $i \in I$. Suppose that $I = I_1 \cup I_2$ with $I_1 \cap I_2 = \emptyset$ and that $V = \bigoplus_{i \in I} W_i$. If $W_{I_1} = \bigoplus_{i \in I_1} W_i$ and $W_{I_2} = \bigoplus_{j \in I_2} W_j$, then $V = W_{I_1} \oplus W_{I_2}$.

Proof. Clearly, $V = W_{I_1} + W_{I_2}$, so we only need to show that $W_{I_1} \cap W_{I_2} = 0$. Suppose $v \in W_{I_1} \cap W_{I_2}$. Write $v = \sum_{i \in I_1} w_i = \sum_{j \in I_2} x_j$. Then, $\sum_{i \in I_1} w_i - \sum_{j \in I_2} x_j = 0$. By uniqueness, $w_i = x_j = 0$ for all i and j . Thus, $v = 0$. ■

Recall that we write $Fv = \text{Span}(v)$. If V is a vector space over F , and $S \subseteq V$ is nonempty, then

$$\sum_{v \in S} Fv = \text{Span}(S).$$

If S is linearly independent, then we further have

$$\bigoplus_{v \in S} Fv = \text{Span}(S).$$

Definition 2.6. Let V be a finite dimensional vector space over F , and let $W \subseteq V$ be a subspace. If there is some subspace $W' \subseteq V$ such that $V = W \oplus W'$, we call W' a complementary subspace of W in V .

If V is a vector space over F and B is a basis for V and $x \in V$, then there are unique $\alpha_v \in F$ for each $v \in B$, where all but finitely many α_v are zero, such that $x = \sum_{v \in B} \alpha_v v$. We write $x = \sum_{v \in B} \alpha_v v$ to mean that α_v is the unique component of x on v , and that $\alpha_v = 0$ for all but finitely many $v \in B$.

Proposition 2.7. *Let V be a finite dimensional vector space over F , and let $W_1, \dots, W_n \subseteq V$ be subspaces with $W_i \cap W_j = \emptyset$ for $i \neq j$. Then the following are equivalent.*

$$(1) \ V = W_1 \oplus \dots \oplus W_n$$

$$(2) \ \text{If } B_i \text{ is a basis for } W_i, \text{ then } B = B_1 \cup \dots \cup B_n \text{ is a basis for } V.$$

Proof. Suppose that $V = W_1 \oplus \dots \oplus W_n$. Clearly, B is spanning, so we only need to show that it is linearly independent. Suppose that

$$\sum_{v \in B} a_v v = 0.$$

Then, we have that

$$\sum_{i=1}^n \sum_{v \in B_i} a_v v = 0.$$

By uniqueness,

$$\sum_{v \in B_i} a_v v = 0$$

for all i . As B_i is a basis, we have that $a_v = 0$ for all $v \in B$, so B is a basis. Now suppose (2), and let $v \in V$. Write

$$v = \sum_{w \in B} a_w w = \sum_{i=1}^n \sum_{w \in B_i} a_w w.$$

If $x_i = \sum_{w \in B_i} a_w w$, then $x_i \in W_i$, and $v = \sum_{i=1}^n x_i$. Thus, $V = W_1 + \dots + W_n$. Now suppose that $\sum_{i=1}^n w_i = 0$ for $w_i \in W_i$. Write $w_i = \sum_{v \in B_i} a_v v$. Then,

$$0 = \sum_{i=1}^n \sum_{v \in B_i} a_v v = \sum_{v \in B} a_v v,$$

so that $a_v = 0$ for all v . Thus, $w_i = 0$ for all i , and $V = W_1 \oplus \dots \oplus W_n$. ■

We now discuss quotients. Let V be a vector space over F , and $W \subseteq V$ be a subspace. For $v \in V$, we write $v + W$ for the set $\{v + w : w \in W\}$.

Proposition 2.8. *Let V be a vector space over F , and $W \subseteq V$ be a subspace. Then $v + W = v' + W$ if and only if $v - v' \in W$.*

Proof. Suppose that $v + W = v' + W$. Then, $v \in v + W$ is of the form $v' + w$ for some $w \in W$. Thus, $v - v' = w \in W$. Now, suppose that $v - v' = w \in W$, and let $v + w_1 \in v + W$. Then, $v + w_1 = v' + (w + w_1) \in v' + W$ so $v + W \subseteq v' + W$. Similarly, if $v' + w_1 \in v' + W$, then $v' + w_1 = v + (w_1 - w) \in v + W$, so $v' + W \subseteq v + W$. Thus, $v + W = v' + W$. ■

2.3 Lecture 3

Notice that

$$V = \bigcup_{v \in V} (v + W),$$

as $v \in v + W$.

Now, we show that $v \sim v' \iff v + W = v' + W$ is an equivalence relation. Clearly, $v \sim v$ and $v \sim v'$ if and only if $v' \sim v$. Finally, suppose that $v \sim v'$ and $v' \sim v''$. Then $v' - v \in W$ and $v'' - v' \in W$, so that $v'' - v' + v' - v = v'' - v \in W$. Thus, $v + W = v'' + W$, so that $v \sim v''$. In particular, the $v + W$'s are disjoint, and thus partition V .

Let $S = \{v + W : v \in V\}$. We make S into a vector space over F . We first check that the obvious operations are well defined.

Proposition 2.9. *Let V be a vector space over F , and $W \subseteq V$ be a subspace. If $v + W = v' + W$ and $w + W = w' + W$, then $(v + w) + W = (v' + w') + W$, and $\alpha v + W = \alpha v' + W$ for $\alpha \in F$.*

Proof. We have that $v - v' \in W$ and that $w - w' \in W$, so that $v - v' + w - w' = (v + w) - (v' + w') \in W$. Thus, $(v + w) + W = (v' + w') + W$. Now, we also have that $\alpha(v - v') = \alpha v - \alpha v' \in W$, so that $\alpha v + W = \alpha v' + W$. ■

Thus, the operations $(v + W) + (v' + W) = (v + v') + W$ and $\alpha(v + W) = \alpha v + W$ are well defined. Further, by the definition of these operations, S is a vector space over F . Notice that, in this space, the zero is $0 + W = W$.

Definition 2.10. Let V be a vector space over F , and $W \subseteq V$ a subspace. Let $S = \{v + W : v \in V\}$, and make S into a vector space over F with

$$(v + W) + (v' + W) = (v + v') + W \quad \alpha(v + W) = (\alpha v) + W.$$

We call this space the quotient of V by W , and denote it V/W .

We will write $\bar{v}$ for $v + W$. If W is not clear, we will write $[v]_W$ for $v + W$. We call the sets $\bar{v}$ cosets of V by W . We also write $x \equiv y \pmod{W}$ if $x + W = y + W$.

2.4 Lecture 4

Consider the linear transformation $\phi : V \rightarrow V/W$ given by $\phi(v) = v + W$. Notice that it is surjective. This is called the canonical surjection. Notice also that $\ker \phi = W$. Suppose that V is finite dimensional. Then, we have that $\dim \operatorname{im} \phi + \dim \ker \phi = \dim V$, so that $\dim V/W = \dim V - \dim W$. We call this value the codimension of W in V .

Proposition 2.11. *Let V be a vector space over F , and $W \subseteq V$ a subspace. Let $\mathcal{B}_1$ be a basis of W , and let $\mathcal{C}$ be a basis of V/W . For each element X in $\mathcal{C}$, choose one (and only one) element $x \in X$ satisfying $x + W = \bar{x} = X$ and set $\mathcal{B}_2$ to be the set of these x 's. Then $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for V .*

Proof. Let $v \in V$, and write $\bar{v} = \sum_{X \in \mathcal{C}} a_X X$. Then,

$$v' := v - \sum_{x \in \mathcal{B}_2} a_{\bar{x}} x \in W.$$

Write $v' = \sum_{x \in \mathcal{B}_1} b_x x$. Then, we have that

$$v = \sum_{x \in \mathcal{B}_1} b_x x + \sum_{x \in \mathcal{B}_2} a_{\bar{x}} x,$$

so $\mathcal{B}$ is spanning. Now, suppose that

$$0 = \sum_{x \in \mathcal{B}} a_x x = \sum_{x \in \mathcal{B}_1} a_x x + \sum_{x \in \mathcal{B}_2} a_x x.$$

Passing to V/W , as have that

$$0 = \sum_{x \in \mathcal{B}_2} a_x \bar{x},$$

so that $a_x = 0$ for all $x \in \mathcal{B}_2$. Now, we have that $\sum_{x \in \mathcal{B}_1} a_x x = 0$. As $\mathcal{B}_1$ is a basis, we have that $a_x = 0$ for all $x \in \mathcal{B}_1$. Thus, $a_x = 0$ for all $x \in \mathcal{B}$, so $\mathcal{B}$ is linearly independent. ■

Recall the Universal Property of Vector Spaces: If V and W are vector spaces over F and $\mathcal{B}$ is a basis for V , and $f : \mathcal{B} \rightarrow W$ is a map of sets, then there exists a unique linear transformation $T : V \rightarrow W$ such that $Tv = f(v)$ for $v \in \mathcal{B}$. Notice the corollary that if V and W are vector spaces over F with bases $\mathcal{B}$ and $\mathcal{C}$ respectively, with $|\mathcal{B}| = |\mathcal{C}|$, then $V \cong W$. To prove this, take a bijection between them f and extend to a linear transformation T . Then T is bijective and so an isomorphism.

2.5 Lecture 5

Theorem 2.12. *Let $T : V \rightarrow W$ be linear. Then there exists a subspace $X \subseteq V$ such that $V = \ker T \oplus X$ and $X \cong \operatorname{im} T$.*

Proof. Let $\mathcal{B}_0$ be a basis for $\ker T$, and extend $\mathcal{B}_0$ to a basis $\mathcal{B}$ for V . Let $\mathcal{B}_1 = \mathcal{B} \setminus \mathcal{B}_0$, so that $\mathcal{B} = \mathcal{B}_0 \sqcup \mathcal{B}_1$. Let

$$X = \bigoplus_{v \in \mathcal{B}_1} Fv.$$

As $\ker T = \bigoplus_{v \in \mathcal{B}_0} Fv$, we have that $V = \ker T \oplus X$. It remains to show that $X \cong \operatorname{im} T$. First, we claim that Tv for $v \in \mathcal{B}_1$ are linearly independent. If $\sum_{v \in \mathcal{B}_1} \alpha_v Tv = 0$, then $\sum_{v \in \mathcal{B}_1} \alpha_v v \in \ker T$, so that we can write this as $\sum_{v \in \mathcal{B}_0} \beta_v v$. As $\mathcal{B}$ is a basis, we have that $\alpha_v = \beta_v = 0$ for all v . If $\mathcal{C} = \{Tv : v \in \mathcal{B}_1\}$, we have that $\mathcal{C}$ is linearly independent, and that $|\mathcal{C}| = |\mathcal{B}_1|$. Now, take $x \in V$. Write $x = \sum_{v \in \mathcal{B}} \alpha_v v = \sum_{v \in \mathcal{B}_0} \alpha_v v + \sum_{v \in \mathcal{B}_1} \alpha_v v$. Then, $Tx = \sum_{v \in \mathcal{B}_1} \alpha_v Tv \in \operatorname{Span} \mathcal{C}$, so that $\mathcal{C}$ spans $\operatorname{im} T$. Thus, $\mathcal{C}$ is a basis for $\operatorname{im} T$, and $X \cong \operatorname{im} T$. ■

Let $T : V \rightarrow Z$ be linear, and write $W = \ker T$. Now, $x + W = y + W$ if and only if $x - y \in W$, if and only if $Tx = Ty$. Thus, if $\bar{T} : V/W \rightarrow Z$ is given by $\bar{T}(v + W) = Tv$, this is well defined and linear. Notice also that it is injective, as if $\bar{T}(x + W) = \bar{T}(y + W)$, then $Tx = Ty$ so that $x - y \in W$, meaning that $x + W = y + W$. If we replace Z with $\operatorname{im} T$, we get that $\bar{T}$ is bijective. We have thus deduced the following.

Theorem 2.13 (The First Isomorphism Theorem). *Let $T : V \rightarrow W$ be linear. Then, $V/\ker T \cong \operatorname{im} T$, with an isomorphism defined by $v + \ker T \mapsto Tv$.*

Now, if $\pi : V \rightarrow V/\ker T$ denotes the projection map, which is surjective, and $\psi : \operatorname{im} T \rightarrow Z$ denotes the inclusion map, which is injective, we can write $T = \psi \circ \bar{T} \circ \pi$, so that T is the composition of surjective, bijective, and injective maps respectively.

Recalling the fact that $V = \ker T \oplus X$, where $X \cong \operatorname{im} T$, we have that $X \cong V/\ker T$.

Notice also that, for every subspace W , there exists a linear transformation T such that $W = \ker T$. Namely, take T to be the projection map from V to V/W . This means that kernels of linear transformations correspond bijectively to subspaces, so that the possible images of linear transformations out of V are completely determined by V .

We now discuss projections. Notice that, if $T : V \rightarrow V$, then $\ker T$ and $\operatorname{im} T$ are T -invariant. Suppose that $V = W_1 \oplus \cdots \oplus W_n$, where each W_i is T -invariant. If $\mathcal{B}_i$ is an ordered basis for W_i , and $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_n$ with the obvious order, then $[T]_{\mathcal{B}}$ is a block matrix, as

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T|_{W_1}]_{\mathcal{B}_1} & 0 & \cdots & 0 \\ 0 & [T|_{W_2}]_{\mathcal{B}_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & [T|_{W_n}]_{\mathcal{B}_n} \end{pmatrix}.$$

Definition 2.14. Let V be a vector space over F , and $P : V \rightarrow V$ be linear. We call P a projection if $P^2 = P \circ P = P$.

For example, the zero and identity maps are projections. Notice also that if P is a projection, then so is $1_V - P$, as $(1_V - P)^2 v = (1_V - P)(v - Pv) = v - Pv - Pv + P^2 v = v - Pv = (1_V - P)v$.

Lemma 2.15. *Let $P : V \rightarrow V$ be a projection. Then, $V = \ker P \oplus \operatorname{im} P$.*

Proof. Let $v \in V$, so that $v - Pv \in \ker P$. Now, $Pv \in \operatorname{im} P$, so that $v = (v - Pv) + Pv \in \ker P + \operatorname{im} P$. Now suppose that $v \in \ker P + \operatorname{im} P$. Then, $Pv = 0$, and $v = Pw$ for some w . Thus, $0 = Pv = P^2 w = Pw = v$, so that $v = 0$. ■

If $\mathcal{B}_1$ is an ordered basis for $\ker P$ and $\mathcal{B}_2$ is an ordered basis for $\operatorname{im} P$, and $\mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2$ is an ordered basis for V , then,

$$\begin{aligned}
 [P]_{\mathcal{B}} &= \begin{pmatrix} [P]_{\mathcal{B}_1} & 0 \\ 0 & [P]_{\mathcal{B}_2} \end{pmatrix} = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \\
 &= \begin{pmatrix} \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} & \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{pmatrix} & \begin{pmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{pmatrix} \end{pmatrix}.
 \end{aligned}$$

2.6 Lecture 6

Lemma 2.16. *Let V be a vector space over F , and let $W, X \subseteq V$ be subspaces. Suppose that $V = W \oplus X$. Then there is a unique projection $P : V \rightarrow V$ with $W = \ker P$ and $X = \operatorname{im} P$. We say such a P is the projection along W onto X .*

Proof. For $v \in V$, write v uniquely as $v = w + x$ for $w \in W$ and $x \in X$. Define P to map v to x . Clearly P is linear. It is also clear that $W = \ker P$ and $X = \operatorname{im} P$, so it remains to check that this is a projection. As $Px = x$ for all $x \in X$, P is a projection. Now, if P satisfies the conditions, for some fixed basis $\mathcal{B}$ of X , P must fix $\mathcal{B}$. Since P sends W to 0, the Universal Property of Vector spaces implies that P is unique. ■

Proposition 2.17. *Let V be a vector space over F , and let $W, X \subseteq V$ be subspaces such that $V = W \oplus X$, and let $P : V \rightarrow V$ be the projection along W onto X , and $T : V \rightarrow V$ be linear. Then the following are equivalent.*

(1) W and X are both T -invariant.

(2) $PT = TP$.

Proof. Suppose that $PT = TP$, and let $w \in W$. Then, $(TP)w = T(0) = 0$, so that $(PT)w = 0$. Thus, $Tw \in \ker P = W$, so W is T -invariant. Now let $x \in X$. Then, $Tx = (TP)x = P(Tx) \in \operatorname{im} P = X$. Now suppose that W and X are both T -invariant. For $v \in V$, uniquely write $v = w + x$ for $w \in W$ and $x \in X$. Then, $(TP)v = (TP)(w + x) = Tx$, and $(PT)v = P(Tw + Tx) = Tx$, so that $TP = PT$. ■

Now, suppose that $V = W_1 \oplus \cdots \oplus W_n$. Then, $V = W_i \oplus (W_1 \oplus \cdots \oplus W_{i-1} \oplus W_{i+1} \oplus \cdots \oplus W_n)$. We can define $P_{W_i} : V \rightarrow V$ to be the projection along $W_1 \oplus \cdots \oplus W_{i-1} \oplus W_{i+1} \oplus \cdots \oplus W_n$ onto W_i . Then each of these are linear and projections, and $P_{W_i}P_{W_j} = \delta_{i,j}P_{W_i}$. We also have that $1_V = P_{W_1} + \cdots + P_{W_n}$. Moreover, if $T : V \rightarrow V$ is linear and each W_i is T -invariant, then $TP_{W_i} = P_{W_i}T$, so that

$$T = TP_{W_1} + \cdots + TP_{W_n} = P_{W_1}T + \cdots + P_{W_n}T.$$

Further, we have that if $P_1, \dots, P_n : V \rightarrow V$ are projections, sum to the identity and satisfy $P_iP_j = \delta_{i,j}P_i$, then $V = W_1 \oplus \cdots \oplus W_n$, where $W_i = \operatorname{im} P_i$. Firstly, $V = W_1 + \cdots + W_n$, as for $v \in V$, we have that $v = P_1v + \cdots + P_nv$. Now, if $w_1 + \cdots + w_n = 0$ for $w_i \in W_i$, then, for $P_iv_i = w_i$, we have that $P_1v_1 + \cdots + P_nv_n = 0$. Applying P_i , we have that $0 = P_i^2v_i = P_iv_i = w_i$, so that we have a direct sum.

Let V be a vector space over F and let $\mathcal{B}$ be a basis for V . Fix $v_0 \in \mathcal{B}$, and let $f_{v_0} : V \rightarrow F$ to be the unique linear transformation so that, for $v \in \mathcal{B}$, $f_{v_0}(v) = \delta_{v,v_0}$. Note that this is surjective. Now, suppose that $x \in V$, and uniquely write $x = \sum_{v \in \mathcal{B}} \alpha_v v$. Then, $f_{v_0}(x) = \alpha_{v_0}$. Thus, we can write

$$x = \sum_{v \in \mathcal{B}} f_v(x)v.$$

We call f_v the coordinate function on v relative to $\mathcal{B}$.

Definition 2.18. Let V be a vector space over F . A linear transformation $f : V \rightarrow F$ is called a linear functional. We write $V^* = L(V, F)$ to be the set of linear functionals. This is called the dual space of V .

Recall that if V and W are finite dimensional vector spaces over F , then $\dim L(V, W) = \dim V \dim W$. Notice that in particular, if V is finite dimensional over F , $\dim V^* = \dim V$. Now, notice that if V is finite dimensional with basis $\mathcal{B} = \{v_1, \dots, v_n\}$ and if

$$\sum_{i=1}^n \alpha_i f_{v_i} = 0,$$

then

$$0 = \sum_{i=1}^n \alpha_i f_{v_i}(v_j) = \alpha_j$$

for all j , so that the f_{v_i} are linearly independent. Thus, they are a basis, as $\dim V = \dim V^*$ in this case.

2.7 Lecture 7

Lemma 2.19. *Let V be a vector space over F , and $\mathcal{B}$ be a basis for V . Let $\mathcal{B}^* = \{f_{v_0} : V \rightarrow F : v_0 \in \mathcal{B}\}$. Then $\mathcal{B}^* \subseteq V^*$ is linearly independent.*

Proof. Suppose that

$$0 = \sum_{v \in \mathcal{B}} \beta_v f_v.$$

Then,

$$0 = \sum_{v \in \mathcal{B}} \beta_v f_v(v_0) = \beta_{v_0}$$

for all $v_0 \in \mathcal{B}$, so that $\mathcal{B}^*$ is linearly independent. ■

Corollary 2.20. *Let V be a vector space over F with basis $\mathcal{B}$. Then the linear transformation $D_{\mathcal{B}}$ from V to V^* induced by sending $v \in \mathcal{B}$ to f_v is injective. In particular, if V is finite dimensional, then $\mathcal{B}^*$ is a basis for V^* and $D_{\mathcal{B}}$ is an isomorphism.*

The problem with the above map is that it requires a basis. However, we can create a natural map if we take the dual space twice. Write $V^{**} = (V^*)^*$. We call this the double dual of V .

Lemma 2.21. *Let V be a vector space over F and $v \in V$. Then, the map $L : V \rightarrow V^{**}$ by $v \mapsto L_v$ for*

$$L_v : V^* \rightarrow F \quad L_v(f) = f(v)$$

is an injective linear transformation.

Proof. We first check that L_v is linear. For $f, g \in V^*$ and $\alpha \in F$, we have that

$$L_v(\alpha f + g) = (\alpha f + g)(v) = \alpha f(v) + g(v) = \alpha L_v(f) + L_v(g).$$

Now, if $f \in V^*$, $\alpha \in F$ and $v, w \in V$, then

$$L_{\alpha v + w}(f) = f(\alpha v + w) = \alpha f(v) + f(w) = \alpha L_v(f) + L_w(f) = (\alpha L_v + L_w)(f),$$

so L is linear. Suppose that $L(v) = 0$, so that $f(v) = 0$ for all $f \in V^*$. If v is nonzero, we can extend it to a basis $\mathcal{B}$, and take $f = f_v$ to get a contradiction. Thus, $v = 0$. ■

In particular, if V is finite dimensional, then this is a natural isomorphism. We show that we can identify $v \in V$ with $L_v \in V^{**}$.

Proposition 2.22. *Let V be a finite dimensional vector space over F , and let $v, w \in V$.*

(1) *$v = w$ if and only if $L_v = L_w$.*

(2) *For all $f \in V^*$, $f(v) = f(w)$ if and only if $L_v(f) = L_w(f)$.*

Moreover, if W is also a finite dimensional vector space over F , then if $T : V \rightarrow W$ is linear, there is a unique linear transformation $\tilde{T} : V^{**} \rightarrow W^{**}$ such that $L(T) = \tilde{T}(L)$, and if $\tilde{T} : V^{**} \rightarrow W^{**}$ is linear, there is a unique linear transformation $T : V \rightarrow W$ such that $L^{-1}(\tilde{T}) = T(L^{-1})$.

Proof.

- (1) Clearly if $v = w$, $L_v = L_w$. Now suppose that $L_v = L_w$. As L is injective, $v = w$.
- (2) This follows from the definition of L_v .

Now, suppose that $T : V \rightarrow W$ is linear. Define $\tilde{T}(L_v) = L_{Tv}$. This proves existence. Now, any such $\tilde{T}$ must have that $\tilde{T}(L_v) = L(Tv) = L_{Tv}$, so this is unique. If $\tilde{T} : V^{**} \rightarrow W^{**}$, define $T(v) = L^{-1}(\tilde{T}(L_v))$. This once again proves existence. If T satisfies this property, then we must have that $T(v) = T(L^{-1}(L_v)) = L^{-1}(\tilde{T}(L_v))$, so T is unique. ■

We use this to identify V with V^{**} by identifying v and L_v .

Now suppose that V is a finite dimensional vector space over F with basis $\mathcal{B} = \{v_1, \dots, v_n\}$, and let $\mathcal{B}^* = \{f_1, \dots, f_n\}$ be defined by $f_i(v_j) = \delta_{i,j}$. Then we call $\mathcal{B}^*$ the dual basis of $\mathcal{B}$. Notice that, $L_{v_i}(f_j) = f_j(v_i) = \delta_{i,j}$, so $\mathcal{B}^{**} = \{L_{v_1}, \dots, L_{v_n}\}$ is the dual basis of the dual basis. Now, we know that we can write $v \in V$ as $\sum_{i=1}^n f_i(v)v_i$. For $g \in V^*$, we can repeat this and write it as $\sum_{i=1}^n L_{v_i}(g)f_i = \sum_{i=1}^n g(v_i)f_i$.

Definition 2.23. Let $T : V \rightarrow W$ be linear. The transpose of T , denoted $T^\top$ is the map $T^\top : W^* \rightarrow V^*$ given by $g \mapsto g \circ T$.

Definition 2.24. Let V be a vector space over F and $\emptyset \neq S \subseteq V$. We define the annihilator of S to be

$$S^0 = \{f \in V^* : f(S) = 0\}.$$

Lemma 2.25. Let V be a vector space over F , and $\emptyset \neq S \subseteq V$.

- (1) $S^0 \subseteq V^*$ is a subspace.
- (2) If V is finite dimensional and we identify V and V^{**} , then $S \subseteq S^{00} = (S^0)^0$.

Proof.

- (1) Suppose that $\alpha \in F$, and $f, g \in S^0$. Then,

$$(\alpha f + g)(s) = \alpha f(s) + g(s) = 0$$

for all $s \in S$, so S^0 is a subspace (it is nonempty: $0 \in S^0$).

- (2) Let $s \in S$ and $f \in S^0$. Then,

$$L_s(f) = f(s) = 0,$$

so $L_s \in S^{00}$. ■

2.8 Lecture 8

Theorem 2.26. *Let V be a finite dimensional vector space over F , and $S \subseteq V$ a subspace. Then,*

$$\dim V = \dim S + \dim S^0.$$

Proof. Let $\mathcal{B}_0 = \{v_1, \dots, v_k\}$ be a basis for S , and extend this to a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V . Let $\mathcal{B}^* = \{f_1, \dots, f_n\}$ be the dual basis of $\mathcal{B}$. We claim that $\mathcal{C} = \{f_{k+1}, \dots, f_n\}$ is a basis for S^0 . Clearly, $\mathcal{C}$ is linearly independent, as $\mathcal{B}^* \supseteq \mathcal{C}$ is linearly independent. Now, let $f \in S^0$ and write

$$f = \sum_{i=1}^n L_{v_i}(f)f_i = \sum_{i=1}^n f(v_i)f_i = \sum_{i=k+1}^n f(v_i)f_i.$$

Thus, $\mathcal{C}$ spans, and so is a basis of S^0 . Thus,

$$\dim S + \dim S^0 = |\mathcal{C}| + |\mathcal{B}_1| = |\mathcal{B} \setminus \mathcal{B}_1| + |\mathcal{B}_1| = |\mathcal{B}| = \dim V.$$

■

Corollary 2.27. *Let V be a finite dimensional vector space over F , and $S \subseteq V$ be a subspace. Identifying V and V^{**} , $S = S^{00}$.*

Proof. We have that $S \subseteq S^{00}$, and

$$\dim S + \dim S^0 = \dim V = \dim V^* = \dim S^0 + \dim S^{00},$$

so $\dim S = \dim S^{00}$.

■

Suppose that we have a fixed linear transformation $T : V \rightarrow W$. Then, for every linear $S : W \rightarrow X$, we have a linear transformation $S \circ T : V \rightarrow X$. Thus, $T : V \rightarrow W$ induces a map $T^* : L(W, X) \rightarrow L(V, X)$ by $S \mapsto S \circ T$. Further, this is linear, as if $S_1, S_2 \in L(W, X)$ and $\alpha \in F$, then $T^*(\alpha S_1 + S_2) = (\alpha S_1 + S_2) \circ T = \alpha S_1 \circ T + S_2 \circ T = \alpha T^*(S_1) + T^*(S_2)$. Now, in the case $X = F$, we obtain the transpose $T^\top$.

Lemma 2.28. *Let $T : V \rightarrow W$ be linear. Then $\ker T^\top = (\operatorname{im} T)^0$.*

Proof. Let $g \in \ker T^\top$. Then, $g \circ T = 0$, so $gv = 0$ for all $v \in \operatorname{im} T$. Now, suppose that $g \in (\operatorname{im} T)^0$. Then, $g \circ T = 0$, so $g \in \ker T^\top$.

■

Theorem 2.29. *Let V be a finite dimensional vector space over F , and let $T : V \rightarrow W$ be linear. Then,*

$$\dim \operatorname{im} T = \dim \operatorname{im} T^\top.$$

Proof. We have that

$$\dim \ker T^\top + \dim \operatorname{im} T^\top = \dim W^* = \dim W = \dim \operatorname{im} T + \dim (\operatorname{im} T)^0.$$

By the lemma, $\ker T^\top = (\operatorname{im} T)^0$, so $\dim \operatorname{im} T = \dim \operatorname{im} T^\top$.

■

Proposition 2.30. *Let V be a finite dimensional vector space over F , and $W_1, W_2 \subseteq V$ be subspaces.*

- (1) $W_1 \subseteq W_2$ if and only if $W_1^0 \supseteq W_2^0$.
- (2) $(W_1 + W_2)^0 = W_1^0 \cap W_2^0$.
- (3) $(W_1 \cap W_2)^0 = W_1^0 + W_2^0$.
- (4) $V^*/W_1^0 \cong W_1^*$.
- (5) Let $T : V \rightarrow W$ be linear. Then $\text{im } T^\top = (\ker T)^0$.

Proof.

- (1) Suppose that $W_1 \subseteq W_2$. Then, if $f \in W_2^0$, $f(v) = 0$ for all $W_2 \supseteq W_1$, so that $f \in W_1^0$. Now, if $W_1^0 \supseteq W_2^0$ and $w \in W_1$, then $f(w) = 0$ for all $f \in W_2^0$. As V is finite dimensional, $W_2 = W_2^{00}$, so that if $L_v(f) = f(v) = 0$ for all $f \in W_2^0$, $v \in W_2$. Thus, $w \in W_2$, so $W_1 \subseteq W_2$.
- (2) Suppose that $f \in W_1^0 \cap W_2^0$. Then, for any $w = w_1 + w_2 \in W_1 + W_2$, we have that $f(w) = f(w_1 + w_2) = f(w_1) + f(w_2) = 0$, so that $f \in (W_1 + W_2)^0$. Now, $W_1, W_2 \subseteq W_1 + W_2$, so that $W_1^0, W_2^0 \supseteq (W_1 + W_2)^0$, meaning $W_1^0 \cap W_2^0 \supseteq (W_1 + W_2)^0$.
- (3) Take a basis $\mathcal{B}$ of V such that $\mathcal{B}' \subseteq \mathcal{B}$ is a basis of $W_1 \cap W_2$ and $\mathcal{B}_i \supseteq \mathcal{B}'$ is a basis of W_i . Take $f \in (W_1 \cap W_2)^0$. Write

$$f_1(v \in \mathcal{B}) = \begin{cases} 0 & v \in \mathcal{B}_1 \\ f(v) & v \in \mathcal{B}_2 \\ 0 & \text{else} \end{cases}$$

and

$$f_2(v \in \mathcal{B}) = \begin{cases} 0 & v \in \mathcal{B}_2 \\ f(v) & v \in \mathcal{B}_1 \\ f(v) & \text{else} \end{cases}.$$

Then, $f_i \in W_i^0$, and $f = f_1 + f_2$. Now, suppose that $f = f_1 + f_2 \in W_1^0 + W_2^0$, and take $w \in W_1 \cap W_2$. Then,

$$f(w) = f_1(w) + f_2(w) = 0.$$

- (4) Write $W = W_1$. Consider the inclusion map $i : W \rightarrow V$. Then, $i^\top(f) = f \circ i = f|_W$. Now, take a basis $\mathcal{B}$ of V such that $\mathcal{B}' \subseteq \mathcal{B}$ is a basis of W . For $f \in W^*$, let

$$f'(v \in \mathcal{B}) = \begin{cases} f(v) & v \in \mathcal{B}' \\ 0 & \text{else} \end{cases}.$$

Then, $i^\top(f') = f$, so $\text{im } i^\top = W^*$. Now, $\ker i^\top = \{f : f|_W = 0\} = W^0$, so the first isomorphism theorem implies $V^*/W_1^0 \cong W_1^*$.

- (5) Suppose that $g = h \circ T \in \text{im } T^\top$. If $v \in \ker T$, then $gv = h \circ Tv = h(0) = 0$, so that $g \in (\ker T)^0$. Now, $\dim(\ker T)^0 = \dim V - \dim \ker T = \dim \text{im } T = \dim \text{im } T^\top$, so $\text{im } T^\top = (\ker T)^0$.

■

Suppose that V and W are finite dimensional vector spaces over F , $\mathcal{B}, \mathcal{B}^*$ are ordered dual bases for V, V^* , and $\mathcal{C}, \mathcal{C}^*$ are ordered dual bases for W, W^* . Suppose that $B = \{v_1, \dots, v_n\}$, $\mathcal{B}^* = \{f_1, \dots, f_n\}$, $\mathcal{C} = \{w_1, \dots, w_m\}$ and $\mathcal{C}^* = \{g_1, \dots, g_m\}$. Let $A = [T]_{\mathcal{B}, \mathcal{C}}$ and $B = [T^\top]_{\mathcal{C}^*, \mathcal{B}^*}$. Then,

$$Tv_k = \sum_{i=1}^m A_{i,k} w_i \quad T^\top g_j = \sum_{i=1}^n B_{i,j} f_i.$$

Now,

$$(T^\top g_j)(v_k) = \left(\sum_{i=1}^n B_{i,j} f_i \right) (v_k) = \sum_{i=1}^m B_{i,j} f_i(v_k) = \sum_{i=1}^n \delta_{i,k} B_{i,j}$$

and

$$(T^\top g_j)(v_k) = g_j(Tv_k) = g \left(\sum_{i=1}^m A_{i,k} w_i \right) = \sum_{i=1}^m A_{i,k} g_j(w_i) = A_{j,k}.$$

Thus,

$$B_{k,j} = A_{j,k}$$

for all j, k . Thus, we have obtained the following.

Theorem 2.31. *Let V and W be finite dimensional vector spaces over F , and $T : V \rightarrow W$ be linear. Let $\mathcal{B}, \mathcal{B}^*$ be ordered dual bases of V, V^* and let $\mathcal{C}, \mathcal{C}^*$ be ordered dual bases of W, W^* . Then,*

$$[T^\top]_{\mathcal{C}^*, \mathcal{B}^*} = ([T]_{\mathcal{B}, \mathcal{C}})^\top.$$

Definition 2.32. Let $A \in F^{m \times n}$. The row (column) rank of A is the dimension of the span of the rows (columns).

Now, viewing A as a linear transformation from $F^{n \times 1}$ to $F^{m \times 1}$, we have that the column rank of A is $\dim \text{im } A$ and the row rank of A is $\dim \text{im } A^\top$. Thus, these are equal. We write $\text{rank } A$ for this value.

2.9 Lecture 9

Let $f \in F[t] \setminus F$. We say that f is irreducible if there are not $g, h \in F[t]$ with $f = gh$ and $\deg g, \deg h < \deg f$. Alternatively, if $f = gh$, then either $g \in F$ or $h \in F$.

Lemma 2.33. *Let $f \in F[t]$ be irreducible and that $f|gh$. Then $f|g$ or $f|h$.*

Proof. Suppose that $f \nmid g$. Then $\gcd(f, g) = 1$, so we can write $1 = fk + g\ell$, so that $h = f(kh) + (gh)\ell$. Since $f|f(kh)$ and $f|(gh)\ell$, we have that $f|f(kh) + (gh)\ell = h$. ■

Now, the fundamental theorem of arithmetic still holds for polynomials replacing primes with irreducible: every polynomial can be factored into the product of irreducibles which is unique up to multiplication by elements in F .

2.10 Lecture 10

Recall that if $T : V \rightarrow V$ is linear, and $f \in F[t]$, then $[f(T)]_{\mathcal{B}} = f([T]_{\mathcal{B}})$ for some ordered basis $\mathcal{B}$, and that if $f = gh$ then $f(T) = g(T)h(T)$.

Lemma 2.34. *Let V be a finite dimensional vector space over F , and let $T : V \rightarrow V$ be linear. Then there exists a monic nonzero polynomial q with $q(T) = 0$.*

Proof. Let $n = \dim V$, so that $\dim L(V, V) = n^2$. Thus, $1_V, T, T^2, \dots, T^{n^2}$ are linearly independent, say

$$a_{n^2}T^{n^2} + \dots + a_1^+a_01_V = 0,$$

where not all of the a_i are zero. Let k be the largest value such that $a_k \neq 0$, and write

$$a_kT^k + \dots + a_1T + a_01_V = 0.$$

Then,

$$T^k + (a_{k-1}a_k^{-1})T^{k-1} + \dots + (a_1a_k^{-1})T + (a_0a_k^{-1})1_V = 0.$$

■

Theorem 2.35. *Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. Then there exists a unique nonzero monic polynomial q_T (called the minimal polynomial of T) with $q_T(T) = 0$ and if $g \in F[t]$ satisfies $g(T) = 0$ then $q_T|g$.*

Proof. By the lemma, there are polynomials that have T as a root. Let q be a monic one of least degree. Suppose that $g(T) = 0$, and write $g = aq + b$, where $\deg b < \deg q$ or $b = 0$. Then, $b(T) = b(T) + a(T)q(T) = g(T) = 0$, so that $b = 0$ by assumption. Thus, $q|g$, and in particular q is unique. ■

Given a matrix A , let q_A similarly denote the minimal polynomial of A . Notice that $q_T = q_{[T]_{\mathcal{B}}}$ for some ordered basis $\mathcal{B}$. In particular, if $A \sim B$, then $q_A = q_B$.

Recall that for $T : V \rightarrow V$, we write $f_T(t) = \det(tI - [T]_{\mathcal{B}})$ for some ordered basis $\mathcal{B}$.

Proposition 2.36. *Let V be finite dimensional vector space over F , $T : V \rightarrow V$ be linear. Then q_T and f_T have the same roots in F .*

Proof. Let λ be a root of q_T . Write $q_T = (t - \lambda)h$ in $F[t]$. Then, $0 = q_T(T) = (T - \lambda 1_V)h(T)$. As $\deg h < \deg q_T$, we have that $h(T) \neq 0$, so that there is some nonzero v with $w = h(T)v \neq 0$. Then $0 = q_T(T)v = (T - \lambda 1_V)h(T)v = (T - \lambda 1_V)w$, so that w is an eigenvector of T with eigenvalue λ . Now suppose that λ is a root of f_T , and let $v \neq 0 \in E_T(\lambda)$. Then $t - \lambda \in F[t]$ satisfies $(T - \lambda)w = 0$ for all $w \in Fv$. Then it is the minimal polynomial of $T|_{Fv} : Fv \rightarrow Fv$. As $q_T(T|_{Fv}) = 0$, we have that $t - \lambda|q_T$. ■

2.11 Lecture 11

Notice that if $T : V \rightarrow V$ is linear, and $W \subseteq V$ is T -invariant, then there is an induced linear transformation $\bar{T} : V/W \rightarrow V/W$ by $\bar{T}(v + W) = T(v) + W$. We claim that $q_{\bar{T}} | q_T$. This is as

$$q_T(\bar{T}) = \overline{q_T(T)} = 0,$$

so that $q_{\bar{T}} | q_T$.

Recall that we say that $T : V \rightarrow V$ is triangularizable if there is an ordered basis $\mathcal{B}$ such that $[T]_{\mathcal{B}}$ is upper triangular (or equivalently lower triangular). Notice that if T is triangularizable, with $A = [T]_{\mathcal{B}}$, then

$$f_T(t) = (t - A_{1,1}) \cdots (t - A_{n,n}),$$

by induction using expansion by minors.

Definition 2.37. We say that $g \in F[t] \setminus F$ splits in $F[t]$ if we can write it as the product of linear polynomials in $F[t]$.

Therefore if T is triangularizable, then f_T splits.

Theorem 2.38. Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ linear. Then T is triangularizable if and only if q_T splits in $F[t]$.

Proof. We show this by induction on $n = \dim V$, with $n = 1$ being trivial. Now, suppose that q_T splits in $F[t]$, and that λ is a root of q_T . Then λ is a root of f_T . Take $v_n \neq 0 \in E_T(\lambda)$ so that $W = Fv_n$ is T -invariant. Now, $\dim V/W = n - 1$, and we have a new operator $\bar{T} : V/W \rightarrow V/W$. Now, $q_{\bar{T}} | q_T$, so $q_{\bar{T}}$ splits. By induction, there are $v_1, \dots, v_{n-1} \in V$ such that the ordered basis $\mathcal{C} = \{\bar{v}_1, \dots, \bar{v}_{n-1}\}$ is an ordered basis for V/W making $[\bar{T}]_{\mathcal{C}}$ lower triangular. Now, $v_1, \dots, v_n$ is a basis of V (it suffices to check linear independence, and quotienting a relation means that the coefficient of v_i must be nonzero when $i \neq n$, meaning that the coefficient of v_n must also be nonzero). Now, if $\mathcal{B} = \{v_1, \dots, v_n\}$, then $[T]_{\mathcal{B}}$ is lower triangular. Now, suppose that T is triangularizable, with $\mathcal{B} = \{v_1, \dots, v_n\}$ being such an ordered basis, and let $A = [T]_{\mathcal{B}}$. Then $f_T = (t - A_{1,1}) \cdots (t - A_{n,n})$ splits in $F[t]$. Thus, $A_{1,1}, \dots, A_{n,n}$ are (not necessarily distinct) roots of q_T . Note however that q_T may have an irreducible factor of degree above 1. Write $\lambda_i = A_{i,i}$. Then,

$$Tv_j = \sum_{i=1}^n A_{i,j}v_i = \lambda_j v_j + \sum_{i=j+1}^n A_{i,j}v_i,$$

for $1 \leq j \leq n - 1$ and $Tv_n = \lambda_n v_n$. Thus,

$$(T - \lambda_j 1_V)v_j = \sum_{i=j+1}^n A_{i,j}v_i \in \text{Span}(v_{j+1}, \dots, v_n).$$

Now,

$$(T - \lambda_n 1_V)v_n = 0,$$

and $(T - \lambda_{n-1}1_V)v_{n-1} \in \text{Span}(v_n)$. Thus, $(T - \lambda_n1_V)(T - \lambda_{n-1}1_V)v_{n-1} = 0$. By induction, $(T - \lambda_n1_V) \cdots (T - \lambda_j1_V)v_j = 0$. Thus,

$$f_T(T)v_i = (T - \lambda_11_V) \cdots (T - \lambda_n1_V)v_i = 0,$$

so that $q_T | f_T$. Thus, q_T splits. ■

Recall that a field is algebraically closed if every polynomial splits.

Corollary 2.39. *Let F be algebraically closed, and let V be a finite dimensional vector space over F , with $T : V \rightarrow V$ being linear. Then T is triangularizable, $q_T | f_T$, and $f_T(T) = 0$.*

2.12 Lecture 12

Suppose that V is a finite dimensional vector space over F , and that $T : V \rightarrow V$ is linear. Let $\mathcal{B}$ be an ordered basis for V and $A = [T]_{\mathcal{B}}$. Then $q_A = q_T$ and $f_A = f_T$. If we let $\tilde{F}$ be the algebraic closure of F , then $A \in \mathbb{M}_n(\tilde{F})$, and so $f_A(A)v = 0$ for all $v \in \tilde{F}^{n \times 1}$.

Lemma 2.40. *Let $f \in F[t]$ and $T : V \rightarrow V$ linear. Then $\ker f(T)$ is T -invariant.*

Proof. If $v \in \ker f(T)$, then

$$f(T)(Tv) = T(f(T)v) = T(0) = 0,$$

so $Tv \in \ker f(T)v$. ■

Lemma 2.41. *Let $g, h \in F[t] \setminus F$ be relatively prime, and write $f = gh$. If $T : V \rightarrow V$ is linear and $f(T) = 0$, then $\ker g(T)$ and $\ker h(T)$ are T -invariant, and $V = \ker g(T) \oplus \ker h(T)$.*

Proof. We only need to show that we have a direct sum here. Write $1 = gk + h\ell$. Thus, $1_V = g(T)k(T) + h(T)\ell(T)$, so we can write $v = g(T)k(T)v + h(T)\ell(T)v$ for $v \in V$. Since $f(T) = 0$, we have that $0 = f(T)k(T)v = h(T)(g(T)k(T)v)$, so that $g(T)k(T)v \in \ker h(T)$. Similarly, $h(T)\ell(T)v \in \ker g(T)$. Thus, $V = \ker g(T) + \ker h(T)$. Now, if $g(T)v = h(T)v = 0$, then $v = g(T)k(T)v + h(T)\ell(T)v = 0$, so we have a direct sum. ■

Proposition 2.42. *Let V be a finite dimensional vector space over F , $T : V \rightarrow V$ linear, that $g, h \in F[t] \setminus F$ are monic and $\gcd(g, h) = 1$. Suppose that $q_T = gh$. Then $\ker g(T)$ and $\ker h(T)$ are T -invariant, $V = \ker g(T) \oplus \ker h(T)$, and $g = q_{T|_{\ker g(T)}}$ and $h = q_{T|_{\ker h(T)}}$.*

Proof. We only need to prove the last statement. Clearly, $g(T)|_{\ker g(T)} = 0$ and $h(T)|_{\ker h(T)} = 0$. Thus, $q_{T|_{\ker g(T)}}|g$ and $q_{T|_{\ker h(T)}}|h$. Now, let $f = \text{lcm}(q_{T|_{\ker g(T)}}, q_{T|_{\ker h(T)}}) = q_{T|_{\ker g(T)}}q_{T|_{\ker h(T)}}$ (we must have that $\gcd(q_{T|_{\ker g(T)}}, q_{T|_{\ker h(T)}}) = 1$, as if not, $\gcd(g, h) \neq 1$). Since $V = \ker g(T) \oplus \ker h(T)$, we have that $f(T) = 0$ on V . Thus, $q_T|f$ by definition. Now, $f|gh = q_T$, so $f = q_T$. Now, we have that $\deg q_{T|_{\ker g(T)}} \leq \deg g$, and $\deg q_{T|_{\ker h(T)}} \leq \deg h$, so that $\deg f \leq \deg q_T$. As we have equality here, we must have that $\deg q_{T|_{\ker g(T)}} = \deg g$, and $\deg q_{T|_{\ker h(T)}} = \deg h$. Thus, $q_{T|_{\ker g(T)}} = g$ and $q_{T|_{\ker h(T)}} = h$. ■

2.13 Lecture 13

Theorem 2.43 (Primary Decomposition Theorem). *Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. Let $q_T = p_1^{e_1} \cdots p_r^{e_r}$, where $p_1, \dots, p_r$ are distinct monic irreducible polynomials in $F[t]$. Then there exists a direct sum decomposition of V into subspaces $V = W_1 \oplus \cdots \oplus W_r$ with the following properties.*

(1) *Each W_i is T -invariant.*

(2) *For each i , $q_{T|_{W_i}} = p_i^{e_i}$.*

Moreover, any direct sum decomposition of V satisfying (1) and (2) is uniquely determined by T and the $p_1, \dots, p_r$ up to order.

Proof. We prove existence by induction on r . If $r = 1$, we are done, so suppose that $r > 1$. Write $g = p_2^{e_2} \cdots p_r^{e_r}$, so that $q_T = p_1^{e_1} g$. Since $\gcd(p_1^{e_1}, g) = 1$, we have that $V = W_1 \oplus V_1$, where $W_1 = \ker p_1^{e_1}(T)$ and $V_1 = \ker g(T)$ are T -invariant. We also have that $q_{T|_{W_1}} = p_1^{e_1}$ and $q_{T|_{V_1}} = p_2^{e_2} \cdots p_r^{e_r}$. Now, V_1 is T -invariant, so $T_1 = T|_{V_1} : V_1 \rightarrow V_1$. By induction, we can write $V_1 = W_2 \oplus \cdots \oplus W_r$, where $W_i = \ker p_i^{e_i}(T_1)$ is T_1 - (and thus T -) invariant (we can assume this by our proof), and $q_{T_1|_{W_i}} = q_{T|_{W_i}} = p_i^{e_i}$. Notice that

$$\ker p_i^{e_i}(T_1) \cap \sum_{\substack{2 \leq j \leq r \\ j \neq i}} \ker p_j^{e_j}(T_1) = 0.$$

We first that $\ker p_i^{e_i}(T) = \ker p_i^{e_i}(T_1)$ for $i > 1$. Suppose that $v \in \ker p_i^{e_i}(T)$. Then, $p_i^{e_i}(T)v = 0$, so $g(T)v = 0$. Thus, $v \in \ker g(T) = V_1$, so $Tv = T|_{V_1}v = T_1v$. Hence, $0 = p_i^{e_i}(T)v = p_i^{e_i}(T_1)v$. Now, suppose that $v \in \ker p_i^{e_i}(T_1)$. Then, $v \in V_1$, so $0 = p_i^{e_i}(T_1)v = p_i^{e_i}(T|_{V_1})v = p_i^{e_i}(T)|_{V_1}v = p_i^{e_i}(T)v$. Thus,

$$V = W_1 \oplus V_1 = W_1 \oplus (W_2 \oplus \cdots \oplus W_r) = \bigoplus_{i=1}^r W_i,$$

where $W_i = \ker p_i^{e_i}(T)$ is T -invariant, and $q_{T|_{W_i}} = p_i^{e_i}$.

We now show uniqueness. Suppose that $V = W_1 \oplus \cdots \oplus W_r$ satisfies (1) and (2). We show that $W_i = \ker p_i^{e_i}(T)$. We show that it suffices to show this for $i = 1$. If it is true for $i = 1$, we have that

$$V = \ker p_1^{e_1}(T) \oplus (W_2 \oplus \cdots \oplus W_r).$$

Now, as $q_{T|_{W_i}} = p_i^{e_i}$, we have that $q_{T|_{W_2 \oplus \cdots \oplus W_r}} = p_2^{e_2} \cdots p_r^{e_r}$. Thus, we have that $W_2 = \ker p_2^{e_2}(T)$, and we can repeat this process. We now show it for $i = 1$. Write $V_1 = W_2 \oplus \cdots \oplus W_r$ so that $V = W_1 \oplus V_1$, and V_1 is T -invariant. Let $g = p_2^{e_2} \cdots p_r^{e_r}$. Now, $q_T = \text{lcm}(q_{T|_{W_1}}, q_{T|_{V_1}})$ and $q_{T|_{V_1}} = p_2^{e_2} \cdots p_r^{e_r} = g$. Thus, we can write $1 = p_1^{e_1}k + g\ell$ in $F[t]$. Thus, $1_V = p_1^{e_1}(T)k(T) + g(T)\ell(T)$. Now, $q_{T|_{W_1}} = p_1^{e_1}$, so that $p_1^{e_1}(T)v = 0$ for all $v \in W_1$. Thus, $W_1 \subseteq \ker p_1^{e_1}(T)$. Now, let $v \in \ker p_1^{e_1}(T) \subseteq V = W_1 \oplus V_1$. Write $v = w_1 + v_1$ for $w_1 \in W_1$ and $v_1 \in V_1$. Since $W_1 \subseteq \ker p_1^{e_1}(T)$, we have that $0 = p_1^{e_1}(T)v = p_1^{e_1}(T)(w_1 + v_1) = p_1^{e_1}(T)v_1$. Now, $v_1 \in V_1 = W_2 \oplus \cdots \oplus W_r$, so $g(T)v_1 = p_2^{e_2}(T) \cdots p_r^{e_r}(T)v_1 = 0$. Thus, $v_1 = p_1^{e_1}(T)k(T)v_1 + g(T)\ell(T)v_1 = 0$, so that $v = w_1 \in W_1$. ■

Notice that the above decomposition also gives

$$q_T = \prod_{i=1}^r p_i^{e_i} = \prod_{i=1}^r q_{T|W_i}.$$

Further, if $\mathcal{B}_i$ is an ordered basis for W_i , then $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r$ is an ordered basis for V with

$$[T]_{\mathcal{B}} = \begin{pmatrix} [T|_{W_1}]_{\mathcal{B}_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & [T|_{W_r}]_{\mathcal{B}_r} \end{pmatrix}.$$

2.14 Lecture 14

Proposition 2.44. *Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. If V is the direct sum of the eigenspaces of T , then T is diagonalizable.*

Proof. Suppose that $V = E_T(\lambda_1) \oplus \cdots \oplus E_T(\lambda_r)$. Let $\mathcal{B}_i$ be a basis for $E_T(\lambda_i)$, and $\mathcal{B} = \mathcal{B}_1 \cup \cdots \cup \mathcal{B}_r = \{v_1, \dots, v_n\}$. Notice that

$$n = \sum_{i=1}^r \dim E_T(\lambda_i) = \dim \left(\bigoplus_{i=1}^r E_T(\lambda_i) \right) = \dim V.$$

Notice further that $\mathcal{B}$ is linearly independent. Thus, $\mathcal{B}$ is a basis consisting of eigenvectors of T . ■

Theorem 2.45. *Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. Then T is diagonalizable if and only if q_T splits in $F[t]$ and has no repeated roots.*

Proof. Suppose that $q_T = \prod_{i=1}^r (t - \lambda_i)$, where $\lambda_1, \dots, \lambda_r$ are the distinct roots of q_T . Let $V_i = \ker(T - \lambda_i 1_V) = E_T(\lambda_i)$. Then, we have that $V = V_1 \oplus \cdots \oplus V_r$, so T is diagonalizable. Now, suppose that T is diagonalizable with basis $\mathcal{B} = \{v_1, \dots, v_n\}$, where $\lambda_1, \dots, \lambda_r$ are the distinct eigenvalues of T . Denote by λ_i the eigenvalue of v_i . (so if $i > r$, there is $j \leq r$ with $\lambda_i = \lambda_j$) Then, we have that $(T - \lambda_i 1_V)v_j = (\lambda_j - \lambda_i)v_j$. Now,

$$\prod_{i=1}^r (T - \lambda_i 1_V)v_j = 0,$$

so that $\prod_{i=1}^r (T - \lambda_i 1_V) = 0$. Thus, $q_T \mid \prod_{i=1}^r (t - \lambda_i)$. In particular, q_T splits in $F[t]$ and has no repeated roots. ■

Notice that, in the previous proof, each λ_i is a root of f_T , and so a root of q_T . We thus have that $q_T = \prod_{i=1}^r (t - \lambda_i)$.

Definition 2.46. Let $J \in \mathbb{M}_n(F)$ is called a Jordan Block Matrix of Eigenvalue λ of Size n if

$$J = J_n(\lambda) := \begin{pmatrix} \lambda & 0 & 0 & \cdots & 0 & 0 & 0 \\ 1 & \lambda & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & \lambda & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & 0 & 0 \\ 0 & 0 & 0 & \cdots & 1 & \lambda & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & \lambda \end{pmatrix}.$$

Notice that, by expansion of minors, $f_{J_n(\lambda)} = (t - \lambda)^n$. Further, $q_{J_n(\lambda)} = (t - \lambda)^n$ ($J_n(\lambda)$ is a root of this, and not a root of $(t - \lambda)^r$ for $r < n$).

Definition 2.47. Let V be a vector space over F . Then $T : V \rightarrow V$ is called nilpotent if $q_T = t^m$ for some m . In other words, there is some $m \in \mathbb{Z}^+$ with $T^m = 0$.

For example, $J_n(0)$ (viewed as a linear operator on $F^{n \times 1}$) is nilpotent.

Lemma 2.48. Let $J = J_n(\lambda)$.

- (1) λ is the only eigenvalue of J .
- (2) $\dim E_J(\lambda) = 1$.
- (3) $f_J = q_J = (t - \lambda)^n$.
- (4) $f_J(J) = 0$.

Proof. We only need to prove (2), as the other results are immediate or have been previously established. Let $N = J - \lambda I$. Then $\ker N = E_J(\lambda)$, and $\dim E_J(\lambda) = \dim \ker N = n - \dim \operatorname{im} N = 1$. ■

Notice that $\dim \operatorname{im} N^r = n - r$, by simply computing the matrix.

Proposition 2.49. Let $A \in \mathbb{M}_n(F)$ be triangularizable. Suppose that $f_A = (t - \lambda)^n$ for some $\lambda \in F$. Then A is diagonalizable if and only if $q_A = t - \lambda$ if and only if $A = \lambda I$.

Proof. If $q_A = t - \lambda$, then $A = \lambda I$, and if $A = \lambda I$, then $q_A = t - \lambda$. Now, if A is diagonalizable, then q_A splits and has no repeated roots. As all roots of q_A are roots of f_A , we have that $q_A = t - \lambda$. If $q_A = t - \lambda$, then it splits and has no repeated roots, so A is diagonalizable. ■

Suppose that V is finite dimensional over F , and that $T : V \rightarrow V$. Let $\mathcal{B}$ be a basis with $[T]_{\mathcal{B}} = J_n(\lambda)$. Then $Tv_i = \lambda v_i + v_{i+1}$ for $1 \leq i < n$, and $Tv_n = \lambda v_n$ (in particular, $E_T(\lambda) = Fv_n$). Then $(T - \lambda 1_V)v_i = v_{i+1}$ for $1 \leq i < n$, so that $(T - \lambda 1_V)^{n-i}v_i = v_n$.

2.15 Lecture 15

Definition 2.50. Let $T : V \rightarrow V$ be linear, and $0 \neq v_n \in E_T(\lambda)$. We say $v_1, \dots, v_n$ is an ordered sequence of generalized eigenvectors of eigenvalue λ of length n if $(T - \lambda 1_V)v_i = v_{i+1}$ for $1 \leq i < n$ and $(T - \lambda 1_V)v_n = 0$.

We let $g_n(\lambda) = g_n(v_n, \lambda) = \{v_1, \dots, v_n\}$. Also note that such a collection is not necessarily unique.

Lemma 2.51. Let V be a vector space and $T : V \rightarrow V$ be linear. Take $0 \neq v_n \in E_T(\lambda)$ and let $v_1, \dots, v_n$ be an ordered sequence of generalized eigenvectors of T of length n , with $g_n(\lambda) = \{v_1, \dots, v_n\}$. Then,

- (1) $g_n(\lambda)$ is linearly independent.
- (2) If V is finite dimensional over F and $\dim V = n$ then $g_n(\lambda)$ is an ordered basis for V .
- (3) $[T]_{g_n(\lambda)} = J_n(\lambda)$.

Proof. We clearly only need to prove (1). Now, we have seen that $(T - \lambda 1_V)^{n-i}v_i = v_n$ for $i < n$, and $(T - \lambda 1_V)v_n = 0$. Thus, $(T - \lambda 1_V)^k v_i = 0$ for $k > n - i$. Now, suppose that $\alpha_1 v_1 + \dots + \alpha_n v_n = 0$ for some $\alpha_i \in F$ not all 0. Choose the least k such that $\alpha_k \neq 0$. Then,

$$0 = (T - \lambda 1_V)^{n-k}(\alpha_k v_k + \dots + \alpha_n v_n) = \alpha_k v_n.$$

Thus, $\alpha_k = 0$, a contradiction. ■

Definition 2.52. A matrix $A \in \mathbb{M}_n(F)$ is called a matrix in Jordan Canonical Form (JCF) if A has the block form

$$A = \begin{pmatrix} J_{r_1}(\lambda_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & J_{r_m}(\lambda_m) \end{pmatrix}$$

for not necessarily distinct $\lambda_1, \dots, \lambda_m$.

Notice that if

$$A = \begin{pmatrix} A_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & A_n \end{pmatrix},$$

then $q_A = \text{lcm}(q_{A_1}, \dots, q_{A_n})$.

Definition 2.53. Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. An ordered basis $\mathcal{B}$ for V is called a Jordan Basis (if it exists) for V relative to T if $\mathcal{B} = g_{r_1}(v_{1,r_1}, \lambda_1) \cup \dots \cup g_{r_m}(v_{m,r_m}, \lambda_m)$, where $g_{r_j}(v_{j,r_j}, \lambda_j)$ is an ordered sequence of generalized eigenvectors of T with eigenvalue λ_j ending at v_{j,r_j} . The λ_i need not be distinct.

Proposition 2.54. Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. Then V has a Jordan Basis relative to T if and only if T has a matrix representation in Jordan Canonical Form.

Suppose that $T : V \rightarrow V$ is linear, and that V has a Jordan basis relative to T . We can gather all the blocks with the same eigenvalue together, and ordering into increasing size, we get the following. First, let $\lambda_1, \dots, \lambda_m$ be the distinct eigenvalues of T . Then, we can write

$$\mathcal{B} = \bigcup_{i=1}^m \bigcup_{j=1}^{n_i} g_{r_{i,j}}(v_{i,j}, \lambda_i)$$

with $r_{i,1} \leq r_{i,2} \leq \dots \leq r_{i,n_i}$ for $1 \leq i \leq m$. Now, let $W_{i,j} = \text{Span } g_{r_i}(v_{i,j}, \lambda_i)$, which is T -invariant. Now,

$$f_T = \prod_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n_i}} (t - \lambda_i)^{r_{i,j}},$$

and

$$q_T = \prod_{i=1}^m \text{lcm}_{1 \leq j \leq n_i} ((t - \lambda_i)^{r_{i,j}}) = \prod_{i=1}^m (t - \lambda_i)^{r_{i,n_i}}.$$

In particular, $q_T | f_T$ and $f_T(T) = 0$. We also have that

$$q_T|_{W_{i,j}} = f_T|_{W_{i,j}} = (t - \lambda_i)^{r_{i,j}}.$$

We call these polynomials elementary divisors of T . Now,

$$V = \bigoplus_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n_i}} W_{i,j}.$$

Let $P_{i,j}$ be the projection onto $W_{i,j}$ along

$$\bigoplus_{\substack{1 \leq k \leq m \\ 1 \leq \ell \leq n_k \\ (k,\ell) \neq (i,j)}} W_{k,\ell}.$$

Then,

$$P_{i,j} P_{k,\ell} = \delta_{i,k} \delta_{j,\ell} P_{i,j}, \quad 1_V = \sum_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n_i}} P_{i,j}, \quad T P_{i,j} = P_{i,j} T, \quad \text{and} \quad T = \sum_{\substack{1 \leq i \leq m \\ 1 \leq j \leq n_i}} T P_{i,j}.$$

Let $W_i = \bigoplus_{j=1}^{n_i} W_{i,j}$. As $r_{i,1} \leq \dots \leq r_{i,n_i}$, we have that

$$(T - \lambda_i 1_V)^{r_{i,n_i}}|_{W_{i,j}} = 0$$

for all $1 \leq j \leq n_i$, but

$$(T - \lambda_i 1_V)^{r_{i,n_i}-1}|_{W_{i,j}} \neq 0$$

for some $1 \leq j \leq n_i$. Thus,

$$q_T|_{W_i} = (t - \lambda_i)^{r_{i,n_i}}.$$

Thus,

$$V = W_1 \oplus \dots \oplus W_m$$

is the unique primary decomposition of V relative to T .

Theorem 2.55. *Let V be a finite dimensional vector space over F , and let $T : V \rightarrow V$ be linear. Suppose that q_T splits. then there exists a Jordan basis $\mathcal{B}$ for V relative to T . Moreover, $[T]_{\mathcal{B}}$ is unique up to the order of the Jordan blocks.*

Lemma 2.56. *It suffices to show existence in Theorem 2.55 in the case $q_T = (t - \lambda)^r$.*

Proof. Suppose that

$$q_T = (t - \lambda_1)^{r_1} \cdots (t - \lambda_m)^{r_m},$$

where $\lambda_1, \dots, \lambda_m$ are distinct. Let

$$W_i = \ker(T - \lambda_i 1_V)^{r_i}.$$

By the Primary Decomposition Theorem, $V = W_1 \oplus \cdots \oplus W_m$, W_i is T -invariant, and $q_{T|_{W_i}} = (t - \lambda_i)^{r_i}$. It therefore suffices to find a Jordan basis for each W_i , so it suffices to show that the theorem holds when $q_T = (t - \lambda)^r$. ■

2.16 Lecture 16

Lemma 2.57. *It suffices to prove existence in Theorem 2.55 in the case when $q_T = t^r$.*

Proof. Suppose that this is proven. Let $S = T - \lambda 1_V$, where $q_T = (t - \lambda)^r$. Then $S^r = (T - \lambda 1_V)^r = 0$ and $S^{r-1} = (T - \lambda 1_V)^{r-1} \neq 0$. Thus, $q_S = t^r$. If $\mathcal{B}$ is a Jordan basis for V relative to S , then $[S]_{\mathcal{B}} = [T]_{\mathcal{B}} - \lambda I$ is a Jordan Canonical Form with diagonal entries 0. Thus, $[T]_{\mathcal{B}} = [S]_{\mathcal{B}} + \lambda I$ is a Jordan Canonical Form with diagonal entries λ , and $\mathcal{B}$ is a Jordan Basis for V relative to T . ■

Lemma 2.58. *If $q_T = t^r$, there exists a Jordan basis $\mathcal{B}$ for V relative to T .*

Proof. We induct on $\dim V$, with $\dim V = 1$ being immediate. Suppose that it holds for $\dim V > 1$. Notice that $0 \subsetneq \ker T$, as there is some $v \neq 0$ with $Tv = 0$ (this follows from the fact that $T^{r-1} \neq 0$ but $T^r = 0$). Thus, $\text{im } T \subsetneq V$. As $\text{im } T$ is T -invariant, we can view $T|_{\text{im } T} : \text{im } T \rightarrow \text{im } T$. As $T^r = 0$, we have that $(T|_{\text{im } T})^r = 0$, so $T|_{\text{im } T}$ is also nilpotent. Now, $q_{T|_{\text{im } T}} | q_T$, so $q_{T|_{\text{im } T}}$ splits in $F[t]$, as $q_{T|_{\text{im } T}} = t^s$ for some $s \leq r$. By induction, there exists a Jordan basis for $\text{im } T$ relative to $T|_{\text{im } T}$. Write $\text{im } T = W_1 \oplus \cdots \oplus W_m$ for some m , with each W_i being $T|_{\text{im } T}$ -invariant, and W_i has a basis of an ordered sequence of generalized eigenvectors for $T|_{W_i}$, and so for $T|_{\text{im } T}$ and T . Write $g_{r_i}(0) = \{w_i, Tw_i, \dots, T^{r_i-1}w_i\}$ for this basis. Thus, $T^{r_i}w_i = 0$ and $q_{T|_{W_i}} = t^{r_i}$. Since $w_i \in W_i$, there is $v_i \in V$ with $Tv_i = w_i$. Now, $T^{r_i+1}v_i = 0$ and $T^{r_i}v_i \neq 0$. Thus, $v_i, Tv_i, \dots, T^{r_i}v_i$ is an ordered sequence of generalized eigenvectors for T in V . Let $V_i = \text{Span}(v_i, Tv_i, \dots, T^{r_i}v_i)$. Then,

$$V_i = \left\{ \sum_{j=0}^{r_i} \alpha_j T^j v_i : \alpha_j \in F \right\} = \{f(T)v_i : f \in F[t]_{r_i}\} = \{f(T)v_i : f \in F[t]\}.$$

Now, the V_i are T -invariant. Notice that if $f \in F[t]$ and $f(T)w_i = 0$, then $f(T) = 0$ on W_i , and if $f \in F[t]$ and $f(T)v_i = 0$, then $f(T) = 0$ on V_i . Set $V' = V_1 + \cdots + V_m$. Then V' is T -invariant. We first claim that $V' = V_1 \oplus \cdots \oplus V_m$. Suppose that $u_i \in V_i$ satisfy $u_1 + \cdots + u_m = 0$. Then there is $f_i \in F[t]$ so that $u_i = f_i(T)v_i$. Then, $f_1(T)v_1 + \cdots + f_m(T)v_m = 0$. Thus, taking T , $f_1(T)w_1 + \cdots + f_m(T)w_m = 0$. Thus, $f_i(T)w_i = 0$, so that $f_i(T) = 0$ on W_i . Thus, $t^{r_i} = q_{T|_{W_i}} | f_i$. Write $f_i = tg_i$, with $\deg g_i < \deg f_i$. Then, $f_1(T)v_1 + \cdots + f_m(T)v_m = g_1(T)w_1 + \cdots + g_m(T)w_m = 0$. Thus, $g_i(T)w_i = 0$, so that $g_i(T) = 0$ on W_i . In particular, we can write $g_i = t^{r_i}h_i$, so that $f_i = t^{r_i+1}h_i$. Thus, $f_i(T)v_i = h_i(T)T^{r_i+1}v_i = 0$, so we have a direct sum. Now, we have that

$$TV' = TV_1 \oplus \cdots \oplus TV_m = W_1 \oplus \cdots \oplus W_m = TV.$$

Thus,

$$T|_{V'} = TP_{V_1} + \cdots + TP_{V_m}.$$

Now, we claim that $V = \ker T + V'$. Let $v \in V$. Since $TV' = TV$, there is $v' \in V$ with $Tv' = Tv$, so that $v - v' \in \ker T$. Thus, $V = V' + \ker T$. By construction, there is a Jordan basis $\mathcal{B}_0$ for the T -invariant subspace V' relative to $T|_{V'}$. Let $\mathcal{C} = \{u_1, \dots, u_k\}$ be a basis for $\ker T = E_T(0)$. If $u_1 \notin \text{Span } \mathcal{B}_0$, let $\mathcal{B}_1 = \mathcal{B}_0 \cup \{u_1\}$. Otherwise, let $\mathcal{B}_1 = \mathcal{B}_0$. Continue this process to construct $\mathcal{B} = \mathcal{B}_k$, a linearly independent and spanning set of

V . Write $u_{j,1}, \dots, u_{j,s}$ for the u_i that we added. Then, $Tu_{j,i} = 0$. Thus, each $Fu_{j,i}$ is T -invariant. Further, $u_{j,i}$ is a basis of $Fu_{j,i}$, and is of the form $g_1(u_{j,i}, 0)$. Now, we have that $V = V' \oplus \text{Span}(u_{j,1}, \dots, u_{j,s}) = V_1 \oplus \dots \oplus V_m \oplus Fu_{j,1} \oplus \dots \oplus Fu_{j,s}$. Thus, $\mathcal{B}$ is in Jordan Canonical Form. ■

2.17 Lecture 17

Lemma 2.59. *The representation of a linear transformation in Jordan Canonical Form is unique.*

Proof. It once again suffices to assume that $q_T = t^r$. Let $\mathcal{B}$ be a Jordan basis for V relative to

T , say $[T]_{\mathcal{B}} = \begin{pmatrix} J_{r_1}(0) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & J_{r_m}(0) \end{pmatrix}$. Now, $\text{rank } J_{r_i}^j(0) = r_i - j$ and $\dim \ker J_{r_i}^j(0) = j$ if $j <$

r , and $\text{rank } J_{r_i}^j(0) = 0$, $\dim \ker J_{r_i}^j(0) = r_i$ if $j \geq r_i$. Clearly, $[T]_{\mathcal{B}}^j = \begin{pmatrix} J_{r_1}(0)^j & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & J_{r_m}(0)^j \end{pmatrix}$.

Thus,

$$m_j = \text{rank}[T]_{\mathcal{B}}^j = \sum_{i=1}^m \text{rank } J_{r_i}(0)^j.$$

Thus,

$$m_{j-1} - m_j = \text{rank}[T]_{\mathcal{B}}^{j-1} - \text{rank}[T]_{\mathcal{B}}^j,$$

which is the number of $\ell \times \ell$ Jordan blocks with $\ell \geq j$. We also have that $m_j - m_{j+1}$ is the number of $\ell \times \ell$ Jordan blocks with $\ell \geq j+1$. Thus, there are $(m_{j-1} - m_j) - (m_j - m_{j+1}) = m_{j+1} - 2m_j + m_{j-1}$ Jordan Blocks of size $j \times j$. Thus, $[T]_{\mathcal{B}}$ is unique up to the order of Jordan Blocks. ■

Two immediate consequences are that if $A \in \mathbb{M}_n(F)$ and q_A splits, then A is similar to a matrix in Jordan Canonical Form that is unique up to the order of the Jordan Blocks, and that every $A \in \mathbb{M}_n(F)$ is similar to a matrix in Jordan Canonical Form unique up to the order of the Jordan Blocks when F is algebraically closed.

Theorem 2.60. *Let F be an algebraically closed field, and let $A, B \in \mathbb{M}_n(F)$. Then the following are equivalent.*

- (1) $A \sim B$.
- (2) A and B have the same Jordan Canonical Form up to block order.
- (3) A and B have the same elementary divisors, counted with multiplicities.

Corollary 2.61. *Let F be algebraically closed, and $A \in \mathbb{M}_n(F)$. Then $A \sim A^{\top}$.*

Proof. We have that $q_A = q_{A^{\top}}$ ($q_A(A^{\top})f = f(q_A(A)) = 0$, so $q_{A^{\top}}|q_A$, so they are equal by symmetry). ■

Definition 2.62. Let $g = t^n + a_{n-1}t^{n-1} + \cdots + a_1t + a_0$ for $n \geq 1$. Then the matrix

$$C(g) = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & -a_{n-2} \\ 0 & 0 & \cdots & 0 & 1 & -a_{n-1} \end{pmatrix}$$

is called the companion matrix of g .

For example, $C(t^n) = J_n(0)$. Now, if f, g are monic, then $f = g$ if and only if $C(f) = C(g)$.

Lemma 2.63. *Let $g \in F[t] \setminus F$ be monic. Then $f_{C(g)} = g$.*

Proof. Let $g = t^n + a_{n-1}t^{n-1} + \cdots + a_1t + a_0$. We prove this by induction on n , with $n = 1$ being trivial. Now, by induction using expansion by minors, we get that $f_{C(g)} = t(t^{n-1} + a_{n-1}t^{n-2} + \cdots + a_1) + (-1)^{n-1}a_0(-1)^{n-1} = g$. ■

2.18 Lecture 18

Lemma 2.64. *Let $g \in F[t] \setminus F$ be monic. Then $q_{C(g)} = f_{C(g)} = g$.*

Proof. Take $C : F^{n \times 1} \rightarrow F^{n \times 1}$ so that $C(g) = [C]_S$. Then $Ce_i = e_{i+1}$ for $1 \leq i \leq n-1$, and $Ce_n = -a_0e_1 - a_1e_2 - \dots = a_{n-1}e_n$. In particular, $e_i = C^{i-1}e_1$. Write $h = b_{n-1}t^{n-1} + \dots + b_1t + b_0$. Then $h(C)e_1 = b_{n-1}C^{n-1}e_1 + \dots + b_1Ce_1 + b_0e_1 = b_{n-1}e_n + \dots + b_1e_2 + b_0e_1$. Thus, $h(C)e_1 = 0$ if and only if $h = 0$. Thus, $\deg q_C \geq n$. Now, since $C^ne_1 = -a_0e_1 - a_1e_2 - \dots - a_{n-1}e_n$, we have that $g(C)e_1 = 0$, so that $f_C(C) = g(C) = 0$. Thus, $q_C|g$, so $q_C = g$. ■

Definition 2.65. Let V be a vector space over F , and $T : V \rightarrow V$ be linear. We say that $v \in V$ is a T -cyclic vector for V and V is T -cyclic if

$$V = \text{Span}(v, Tv, \dots, T^nv, \dots).$$

Proposition 2.66. *Let V be a finite dimensional vector space over F , and let $T : V \rightarrow V$ be linear. Suppose that V is T -cyclic, and write $n = \dim V$. Then,*

- (1) $\mathcal{B} = \{v, Tv, \dots, T^{n-1}v\}$ is a basis for V .
- (2) $[T]_{\mathcal{B}} = C(f_T)$.
- (3) $f_T = q_T$.

Proof.

- (1) We must have that $\{v, Tv, \dots, T^nv\}$ is linearly dependent. Let j be the first positive integer with $T^jv \in \text{Span}\{v, Tv, \dots, T^{j-1}v\}$, say

$$T^jv = \alpha_{j-1}T^{j-1}v + \alpha_{j-2}T^{j-2}v + \dots + \alpha_1Tv + \alpha_0v.$$

Then,

$$T^{j+1}v = \alpha_{j-1}T^jv + \alpha_{j-2}T^{j-1}v + \dots + \alpha_1T^2v + \alpha_0Tv \in \text{Span}(T^{j-1}v, T^{j-2}v, \dots, Tv, v).$$

Iterating this, we get that $V = \text{Span}(T^{j-1}v, T^{j-2}v, \dots, Tv, v)$, so $j \geq n$. We must also have that $j \leq n$ by the definition of j .

- (2) We have that $[T]_{\mathcal{B}}$ is of the form

$$C(g) = \begin{pmatrix} 0 & 0 & \dots & 0 & 0 & * \\ 1 & 0 & \dots & 0 & 0 & * \\ 0 & 1 & \dots & 0 & 0 & * \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & 0 & * \\ 0 & 0 & \dots & 0 & 1 & * \end{pmatrix}.$$

This is $C(g)$ for some g , and taking the fundamental polynomial gives that it is $C(f_T)$.

- (3) This follows from (2).

■

Theorem 2.67 (Cayley-Hamilton). *Let V be a finite dimensional vector space over F , and let $T : V \rightarrow V$ be linear. Then $q_T | f_T$ in $F[t]$, and $f_T(T) = 0$.*

Proof. Let $v \in V$. It suffices to show that $f_T(T)v = 0$. Suppose that $v \neq 0$, as this is trivial when $v = 0$. Let W be the T -cyclic subspace of V generated by v , $W = F[T]v$. Let $r = \dim W > 0$. Then, $\mathcal{B} = \{v, Tv, \dots, T^{r-1}v\}$ is a basis for W . Extend this to an ordered basis $\mathcal{C} = \mathcal{B} \cup \mathcal{B}'$ for V . Then,

$$[T]_{\mathcal{C}} = \begin{pmatrix} [T|_W]_{\mathcal{B}} & A \\ 0 & B \end{pmatrix},$$

so $f_T = f_{[T]_{\mathcal{C}}} = f_{T|_W} \cdot f_B$. Now, $f_{T|_W} = q_{T|_W}$, so $f_{T|_W}(T|_W) = 0$. Thus, as $v \in W$, $f_{T|_W}(T)v = 0$. Hence,

$$f_T(T)v = f_{[T]_{\mathcal{C}}}([T]_{\mathcal{C}})[v]_{\mathcal{C}} = f_{T|_W}([T]_{\mathcal{C}})[v]_{\mathcal{C}} \cdot f_B([T]_{\mathcal{C}})[v]_{\mathcal{C}} = 0.$$

■

We say that $A \in F[t]^{m \times n}$ is in Smith Normal Form if $A = 0$ or if A is of the form

$$\begin{pmatrix} q_1 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ 0 & q_2 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & q_3 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & q_r & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 & \cdots & 0 & 0 \end{pmatrix},$$

where $q_1 | q_2 | \cdots | q_r$ are monic. We generalize Gaussian elimination. We say that $A \in \mathbb{M}_n(F[t])$ is an elementary matrix of Type I if there exists $\lambda \in F[t]$ and $\ell \neq k$ such that

$$A_{i,j} = \begin{cases} 1 & i = j \\ \lambda & (i, j) = (k, \ell) \\ 0 & \text{else} \end{cases}.$$

We say that it is an elementary matrix of Type II if there exists $k \neq \ell$ such that

$$A_{i,j} = \begin{cases} 1 & i = j \neq \ell \text{ or } i = j \neq k \\ 0 & i = j = \ell \text{ or } i = j = k \\ 1 & (k, \ell) = (i, j) \text{ or } (k, \ell) = (j, i) \\ 0 & \text{else} \end{cases}.$$

We say that it is an elementary matrix of Type III if there is $0 \neq u \in F$ and ℓ such that

$$A_{i,j} = \begin{cases} 1 & i = j \neq \ell \\ u & i = j = \ell \\ 0 & \text{else} \end{cases}.$$

Type I is equivalent to adding a multiple of a row (column) of A to another row (column) of A . Type II is equivalent to interchanging two rows (columns) of A . Type III is equivalent to multiplying a row (column) by an element in $F[t]$ having a multiplicative inverse. Notice that all elementary matrices are invertible.

Recall that we write $\text{GL}_n(F[t])$ for the set of $A \in \mathbb{M}_n(F[t])$ that are invertible.

Definition 2.68. Let $A, B \in F[t]^{m \times n}$. We say that A is equivalent to B , and write $A \approx B$ if there exist $P \in \text{GL}_m(F[t])$ and $Q \in \text{GL}_n(F[t])$ such that $B = PAQ$.

Theorem 2.69. Let $A \in F[t]^{m \times n}$. Then A is equivalent to a matrix in Smith Normal Form. Moreover, there exist matrices $P \in \text{GL}_m(F[t])$ and $Q \in \text{GL}_n(F[t])$, each a product of matrices of Type I, Type II and Type III, such that PAQ is in Smith Normal Form.

Proof. We provide an algorithm to find a Smith Normal Form for A . If $A = 0$, we are done, so suppose that $A \neq 0$. Write $A = (a_{i,j})$.

First, choose $a = a_{i,j} \neq 0$ such that $\deg a$ is minimal among all $\deg a_{\ell,k}$ for $a_{\ell,k} \neq 0$. Put a in the $(1,1)$ spot using matrices of Type II, so that we may assume $a = a_{1,1}$. Further, use a Type III matrix to make a monic.

Now, if $a|a_{i,j}$ in $F[t]$ for all i and j , use Type I matrices to transform A into a matrix of the form $\begin{pmatrix} a & 0 \\ 0 & A_1 \end{pmatrix}$. In this case, a divides every entry of A_1 , and every non-zero entry of A_1 has degree greater than $\deg a$. In particular, $\deg a$ is still minimal. Now take A_1 and go back to the beginning.

If the previous step does not apply and there is $b = a_{i,j}$ in either the first row or first column such that $a \nmid b$, then write $b = qa + r$ with $r \neq 0$ and $\deg r < \deg a$. Use Type I matrices to change A into a matrix containing r . Since $\deg a$ is no longer minimal, go back to the beginning.

If neither of the previous steps apply, then $a|a_{i,j}$ whenever $i = 1$ or $j = 1$. Use Type I matrices to convert A to $\begin{pmatrix} a & 0 \\ 0 & A_1 \end{pmatrix}$. Now, if there is a non-zero entry b in A_1 with $\deg b < \deg a$, then $\deg a$ is no longer minimal, so go back to the beginning. If $a|b$ for all b in A_1 , then go back to the beginning. If there is $b \in A_1$ with $a \nmid b$, use the division algorithm to write $b = qa + r$ with $r \neq 0$ and $\deg r < \deg a$. Use a Type I matrix to get b into the first column (this does not change the $(1,1)$ entry a). Now use Type I matrices to change b to r . Since $\deg a$ is no longer minimal, go back to the beginning. ■

The Smith Normal Form derived is unique. We call the polynomials $q_1|q_2|\cdots|q_r$ the invariant factors of A . It follows that, if $A, B \in F[t]^{m \times n}$, then $A \approx B$ if and only if they have the same Smith Normal Form, if and only if they have the same invariant factors.

2.19 Lecture 19

Consider the Smith Normal Form of the matrix $tI - [T]_{\mathcal{C}}$ for some ordered basis $\mathcal{C}$ for V . Taking determinants, the Smith Normal Form of $tI - [T]_{\mathcal{C}}$ cannot have a 0 on the diagonal, as $f_T \neq 0$.

Let V be a finite dimensional vector space over F . We want to show that V can be decomposed as a direct sum of T -cyclic subspaces of V . Further, we want to show there is an ordered basis $\mathcal{B}$ for V with $V = W_1 \oplus \cdots \oplus W_r$, where each $W_i = \ker q_{T|_{W_i}}(T|_{W_i})$ is T -cyclic. We further want that $q_i = q_{T|_{W_i}} = f_{T|_{W_i}}$ (where q_i comes from the Smith Normal Form of $tI - [T]_{\mathcal{C}}$ as above), that $q_T = q_r$, that $f_T = q_1 \cdots q_r = q_{T|_{W_1}} \cdots q_{T|_{W_r}}$, and that

$$[T]_{\mathcal{B}} = \begin{pmatrix} C(q_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & C(q_r) \end{pmatrix}.$$

Notice that if $A \in \mathbb{M}_n(F[t])$ is in Smith Normal Form, then $A \in \text{GL}_n F[t]$ if and only if $A = I$, since

$$\begin{pmatrix} q_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & q_r & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix} \in \text{GL}_n F[t]$$

means that $0 \cdots \cdots 0 \cdot q_1 \cdots \cdots q_r = \det A \in F \setminus \{0\}$. This further means that $A \in \mathbb{M}_n(F[t])$ is in $\text{GL}_n F[t]$ if and only if its Smith Normal Form is I .

Definition 2.70. Let $A \in \mathbb{M}_n(F)$. The invariant factors of A are the invariant factors of the matrix $tI - A$. If $T : V \rightarrow V$ is a linear transformation, the invariant factors of T are the invariant factors of $tI - [T]_{\mathcal{B}}$ for some ordered basis $\mathcal{B}$. In both cases, we omit all '1's appearing.

Lemma 2.71. Let $g \in F[t] \setminus F$ be monic of degree n . Then

$$It - C(g) \approx \begin{pmatrix} 1 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \\ 0 & \cdots & 0 & g \end{pmatrix}.$$

Proof. We have that

$$C(g) = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 & -a_0 \\ 1 & 0 & \cdots & 0 & 0 & -a_1 \\ 0 & 1 & \cdots & 0 & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 & -a_{n-2} \\ 0 & 0 & \cdots & 0 & 1 & -a_{n-1} \end{pmatrix},$$

so

$$It - C(g) = \begin{pmatrix} t & 0 & \cdots & 0 & 0 & a_0 \\ -1 & t & \cdots & 0 & 0 & a_1 \\ 0 & -1 & \cdots & 0 & 0 & a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & t & a_{n-2} \\ 0 & 0 & \cdots & 0 & -1 & t + a_{n-1} \end{pmatrix}.$$

Take the n th row, multiply by t , and add it to the $(n-1)$ st row. Now multiply the $(n-1)$ st row by t and add it to the $(n-2)$ nd row. Continue this process to get

$$It - C(g) \approx \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 & t^n + a_{n-1}t^{n-1} + \cdots + a_0 \\ -1 & 0 & \cdots & 0 & 0 & t^{n-1} + a_{n-1}t^{n-2} + \cdots + a_1 \\ 0 & -1 & \cdots & 0 & 0 & t^{n-2} + a_{n-1}t^{n-3} + \cdots + a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 0 & t^2 + a_{n-1}t + a_{n-2} \\ 0 & 0 & \cdots & 0 & -1 & t + a_{n-1} \end{pmatrix}.$$

Now, use column operations to get

$$It - C(g) \approx \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 & g \\ -1 & 0 & \cdots & 0 & 0 & 0 \\ 0 & -1 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & -1 & 0 & 0 \\ 0 & 0 & \cdots & 0 & -1 & 0 \end{pmatrix}.$$

Bringing the top row to the bottom, and multiplying all other rows by -1 , we get

$$It - C(g) \approx \begin{pmatrix} 1 & \cdots & 0 & 0 \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \cdots & 1 & 0 \\ 0 & \cdots & 0 & g \end{pmatrix}.$$

■

The above lemma gives the Smith Normal Form of $C(g)$, which we assume to be unique.

Corollary 2.72. *Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. Let $q_1 | \cdots | q_r$ be the invariants of T in $F[t]$. Then*

$$tI - \begin{pmatrix} C(q_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & C(q_r) \end{pmatrix} \approx \begin{pmatrix} 1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & q_1 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & q_r \end{pmatrix}.$$

Lemma 2.73. *Let $A \approx B$ in $\mathbb{M}_n(F[t])$. Then there are $P, Q \in \mathbb{M}_n(F[t])$, each the products of elementary matrices, such that $A = PBQ$.*

Let $A_i \in \mathbb{M}_r(F)$ for $0 \leq i \leq n$, and let

$$A_n t^n + A_{n-1} t^{n-1} + \cdots + A_0$$

denote

$$A_n(t^n I) + A_{n-1}(t^{n-1} I) + \cdots + A_0(I).$$

Then if $A = (\alpha_{i,j})$, then $At^n = (\alpha_{i,j}t^n)$. This gives us that $(\mathbb{M}_r(F))[t] = \mathbb{M}_r(F[t])$.

2.20 Lecture 20

Lemma 2.74. *Let $A_0 \in \mathbb{M}_n(F)$, let $A = tI - A_0$, and $0 \neq P \in \mathbb{M}_n(F[t])$. Then there are matrices $M, N \in \mathbb{M}_n(F[t])$ and $R, S \in \mathbb{M}_n(F)$ with $P = AM + R$ and $P = NA + S$.*

Proof. Let $m = \max_{\ell,k} \deg P_{\ell,k}$, where $P_{\ell,k} \neq 0$. For all i, j , let $\alpha_{i,j}$ denote the leading coefficient of $P_{i,j}$ if $\deg P_{i,j} = m$, and 0 otherwise. Then, $P_{i,j} = \alpha_{i,j}t^m + f_{i,j}$, where $\deg f_{i,j} < m$. Let $(\alpha_{i,j}) \in \mathbb{M}_n(F)$ and set $P_{m-1} = (\alpha_{i,j})t^{m-1} = (\alpha_{i,j}t^{m-1})$. Then, $AP_{m-1} = (tI - A_0)(\alpha_{i,j})t^{m-1} = (\alpha_{i,j})t^m - A_0(\alpha_{i,j})t^{m-1}$, so this is 0 or has degree m . Thus, the t^m -coefficient of $(AP_{m-1})_{i,j}$ is $\alpha_{i,j}$. Thus, $P - AP_{m-1}$ has polynomial entries of degree at most $m-1$. Apply the same argument to this to get P_{m-2} with $(P - AP_{m-1}) - AP_{m-2}$ having degree at most $m-2$ or 0. Continuing, obtain $P_{m-3}, \dots, P_0$, and let $M = P_{m-1} + P_{m-2} + \dots + P_0$ and $R = P - AM$. This contains only constant entries by construction and $P = AM + R$. The other equation is proven in an analogous way. ■

Theorem 2.75. *Let $A_0, B_0 \in \mathbb{M}_n(F)$, and let $A = tI - A_0$ and $B = tI - B_0$. Then the following are equivalent.*

- (1) $A_0 \sim B_0$.
- (2) $A \approx B$.
- (3) A and B have the same Smith Normal Form.
- (4) A_0 and B_0 have the same invariant factors.

Proof. We only need to show that (1) $\iff$ (2) ((3) and (4) are equivalent, as $tI - A_0$ and $tI - B_0$ cannot have zeros along the diagonal in their Smith Normal Forms). Suppose that $A_0 \sim B_0$, say with $B_0 = PA_0P^{-1}$ for $P \in \text{GL}_n(F)$. Then $P(tI - A_0)P^{-1} = PtP^{-1} - PA_0P^{-1} = tI - B_0 = B$. Now suppose that there are $P_1, Q_1 \in \text{GL}_n(F[t])$ each the product of elementary matrices with $B = P_1AQ_1$. Write $P_1 = BP_2 + R$ for $P_2 \in \mathbb{M}_n(F[t])$ and $R \in \mathbb{M}_n(F)$. Write $Q_1 = Q_2B + S$ for $Q_2 \in \mathbb{M}_n(F[t])$ and $S \in \mathbb{M}_n(F)$. Now, we have that $P_1A = BQ_1^{-1}$ and that $AQ_1 = P_1^{-1}B$. Then,

$$B = (BP_2 + R)AQ_1 = BP_2AQ_1 + RAQ_1 = BP_2P_1^{-1}B + RAQ_1 = BP_2P_1^{-1}B + RAQ_2B + RAS.$$

Now, $R = P_1 - BP_2$, so we have that

$$\begin{aligned} RAQ_2B &= (P_1 - BP_2)AQ_2B = P_1AQ_2B - BP_2AQ_2B \\ &= BQ_1^{-1}Q_2B - BP_2AQ_1B = B[Q_1^{-1}Q_2 - P_2AQ_2]B. \end{aligned}$$

Thus,

$$B = BP_2P_1^{-1}B + B[Q_1^{-1}Q_2 - P_2AQ_2]B + RAS = B[P_2P_1^{-1} + Q_1^{-1}Q_2 - P_2AQ_2]B + RAS.$$

Let $T = P_2P_1^{-1} + Q_1^{-1}Q_2 - P_2AQ_2$. Then $B = BTB + RAS$. Now, every entry of B is 0 or has degree at most 1, and every entry of RAS also is 0 or has degree at most 1. Now, let

$$T = T_mt^m + T_{m-1}t^{m-1} + \cdots + T_0$$

with $T_0, \dots, T_m \in \mathbb{M}_n(F)$. Then $BTB = (tI - B_0)(T_mt^m + \cdots + T_0)(tI - B_0)$. Comparing coefficients of this with $B - RAS$, we get that $T_m = 0$, so $T = 0$. Thus, $B = RAS = RSt + RA_0S$. Thus, $RS = I$ and $B_0 = RA_0S$. In other words, $B_0 = RA_0R^{-1}$. ■

Suppose that V is a finite dimensional vector space, and $T : V \rightarrow V$ is linear with invariant factors $q_1, q_2, \dots, q_r$. Then $\begin{pmatrix} C(q_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & C(q_r) \end{pmatrix}$ has the same invariant factors as

T , so there is some basis $\mathcal{B}$ for V with $[T]_{\mathcal{B}} = \begin{pmatrix} C(q_1) & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & C(q_r) \end{pmatrix}$. This further gives us

$V = W_1 \oplus \cdots \oplus W_r$, and these have the following properties.

- (1) The W_i are T -cyclic.
- (2) $q_i = q_{T|_{W_i}} = f_{T|_{W_i}}$: this follows from properties of companion matrices.
- (3) $q_T = q_r$: $q_T(C(q_r)) = 0$, as this is a block of T , and $q_R(T) = 0$, as the minimal polynomial of each block divides q_R .
- (4) $f_T = q_1 \cdots q_r = q_{T|_{W_1}} \cdots q_{T|_{W_r}}$: the first equality comes from the fact that $f_{C(q_i)} = q_i$, and the second equality is trivial.

Now, as $q_i|q_T$ for $1 \leq i \leq r$, we in particular have the following.

Corollary 2.76. *Let V be a finite dimensional vector space over F , and $T : V \rightarrow V$ be linear. Then $q_T|f_T$ in $F[t]$ and q_T and f_T have the same irreducible factors.*

Corollary 2.77. *Let $A, B \in \mathbb{M}_n(F)$ and let $F \subseteq K$. Then $A \sim B$ in $\mathbb{M}_n(F)$ if and only if $A \sim B$ in $\mathbb{M}_n(K)$.*

Proof. They have the same invariants over F if and only if they have the same invariants over K . ■

2.21 Lecture 21

Given $F \subseteq \mathbb{C}$, we define $\overline{F} = \{\overline{\alpha} : \alpha \in F\}$.

Definition 2.78. Let $F \subseteq \mathbb{C}$ satisfy $F = \overline{F}$, and let V be a vector space over F . Then V is called an inner product space over F relative to $\langle \cdot, \cdot \rangle : V \times V \rightarrow F$ if this satisfies $\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle$, $\langle \alpha v, w \rangle = \alpha \langle v, w \rangle$, $\langle v, w \rangle = \overline{\langle w, v \rangle}$, $\langle v, v \rangle \geq 0$, and $\langle v, v \rangle = 0$ if and only if $v = 0$, for all $v, v_1, v_2, w \in V$ and $\alpha \in F$.

Notice that if $v \in V$ satisfies $\langle w, v \rangle = 0$ for all $w \in V$, then $v = 0$.

Definition 2.79. A map $f : V \rightarrow W$ if inner product spaces over F is called sesquilinear if, for all $v_1, v_2 \in V$ and $\alpha \in F$, $f(v_1 + \alpha v_2) = f(v_1) + \overline{\alpha} f(v_2)$. We write $V^\dagger$ for the set of sesquilinear maps from V to F .

For example, if $F \subseteq \mathbb{R}$, then any sesquilinear map is linear, and $V^\dagger = V^*$. The map from V to V^* given by $v \mapsto \rho_v = \langle \cdot, v \rangle$ is sesquilinear. The map $\lambda_v = \langle v, \cdot \rangle$ is a sesquilinear map from V to F . This also gives an injective map from V to $V^\dagger$.

We say that a sesquilinear map f is a sesquilinear isomorphism if it is bijective and f^{-1} is sesquilinear. By the same proof as the linear case, f is a sesquilinear isomorphism if and only if it is bijective.

We write $\|v\| = \sqrt{\langle v, v \rangle} \geq 0$, and call this the length of V . However, $\|v\|$ may not lie in F .

Recall all definitions and properties given in the previous course.

If $V = W_1 \oplus \cdots \oplus W_n$, and $W_i \perp W_j$ for $i \neq j$, we write $V = W_1 \perp \cdots \perp W_n$. By the orthogonal decomposition theorem, we have that $W_i^\perp = W_1 \perp \cdots \perp W_{i-1} \perp W_{i+1} \perp \cdots \perp W_n$. If $P_i : V \rightarrow V$ is the projection along $W_i^\perp$ onto W_i , then $\ker P_i = W_i^\perp$, $\text{im } P_i = W_i$, $P_i P_j = \delta_{i,j} P_j$, and $1_V = P_1 + \cdots + P_n$. We call the P_i the orthogonal projections. We have that $P_i(v) = v_{W_i}$. In particular, $v = v_{W_1} + \cdots + v_{W_n}$.

Definition 2.80. Let V and W be inner product spaces over F , and let $T : V \rightarrow W$ be linear. A linear transformation $T^* : W \rightarrow V$ is called the adjoint of T if

$$\langle Tv, w \rangle_W = \langle v, T^*w \rangle_V$$

for all $v \in V$ and $w \in W$.

Notice that if such a T^* exists, it is unique. If R and S both satisfy this, then $\langle v, Sw - Rw \rangle = \langle Tv, w \rangle - \langle Tv, w \rangle = 0$ for all v and w , so $R - S = 0$.

2.22 Lecture 22

Theorem 2.81. *Let V and W be finite dimensional inner product spaces over F , and let $T : V \rightarrow W$ be linear. Then the adjoint $T^* : W \rightarrow V$ exists.*

Proof. Let $w \in W$. We define T^* as follows. Let $f_w : V \rightarrow F$ be given by $f_w(v) = \langle Tv, w \rangle$. Then $f_w \in V^*$. Now, we have a bijective sesquilinear map $\rho : V \rightarrow V^*$ by $v \mapsto \rho_v : x \mapsto \langle x, v \rangle$. Thus, there is a unique $x_w \in V$ with $f_w = \rho_{x_w}$. Define $T^*w = x_w$. This satisfies the adjoint property. If $v \in V$, $w_1, w_2 \in W$, and $\alpha \in F$, then

$$\begin{aligned} \langle v, T^*(\alpha w_1 + w_2) \rangle &= \langle Tv, \alpha w_1 + w_2 \rangle = \bar{\alpha} \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle \\ &= \bar{\alpha} \langle v, T^*w_1 \rangle + \langle v, T^*w_2 \rangle = \langle v, \alpha T^*w_1 + T^*w_2 \rangle. \end{aligned}$$

As this holds for all v , T^* is linear. ■

We therefore also have that $T = T^{**} = (T^*)^*$, by uniqueness.

Computing matrices, we get that the matrix representation of T^* is the adjoint of the matrix representation of T , when we use orthonormal bases.

Definition 2.82. Let V and W be inner product spaces over F , and let $T : V \rightarrow W$ be linear. Then T is called an isometry if T is an isomorphism, and if $\langle Tv, Tv' \rangle = \langle v, v' \rangle$ for all $v, v' \in V$.

Theorem 2.83. *Let V and W be finite dimensional inner product spaces over F with $\dim V = \dim W$ and $T : V \rightarrow W$ linear. Then the following are equivalent.*

- (1) $\langle Tv, Tv' \rangle = \langle v, v' \rangle$ for all $v, v' \in V$.
- (2) T is an isometry.
- (3) If $\mathcal{B} = \{v_1, \dots, v_n\}$ is an orthogonal basis for V , then $\mathcal{C} = \{Tv_1, \dots, Tv_n\}$ is an orthogonal basis for W and $\|Tv_i\| = \|v_i\|$ for all i .
- (4) There exists an orthonormal basis $\mathcal{B} = \{v_1, \dots, v_n\}$ for V such that $\mathcal{C} = \{Tv_1, \dots, Tv_n\}$ is an orthonormal basis for W with $\|Tv_i\| = \|v_i\|$ for all i .

In particular, if V has orthonormal bases, then T is an isometry if and only if T takes some (all) orthonormal basis (bases) of V to an orthonormal basis (bases) for W .

Proof. Suppose that $\langle Tv, Tv' \rangle = \langle v, v' \rangle$ for all $v, v' \in V$. Then, $Tv = 0$ if and only if $\langle Tv, Tv \rangle = 0$, if and only if $\langle v, v \rangle = 0$ if and only if $v = 0$. Thus, (1) $\implies$ (2). Now suppose that T is an isometry. That (2) $\implies$ (3) and that (3) $\implies$ (4) is immediate. Now suppose (4). Writing out vectors inside of inner products in terms of $\mathcal{B}$ and $\mathcal{C}$, we obtain (1). ■

Corollary 2.84. *Let V and W be finite dimensional inner product spaces over F both having orthonormal bases. Then V is isometric to W if and only if $\dim V = \dim W$.*

Proof. Use the Universal Property of Vector Spaces and the theorem. ■

Theorem 2.85. *Let V and W be inner product spaces over F . Then, for $T : V \rightarrow W$, $\langle Tv, Tv' \rangle = \langle v, v' \rangle$ for all $v, v' \in V$ if and only if $\|Tv\| = \|v\|$ for all $v \in V$.*

Proof. We only need to show that preserving lengths implies preserving inner products. Let $x, y \in V$, and let $\langle x, y \rangle = \alpha + \beta i$ and $\langle Tx, Ty \rangle = \gamma + \delta i$. Considering $\langle Tx + Ty, Tx + Ty \rangle$, we get that $\alpha = \gamma$. If $F \subseteq \mathbb{R}$, we are done, so suppose $F \not\subseteq \mathbb{R}$. Now, as $F = \overline{F}$, we have that there is some nonzero $\mu \in \mathbb{R}$ with $\mu i \in F$. Now,

$$\langle x, (\mu i)y \rangle = (-\mu i)\langle x, y \rangle = (-\mu i)(\alpha + i\beta) = -\mu i\alpha + \beta\mu$$

and

$$\langle Tx, (\mu i)Ty \rangle = (-\mu i)\langle Tx, Ty \rangle = (-\mu i)(\gamma + \delta i) = -\mu i\gamma + \delta\mu.$$

We thus have that $\beta\mu = \delta\mu$ for the same reason that $\alpha = \gamma$, so $\beta = \delta$. ■

2.23 Lecture 23

Definition 2.86. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. We say that T is a unitary operator if T is an isometry. If $F \subseteq \mathbb{R}$, we say T is an orthogonal operator.

Proposition 2.87. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. Suppose that T^* exists. Then T is an isometry if and only if $T^* = T^{-1}$.

Proof. Suppose that T is an isometry, so that it is in particular invertible. Then,

$$\langle Tx, y \rangle = \langle Tx, 1_V y \rangle = \langle Tx, TT^{-1}y \rangle = \langle x, T^{-1}y \rangle.$$

By uniqueness of the adjoint, $T^* = T^{-1}$. Now suppose that $T^* = T^{-1}$. Then,

$$\langle Tv, Tw \rangle = \langle v, T^*Tw \rangle = \langle v, T^{-1}Tw \rangle = \langle v, w \rangle.$$

As T is invertible, it is an isomorphism, and so an isometry. ■

Notice that we have further showed that adjoints exist for isometries. We have also shown that T is unitary if and only if T^* is unitary, as $(T^*)^* = T$.

Definition 2.88. Let $F \subseteq \mathbb{C}$ with $\overline{F} = F$. We say $A \in \mathbb{M}_n(F)$ is unitary if $A^*A = I$, or equivalently if $AA^* = I$. We write $U_n(F) = \{A \in \text{GL}_n(F) : AA^* = I\}$.

As A can be viewed as a linear transformation from $F^{n \times 1}$ to $F^{n \times 1}$, we get that A is unitary if and only if the columns of A form an ordered orthonormal basis for $F^{n \times 1}$, which happens if and only if the rows of A form an ordered orthonormal basis for $F^{1 \times n}$.

If $T : V \rightarrow V$ is linear, $\dim V = n$, and $\mathcal{B}, \mathcal{C}$ are orthonormal bases for V , then T is unitary if and only if $[T]_{\mathcal{B}, \mathcal{C}}$ is unitary.

Lemma 2.89. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. Let $W \subseteq V$ be a subspace, and suppose that T^* exists. Then if W is T -invariant, $W^\perp$ is T^* -invariant.

Proof. Let $v \in W^\perp$ and $w \in W$. Then,

$$\langle w, T^*v \rangle = \langle Tw, v \rangle = 0.$$
■

Lemma 2.90. Let V be a finite dimensional inner product space over F , and let $T : V \rightarrow V$ be linear. If λ is an eigenvalue of T , then $\bar{\lambda}$ is an eigenvalue of T^* .

Proof. Let $S = T - \lambda 1_V$. Then $S^* = T^* - \bar{\lambda} 1_V$. Let $0 \neq v \in E_T(\lambda)$. For all $w \in V$, we have that

$$0 = \langle 0, w \rangle = \langle Sv, w \rangle = \langle v, S^*w \rangle.$$

Thus, $v \perp \text{im } S^*$ and $v \notin \text{im } S^*$, as $v \neq 0$. Thus, $0 \subsetneq \ker S^*$, so that $E_{T^*}(\bar{\lambda}) \neq 0$. ■

Theorem 2.91 (Schur). *Let V be a finite dimensional inner product space over F , with $F = \mathbb{R}$ or $\mathbb{C}$. Let $T : V \rightarrow V$ be linear. Suppose that f_T splits in $F[t]$. Then there exists an ordered orthonormal basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}$ is upper triangular.*

Proof. We induct on $n = \dim V$, with $n = 1$ being immediate. Let λ be a root of f_T . Then there is $0 \neq v \in E_{T^*}(\bar{\lambda})$. Now, $V = (Fv) \perp (Fv)^{\perp}$, and $\dim(Fv)^{\perp} = n - 1$. Now Fv is T^* -invariant, so $(Fv)^{\perp}$ is T -invariant. Let $\mathcal{C}_0$ be an ordered basis for $(Fv)^{\perp}$. Then $\mathcal{C} = \mathcal{C}_0 \cup \{v\}$ is an ordered basis for V , and we have

$$[T]_{\mathcal{C}} = \begin{pmatrix} [T|_{(Fv)^{\perp}}]_{\mathcal{C}_0} & [Tv]_{\mathcal{C}_0} \\ 0 & | \end{pmatrix}.$$

Now, $f_{T|_{(Fv)^{\perp}}} | f_T$, so it splits. By induction, there is an orthonormal basis $\mathcal{B} = \{v_1, \dots, v_{n-1}\}$ for $(Fv)^{\perp}$ such that $[T|_{(Fv)^{\perp}}]_{\mathcal{B}_0}$ is upper triangular. Then, $\mathcal{B} = \mathcal{B}_0 \cup \{v/||v||\}$ is an orthonormal basis for V such that $[T]_{\mathcal{B}}$ is upper triangular. ■

2.24 Lecture 24

Definition 2.92. Let V be an inner product space over F , and let $T : V \rightarrow V$ be linear. Suppose that T^* exists. We say that T is normal if $TT^* = T^*T$, and T is hermitian (or self-adjoint) if $T = T^*$.

Lemma 2.93. Let V be an inner product space over F , $\lambda \in F$, $0 \neq v \in V$, and $T : V \rightarrow V$ be normal. Then $v \in E_T(\lambda)$ if and only if $v \in E_{T^*}(\bar{\lambda})$.

Proof. Let $S = T - \lambda 1_V$, so that $S^* = T^* - \bar{\lambda} 1_V$. It follows that S is normal. Now,

$$\|Sv\|^2 = \langle Sv, Sv \rangle = \langle v, S^*Sv \rangle = \langle v, SS^*v \rangle = \langle S^*v, S^*v \rangle = \|S^*v\|^2.$$

Thus, $v \in E_T(\lambda)$ if and only if $Sv = 0$ if and only if $S^*v = 0$ if and only if $v \in E_{T^*}(\bar{\lambda})$. ■

Corollary 2.94. Let V be an inner product space over F , $T : V \rightarrow V$ be normal, and $\lambda \neq \mu$ be eigenvalues of T . Then $E_T(\lambda)$ and $E_T(\mu)$ are orthogonal.

Proof. Let $0 \neq v \in E_T(\lambda)$ and $0 \neq w \in E_T(\mu)$. Then,

$$\lambda \langle v, w \rangle = \langle \lambda v, w \rangle = \langle Tv, w \rangle = \langle vT^*w \rangle = \langle v, \bar{\mu}w \rangle = \bar{\mu} \langle v, w \rangle.$$

As $\lambda \neq \mu$, $\langle v, w \rangle = 0$. ■

Proposition 2.95. Let V be a finite dimensional inner product space over $F = \mathbb{R}$ or $\mathbb{C}$, and let $T : V \rightarrow V$ be linear. Suppose that $\mathcal{B}$ is an ordered orthonormal basis for V such that $[T]_{\mathcal{B}}$ is upper triangular. Then T is normal if and only if $[T]_{\mathcal{B}}$ is diagonal.

Proof. Suppose that $[T]_{\mathcal{B}}$ is diagonal, say with diagonal entries $\lambda_1, \dots, \lambda_n$ (in that order). Then $[T^*]_{\mathcal{B}}$ has diagonal entries $\bar{\lambda}_1, \dots, \bar{\lambda}_n$. Hence, $[TT^*]_{\mathcal{B}}$ and $[T^*T]_{\mathcal{B}}$ are diagonal, and have entries $|\lambda_1|^2, \dots, |\lambda_n|^2$, so $TT^* = T^*T$. Now suppose that T is normal, and let $A = [T]_{\mathcal{B}}$ be upper triangular. Write $\mathcal{B} = \{v_1, \dots, v_n\}$. Then $Tv_1 = A_{1,1}v_1$, and $T^*v_1 = \bar{A}_{1,1}v_1$. We also have that $T^*v_1 = \sum_{i=1}^n \bar{A}_{1,i}v_i$, so that $A_{1,i} = 0$ for $i > 1$. In particular $A_{1,2} = 0$, so $Tv_2 = A_{2,2}v_2$. Continue this process to obtain that A is diagonal. ■

Corollary 2.96 (Spectral Theorem for Normal Operators). Let V be a finite dimensional inner product space over $\mathbb{C}$, and let $T : V \rightarrow V$ be linear. Then T is normal if and only if there exists an orthonormal basis $\mathcal{B}$ for V consisting of eigenvectors of T .

Proof. f_T splits by the fundamental theorem of algebra, so by Schur's Theorem, T is triangularizable. The rest follows from the previous proposition. ■

This is also true in $\mathbb{R}$ assuming that f_T splits.

Suppose that V is a finite dimensional inner product space over $F = \mathbb{R}$ or $\mathbb{C}$, $T : V \rightarrow V$ is linear, f_T splits, and T is normal. Let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of T in F , and let $\mathcal{C}$ be an orthonormal basis for V . We know that $v \in E_T(\lambda_i)$ if and only if $v \in E_{T^*}(\bar{\lambda}_i)$. Let P_i be the orthogonal projection along $E_T(\lambda_i)^\perp$ onto $E_T(\lambda_i)$. Then this is also the orthogonal projection along $E_{T^*}(\bar{\lambda}_i)^\perp$ onto $E_{T^*}(\bar{\lambda}_i)$. This is unique with $E_T(\lambda_i)$ and $E_{T^*}(\bar{\lambda}_i)$ being T - and T^* -invariant respectively. Thus, $TP_i = P_iT$ and $T^*P_i = P_iT^*$. We also have that $1_V = P_1 + \dots + P_r$ and that $P_iP_j = P_i\delta_{i,j}$, $T = \lambda_1P_1 + \dots + \lambda_rP_r$, and $T^* = \bar{\lambda}_1P_1 + \dots + \bar{\lambda}_rP_r$. If $\mathcal{B}_i$ is an ordered orthonormal basis for $E_T(\lambda_i)$, then $\mathcal{B} = \mathcal{B}_1 \cup \dots \cup \mathcal{B}_r$ is an orthonormal basis for V with $[T]_{\mathcal{B}}$ and $[T^*]_{\mathcal{B}}$ being diagonal. If $Q = [1_V]_{\mathcal{B},\mathcal{C}}$, then Q is unitary, as it takes an orthonormal basis to an orthonormal basis, so $Q^{-1} = Q^*$, $[T]_{\mathcal{B}} = Q^*[T]_{\mathcal{C}}Q$, and $[T^*]_{\mathcal{B}} = Q^*[T^*]_{\mathcal{C}}Q$.

2.25 Lecture 25

Lemma 2.97. *Let V be an inner product space over F , and let $T : V \rightarrow V$ be hermitian. If λ is an eigenvalue of T , then λ is real.*

Proof. Let $0 \neq v \in E_T(\lambda)$. Then

$$\lambda \|v\|^2 = \lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle Tv, v \rangle = \langle v, Tv \rangle = \langle v, \lambda v \rangle = \bar{\lambda} \langle v, v \rangle = \bar{\lambda} \|v\|^2.$$

Thus, $\lambda = \bar{\lambda}$, as $v \neq 0$. ■

Lemma 2.98. *Let V be a finite dimensional inner product space over $F = \mathbb{R}$ or $\mathbb{C}$, and let $T : V \rightarrow V$ be hermitian. Then f_T splits in $F[t]$.*

Proof. We may assume $F = \mathbb{R}$. Let $\mathcal{B}$ be an orthonormal basis for V . Then $A = [T]_{\mathcal{B}} = [T^*]_{\mathcal{B}} = [T]_{\mathcal{B}}^* = A^*$ in $\mathbb{M}_n(\mathbb{R}) \subseteq \mathbb{M}_n(\mathbb{C})$. As $A : \mathbb{C}^{n \times 1} \rightarrow \mathbb{C}^{n \times 1}$ is hermitian, $f_A \in \mathbb{C}[t]$ splits with real roots by the previous lemma. Thus, it splits in $\mathbb{R}[t]$. ■

Theorem 2.99 (Spectral Theorem for Hermitian Operators). *Let V be a finite dimensional inner product space over $F = \mathbb{R}$ or $\mathbb{C}$, and let $T : V \rightarrow V$ be hermitian. Then there exists an orthonormal basis for V of eigenvalues of T , where all eigenvalues are real.*

Proof. If $F = \mathbb{C}$, we are done, so suppose $F = \mathbb{R}$. Since f_T splits in $\mathbb{R}[t]$, there exists an orthonormal basis $\mathcal{B}$ for V such that $[T]_{\mathcal{B}}$ is upper triangular. Since T is normal, $[T]_{\mathcal{B}}$ is diagonal. ■

The converse of this is also true.

Lemma 2.100. *Let $q = t^2 + bt + c$ in $\mathbb{R}[t]$, with $b^2 < 4c$. If V is a finite dimensional inner product space over $\mathbb{R}$ and $T : V \rightarrow V$ is hermitian, then $q(T)$ is an isomorphism.*

Proof. It suffices to show that $q(T)$ is injective. Let $0 \neq v \in V$. We have that

$$\begin{aligned} \langle q(T)v, v \rangle &= \langle T^2v, v \rangle + b \langle Tv, v \rangle + c \langle v, v \rangle = \langle Tv, Tv \rangle + b \langle Tv, v \rangle + c \langle v, v \rangle \\ &= \|Tv\|^2 + b \langle Tv, v \rangle + c \|v\|^2 \geq \|Tv\|^2 - |b| \cdot \|Tv\| \cdot \|v\| + c \|v\|^2 \\ &= \left(\|Tv\| - \frac{|b|}{2} \|v\| \right)^2 + \left(c - \frac{b^2}{4} \right) \|v\|^2 > 0. \end{aligned}$$

Thus $q(T)v \neq 0$. ■

Let V be a finite dimensional inner product space over $\mathbb{R}$, and $T : V \rightarrow V$ be hermitian. Let $q \in \mathbb{R}[t]$ be monic and satisfy $q(T) = 0$. Write $q = (t - \alpha_1)^{e_1} \cdots (t - \alpha_r)^{e_r} q_1^{f_1} \cdots q_s^{f_s}$, where the q_i are irreducible monic quadratics. Then

$$0 = q(T) = (T - \alpha_1 1_V)^{e_1} \cdots (T - \alpha_r 1_V)^{e_r} q_1(T)^{f_1} \cdots q_s(T)^{f_s}.$$

As the $q_i(T)$ are isomorphisms, at least one of the $T - \alpha_i 1_V$'s is not injective, so T must have at least one eigenvalue.

2.26 Lecture 26

We restrict to $F = \mathbb{R}$ or $\mathbb{C}$.

Let V be a finite dimensional inner product space over F , $T : V \rightarrow V$ be hermitian, $\mathcal{B} = \{v_1, \dots, v_n\}$ be an orthonormal basis of eigenvalues of T , and $Tv_i = \lambda_i v_i$. Then $\lambda_i \in \mathbb{R}$ for all i . Suppose that $v \in V$ with $v = \sum_{i=1}^n \alpha_i v_i$. Then,

$$\langle Tv, v \rangle = \left\langle \sum_{i=1}^n T(\alpha_i v_i), \sum_{j=1}^n \alpha_j v_j \right\rangle = \left\langle \sum_{i=1}^n \lambda_i \alpha_i v_i, \sum_{j=1}^n \alpha_j v_j \right\rangle = \sum_{1 \leq i, j \leq n} \lambda_i \alpha_i \overline{\alpha_j} \langle v_i, v_j \rangle = \sum_{i=1}^n \lambda_i |\alpha_i|^2.$$

Definition 2.101. Let V be a finite dimensional inner product space over F , and $T : V \rightarrow V$ be hermitian. We say that T is positive or positive definite (nonnegative or positive semidefinite) if $\langle Tv, v \rangle > 0$ (≥ 0) for all $0 \neq v \in V$. We say that it is negative or negative definite (nonpositive or negative semidefinite) if $\langle Tv, v \rangle < 0$ (≤ 0) for all $0 \neq v \in V$.

Proposition 2.102. Let V be a finite dimensional inner product space over F , and $T : V \rightarrow V$ be hermitian. Then T is positive semidefinite (positive) if and only if all eigenvalues of T are nonnegative (positive).

Theorem 2.103. Let V be a finite dimensional inner product space over F , and $T : V \rightarrow V$ be hermitian. Then T is nonnegative (positive) if and only if there is a nonnegative (positive) $S : V \rightarrow V$ such that $T = S^2$.

Proof. Suppose that T is hermitian, and let $\mathcal{B}$ be an ordered orthonormal basis for V of eigenvectors of T . Write $Tv_i = \lambda_i v_i$ for $\lambda_i \geq 0$ in $\mathbb{R}$. Then there are $\mu_i \geq 0$ with $\lambda_i = \mu_i^2$. Let S be defined by $Sv_i = \mu_i v_i$. Further, $S^2 = T$. Now, $[S^*]_{\mathcal{B}} = [S]_{\mathcal{B}}^* = [S]_{\mathcal{B}}$, so S is hermitian. Now suppose that such an S exists. Then, as S is hermitian, there is an ordered orthonormal basis $\mathcal{B}$ for V of eigenvectors of S . Then $[T]_{\mathcal{B}} = [S]_{\mathcal{B}}^2$ is diagonal. Thus, $\mathcal{B}$ is an orthonormal basis for V of eigenvectors of T , each with real eigenvalues, so T is hermitian. ■

Theorem 2.104. Let V be a finite dimensional inner product space over F , and $T : V \rightarrow V$ be hermitian. Suppose that T is nonnegative. Then T has a unique square root S .

Proof. Let $S^2 = T$, where $S : V \rightarrow V$ is nonnegative. Write $V = E_T(\lambda_1) \perp \dots \perp E_T(\lambda_r) = E_S(\mu_1) \perp \dots \perp E_S(\mu_s)$. Now, $S = \mu_1 P_{\mu_1} + \dots + \mu_s P_{\mu_s}$, so

$$S^2 = (\mu_1 P_{\mu_1} + \dots + \mu_s P_{\mu_s})(\mu_1 P_{\mu_1} + \dots + \mu_s P_{\mu_s}) = \mu_1^2 P_{\mu_1} + \dots + \mu_s^2 P_{\mu_s}.$$

Thus,

$$\mu_1^2 P_{\mu_1} + \dots + \mu_s^2 P_{\mu_s} = \lambda_1 P_{\lambda_1} + \dots + \lambda_r P_{\lambda_r}.$$

By uniqueness, we have $s = r$ and, rearranging if necessary, $\mu_i^2 = \lambda_i$ and $P_{\mu_i} = P_{\lambda_i}$ for all i . ■

Lemma 2.105. Let V and W be finite dimensional inner product spaces over F , and let $T : V \rightarrow W$ be linear. Then $T^*T : V \rightarrow V$ is hermitian and nonnegative.

Proof. Let $x, y \in V$. Then,

$$\langle T^*Tx, y \rangle = \langle Tx, Ty \rangle = \langle x, T^*Ty \rangle.$$

Let λ be an eigenvalue of T^*T , and $0 \neq v \in V$ such that $T^*Tv = \lambda v$. Then

$$\lambda \|v\|^2 = \lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle T^*Tv, v \rangle = \langle Tv, Tv \rangle = \|Tv\|^2,$$

so

$$\lambda = \frac{\|Tv\|^2}{\|v\|^2} \geq 0.$$

■

Corollary 2.106. *Let V be a finite dimensional inner product space over F , and $T : V \rightarrow V$ be linear. Then T is nonnegative (positive) if and only if there is a linear (isomorphism) $S : V \rightarrow V$ such that $T = S^*S$.*

Proof. If T is nonnegative, this follows from the theorem. The other direction follows from the lemma. ■

Given $A \in F^{m \times n}$, we write $A^{(i)}$ for the i th column of A , and we will use $\langle \cdot, \cdot \rangle$ to denote the dot product on F^n .

Definition 2.107. Let $D \in F^{m \times n}$. We say that D is pseudodiagonal if $D_{i,j} = 0$ for all $i \neq j$.

Theorem 2.108 (Singular Value Theorem). *Let $A \in F^{m \times n}$. Then there are $U \in U_n(F)$ and $X \in U_m(F)$ such that $D = X^*AU$ is a pseudodiagonal matrix satisfying $D_{1,1} \geq D_{2,2} \geq \dots \geq D_{r,r} > 0$, where $r = \text{rank } A$, and $D_{i,i} = 0$ for $i > r$.*

Proof. We know that A^*A is hermitian and has nonnegative eigenvalues. Let $\lambda_1, \dots, \lambda_r$ be the positive eigenvalues ordered such that $\lambda_1 \geq \dots \geq \lambda_r > 0$. Then, there exists a unitary operator I such that

$$(AU)^*(AU) = U^*A^*AU = \begin{pmatrix} \lambda_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \dots & \lambda_r & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix} \in \mathbb{M}_n(F).$$

Let $C = AU$. Then $C^*C = (AU)^*(AU)$. Write $\lambda_i = \mu_i^2$ for $\mu_i > 0$. Set

$$B = \begin{pmatrix} \mu_1 & \dots & 0 & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \dots & \mu_r & 0 & \dots & 0 \\ 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & 0 & \dots & 0 \end{pmatrix} \in \mathbb{M}_n(F).$$

Write $\lambda_i = 0$ for $i > r$. Then,

$$\lambda_i \delta_{i,j} = (C^* C)_{i,j} = \sum_{\ell=1}^n (C^*)_{i,\ell} C_{\ell,j} = \sum_{\ell=1}^n \overline{C_{\ell,i}} C_{\ell,j} = \langle C^{(j)}, C^{(i)} \rangle.$$

Thus, we have $C = [C^{(1)} \ \dots \ C^{(r)} \ 0 \ \dots \ 0]$. We also have that $\{C^{(1)}, \dots, C^{(r)}\}$ is orthogonal, and so linearly independent. In particular, $\text{rank } C = r$. Now, $\|C^{(i)}\|^2 = \langle C^{(i)}, C^{(i)} \rangle = \lambda_i = \mu_i^2$. As U is invertible, $\text{rank } A = \text{rank } AU = \text{rank } C = r$. Let $X^{(i)} = \frac{1}{\mu_i} C^{(i)}$ for $1 \leq i \leq r$. Then $\{X^{(1)}, \dots, X^{(r)}\}$ is an orthonormal set. Extend this to an orthonormal basis $\mathcal{B} = \{X^{(1)}, \dots, X^{(m)}\}$. Then $X = [X^{(1)} \ \dots \ X^{(m)}] = [1_{F^{m \times 1}}]_{\mathcal{S}, \mathcal{B}}$. As both $\mathcal{S}$ and $\mathcal{B}$ are orthonormal bases, $X \in U_m(F)$. Let D be the pseudodiagonal matrix

$$\begin{pmatrix} \mu_1 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & 0 \\ 0 & \cdots & \mu_r & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix} \in F^{m \times n}.$$

Then $XD = [\mu_1 X^{(1)} \ \dots \ \mu_r X^{(r)} \ 0 \ \dots \ 0] = C = AU$. Thus, $D = X^* AU$. ■

2.27 Lecture 27

We call this the Singular Value Decomposition of A , and the $\mu_1, \dots, \mu_r$ the singular values of A .

Corollary 2.109. *The singular values of $A \in F^{m \times n}$ are unique (including multiplicity) up to order.*

Proof. Let $A = XDU^*$ be a singular value decomposition of A . Then, $A^*A = (XDU^*)^*(XDU^*) = UD^*X^*XDU^* = UD^*DU^*$. Thus, $A^*A \sim D^*S$, so the eigenvalues of A^*A are $D_{1,1}^2, \dots, D_{r,r}^2$. ■

Corollary 2.110. *The singular values of $A \in F^{m \times n}$ are the same as the singular values of $A^* \in F^{n \times m}$.*

Proof. $(XDU^*)^* = UD^*X^*$, and D and D^* have the same nonzero diagonal entries. ■

Theorem 2.111 (Singular Value Theorem for Linear Transformations). *Let V, W be finite dimensional inner product spaces over F , and let $T : V \rightarrow W$ be linear of rank r . Then there exists orthonormal bases $\mathcal{B} = \{v_1, \dots, v_n\}$ for V and $\mathcal{C} = \{w_1, \dots, w_m\}$ for W , along with $\mu_1 \geq \dots \geq \mu_r > 0$ satisfying*

$$Tv_i = \begin{cases} \mu_i w_i & 1 \leq i \leq r \\ 0 & i > r \end{cases}.$$

*Conversely, if the above are satisfied, then v_i is an eigenvector for T^*T with eigenvalue μ_i^2 for $1 \leq i \leq r$, and eigenvalue 0 for $i > r$.*

The proof of this follows from the Matrix Theory Theorem.

Theorem 2.112 (Polar Decomposition). *Let $A \in \mathbb{M}_n(F)$. Then there is $\tilde{U} \in U_n(F)$ and $N \in \mathbb{M}_n(F)$ hermitian with real nonnegative eigenvalues satisfying $A = \tilde{U}N$.*

Proof. Let $A = XDU^*$ be a singular value decomposition, for $X, U \in U_n(F)$. We have that $D = D^*$ is hermitian with nonnegative eigenvalues, so

$$A = XDU^* = XU^*UDU^* = (XU^*)(UDU^*).$$

Since $(XU^*)^*(XU^*) = UX^*XU^* = I$, $XU^* \in U_n(F)$. If $\tilde{U} = XU^*$ and $N = UDU^*$, then $A = \tilde{U}N$. ■

Suppose that V and X are finite dimensional inner product spaces over F , and $T : X \rightarrow V$ is linear. Let $W = \text{im } T$. For $v \in V$ and $x \in X$, we call x a best approximation to v via T if $Tx = v_W$. We call x an optimal approximation to v via T if it is a best approximation to v via T and $\|x\|$ is minimal among all best approximations to v via T . Now, we have that

$$\langle x, T^*y \rangle = \langle Tx, y \rangle,$$

we have that $W^\perp = (\text{im } T)^\perp = \ker T^*$. Since $v - v_W \in W^\perp$, we have that $T^*v = T^*v_W$. Thus, if x is a best approximation of v via T , then $T^*Tx = T^*v$. Conversely, suppose that

this holds. Then, $Tx - v \in \ker T^* = W^\perp$. In particular, $v_W = (Tx - (Tx - v))_W = (Tx)_W - (Tx - v)_W = Tx$. Thus, x is a best approximation to v via T if and only if $T^*Tx = T^*v$.

Suppose that T is injective. Then T^*T is an isomorphism, and $P_W = T(T^*T)^{-1}T^*$, where $P_W(v) = v_W$. If $T^*Tx = 0$, then $0 = \langle T^*Tx, x \rangle = \langle Tx, Tx \rangle$, so $Tx = 0$, meaning $x = 0$. Thus, T^*T is injective, so an isomorphism. Let $v \in V$. Now, since T^*T is an isomorphism, there exists an $x \in X$ such that $T^*Tx = T^*v$, so $T(T^*T)^{-1}T^*v = T(T^*T)^{-1}(T^*T)x = Tx = v_W = P_W(v)$.

2.28 Lecture 28

We now claim that, for $v \in V$, there exists a unique $x \in X$ that is an optimal approximation to v via T . For any two best approximations x, x' to v via T , $T^*Tx = T^*Tx'$, so that $x - x' \in \ker T^*T$. Thus, if x is a best approximation to v via T , then any other is of the form $x + y$ for $y \in Y = \ker T^*T$. For such $x + y$, $P_Y(x + y) = P_Y(x) + y$. Write $x'' = x - P_Y(x)$. Then $P_Y(x'') = 0$, so x'' is orthogonal to Y . Thus, $\|x'' + y\|^2 = \|x''\|^2 + \|y\|^2 \geq \|x''\|^2$. Thus, $x'' = P_{Y^\perp}(x)$ is the unique optimal approximation. This further shows that when T is injective, $x'' = (T^*T)^{-1}T^*v$, as $(T^*T)x'' = T^*v$, and $\ker T^*T = 0$.